

The Price of Delay: infrastructure lock-in and the carbon tax lock-in premium in the oil sector

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Abstract

Development costs are sunk once an oil field is built, so a carbon tax high enough to have prevented its development can be too low to stop its production, which only needs to cover operating costs. Delaying climate policy therefore locks in emissions that the delayed policy can no longer remove. Existing models of oil supply give each field a single cost per barrel, which rules this lock-in out. I separate the decision to develop a field from the decision to keep it producing for 25,336 oil fields, which differ widely in costs, production profiles and carbon intensities. Delaying a 1.5°C-aligned carbon tax from 2016 to 2030 overshoots the oil carbon budget by 19%, and a quarter of the overshoot comes after 2030 from fields that a timely tax would have deterred. A planner who ignores lock-in sets the 2030 tax too low, overshooting the remaining budget by 12%, and the additional tax needed to close this gap is the lock-in premium. Because lock-in is concentrated in a few hundred projects resilient to a higher tax, its cost falls on low-cost producers elsewhere, thus redistributing stranded assets, and early targeted restrictions on these projects can remove much of it.

KEYWORDS: Climate change, fossil fuels, oil, carbon tax, climate policy delay, carbon lock-in, stranded assets, infrastructure lock-in, supply-side climate policy.

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1 Introduction

In 2015, the Paris Agreement set the objective of holding global warming well below 2°C and pursuing efforts to limit it to 1.5°C. Meeting this objective requires leaving a large share of known oil reserves in the ground (McGlade and Ekins, 2015; Welsby et al., 2021), and carbon pricing is widely regarded as the most effective way to get there. In the decade since, however, carbon prices have remained limited in coverage and far below the levels these objectives require (Metcalfe, 2019). Oil companies do expect some form of climate policy, but they screen new projects with internal carbon prices well below what the objectives imply, often only to stress-test existing assets (Ahluwalia, 2017). Recent evidence suggests why: firms invest according to the carbon price they find credible, and under policy uncertainty this price sits far below the announced one (Dolls et al., 2026). The years since Paris are therefore best described as a delay in carbon pricing, and one that investors largely did not expect to end.

A delay of this kind has two consequences for the oil sector. The first is overproduction: fields already in operation keep producing barrels that a timely tax would have retired. The second is investment: firms develop new fields that a timely tax would have deterred. The two differ in when their emissions occur. Overproduction stops when the tax arrives, since costlier fields are then phased out. Investment has long-lasting effects, because development costs are sunk once a field is built: from then on, the field only needs to cover its operating costs, and a field that is cheap to operate can keep producing for decades. A tax high enough to have prevented the development of a field is therefore not necessarily high enough to stop its production once it exists. The delay leaves behind a stock of committed emissions that the delayed tax cannot remove, and the tax must rise beyond what overproduction alone would justify to meet the same carbon budget.

Existing models of oil supply cannot capture this second consequence, because they collapse the decision to develop a field and the decision to keep it producing into a single cost per barrel. Integrated assessment models represent fossil fuel supply as a cost curve in cumulative extraction, with no memory of past investment (Grubb et al., 2021). Asset-level models of oil supply, which have recently brought field-level detail into the analysis of climate policy, assign each field a single cost per barrel that spreads its development cost over its lifetime (Heal and Schlenker, 2019; Asker et al., 2019; Coulomb et al., 2026). Under this convention, a field occupies the same position on the supply curve whether or not its development capital has already been spent, so that any tax high enough to have deterred a project is also high enough to shut it down once built, and lock-in cannot arise. Applied to the cost of inaction, these models count the overproduction that occurs during the delay

and miss the emissions the delay locks in.

This paper brings lock-in into an asset-level model of oil supply, by separating, for each field, the decision to develop it from the decision to keep it producing. I use this model to measure how much of the cost of the delay since Paris comes from lock-in and what tax would make up for it, and to show how lock-in changes who bears the cost of the delay.

I first develop a simple theoretical model, in which a field sells its oil at a given constant price and faces a carbon tax that is implemented late. Before development, a field must expect to cover both its development cost and its operating costs, which rise with age. Once the development cost is sunk, only operating costs remain, and the highest tax the field can bear jumps upward. The taxes in between, high enough to have prevented the field but too low to stop it once built, form its *lock-in zone*. The depth of this zone is set by capital intensity, and its position by operating costs and carbon intensity. A field developed during the delay locks in emissions if the tax that is eventually implemented falls inside its zone, and if the field is still young enough to cover its operating costs at that date. I then compare the tax needed after the delay to meet the carbon budget with the tax set by a planner who, as existing models do, gives each field a single cost per barrel, which I call the *naive planner*. This planner treats locked-in fields as if they could still be deterred, and sets the tax too low. Its error can be measured in tonnes, as the emissions it leaves above the budget, which I call the *residual overshoot*, or in dollars, as the additional tax needed to close the gap, the *carbon tax lock-in premium*. The model extends to a tax rising over time, in which case the development threshold of a field depends on the year it was developed.

I take the model to asset-level data from Rystad Energy’s UCube database, covering 25,336 oil fields with positive oil reserves in 2016. Oil fields differ along every dimension that matters for lock-in, and I take into account the heterogeneity across fields in each of these dimensions. For each field, development costs are separated from operating costs, the latter increasing stage by stage as the field depletes. The production profile of a given field is fixed once it is built, following recent evidence on geological constraints on fields production. Its lifecycle carbon intensity, covering extraction, refining and final use, is estimated from engineering models and satellite observations of flaring, and its oil is sold at a discount to Brent that reflects its quality. These dimensions vary widely. Among the fields developed during the delay, the median shale field needs \$3 of development capital per barrel of annual capacity and the median deepwater field \$121, lifecycle carbon intensities have a standard deviation of about a quarter of their mean, and price discounts range from 3% for light crudes to 29% for bitumen. Whether a field locks in emissions depends on how these dimensions combine within the field, which a supply curve built on averages would hide, and the results indeed show very different patterns for shale, oil sands and deepwater fields.

Investments observed up to 2025 and Rystad’s forecast thereafter provide the business-as-usual counterfactual. The reference Brent price is \$80/bbl and the discount rate 10%.

The model focuses on supply: the oil price does not respond to the tax, which is therefore borne entirely by producers. This assumption matters less than it may seem, for three reasons. First, the existence of lock-in does not depend on demand. Once its development cost is sunk, a field keeps producing as long as its operating margin covers the tax, whatever the path of prices, so that a price response would change how much is locked in while leaving the mechanism intact. Second, the sensitivity analysis considers a partial pass-through of the tax to consumers and alternative oil prices, which move the residual overshoot by less than 1.5 GtCO₂eq in either direction. Third, lock-in sits deep inside the fields that carry it. A moderate price response would therefore leave the identity of locked-in fields, and the incidence of the delay, largely unchanged.

The first exercise compares a \$57.2/tCO₂eq carbon tax aligned with the oil carbon budget of [Welsby et al. \(2021\)](#), implemented in 2016 and rising at 2% per year, with the same tax delayed to 2030 and starting at the level the timely tax would have reached by then. The delay overshoots the oil carbon budget by 66.7 GtCO₂eq, 19% of the budget and about four years of global oil emissions at their current rate.¹ The intensive margin accounts for 62% of this overshoot, and the extensive one, from fields developed only because of the delay, for 38%. Almost two thirds of the extensive margin is released after the tax is implemented. Shale and oil sands account for most of the overshoot before 2030, while offshore fields concentrate three quarters of the emissions released after 2030, and deepwater fields alone 39%. The model explains this pattern: shale projects are cheap to develop and expensive to operate, so their lock-in zones are shallow and low, while deepwater projects are capital-intensive and cheap to operate once built.

The second exercise asks what tax is needed in 2030 to make up for the delay, and how far from it the naive planner ends up. Meeting the 144.9 GtCO₂eq left in the budget in 2030 requires a tax of \$70.5, while the naive planner sets \$66.8.² The lock-in premium is therefore \$3.6/tCO₂eq, 5.4% of the naive tax. At the naive tax, the fields locked in by the delay leave a residual overshoot of 17.1 GtCO₂eq, 12% of the remaining budget. Both measures survive realistic investor expectations of future climate policy: 80% remain when investors screen their projects with half the required tax, the level closest to the internal carbon prices firms report ([Ahluwalia, 2017](#)).

¹Total lifecycle emissions of oil amount to 16.5 GtCO₂eq in 2025, using estimated carbon intensities and Rystad data on production.

²All tax levels in the paper are expressed as 2016-anchored levels. The tax increasing by 2% annually, these levels correspond to \$88.2 and \$93.0 in 2030.

Lock-in is concentrated and resilient. The 17.1 GtCO₂eq committed beyond 2030 come from 694 fields, 41 of which carry half of the total, and raising the tax to the level the budget requires only removes 8% of this stock. The premium therefore strands production elsewhere, so that the delay redistributes stranded assets through two channels. Overproduction during the delay must be offset by stranding other fields after it, as the tax rises from \$57.2 to \$66.8, and locked-in fields, which barely respond to a higher tax, require stranding additional fields as the tax rises from \$66.8 to \$70.5. North America gains through both channels, since its shale and oil sands fields overproduce during the delay and its deepwater projects lock in production afterwards. The Middle East and Russia lose through both, because a higher tax retires their low-cost fields earlier without affecting them during the delay. More interestingly, lock-in can reverse the incidence of the delay: Brazil and Africa lose through overproduction but gain more through lock-in, and end up producing more than under a timely tax.

This concentration makes targeted policy effective against lock-in. A ban on new deepwater approvals from 2016 would have cut the residual overshoot by 40% and brought the premium down to \$2.5, while a ban on new shale approvals would have left both close to their initial level. Banning the fifty projects that carry the most locked-in emissions would have cut the residual overshoot by more than half. Acting later loses most of the gain: applied from 2025, the deepwater ban cuts the residual overshoot by only 11%, since most of the projects concerned have been approved by then.

Related literature. This paper relates to three literatures. The first models the supply of fossil fuels. A theoretical strand departs from the Hotelling framework by treating supply as a sequence of fixed investments, each followed by a production path that cannot be freely adjusted (Venables, 2014; Okullo and Reynès, 2016; Kellogg, 2026). Anderson et al. (2018) provide the empirical support for this view, showing that producers respond to prices by drilling more or fewer wells rather than by adjusting the flow from wells already drilled, and Wachtmeister and Höök (2020) illustrate its consequences with representative profiles of conventional fields and tight oil wells. An empirical strand builds asset-level models of oil supply from Rystad data, to study the incidence of a carbon tax (Heal and Schlenker, 2019), the misallocation of extraction across fields and the role of market power (Asker et al., 2019; De Cannière, 2025), and the excess emissions caused by extracting carbon-intensive barrels before cleaner ones (Asker et al., 2024; Coulomb et al., 2026). These asset-level models assign each field a single cost per barrel, a reasonable simplification when the question is how a given quantity of oil is allocated across fields, and a restrictive one when the question is the timing of policy. This paper bridges the two strands. It takes the theoretical view to the data, with each field’s own costs, carbon intensity and production path, and it brings

fixed investments and fixed production paths into an asset-level model, which makes lock-in emerge when climate policy arrives late.

The second studies the cost of delaying climate policy, mostly with integrated assessment models. These studies find that delaying ambitious policy until 2030 raises its cost substantially (Riahi et al., 2015; Kotlikoff et al., 2021), and that the models with the richest representation of inertia can put the objectives out of reach (Jakob et al., 2012; Luderer et al., 2013; Bertram et al., 2015). Inertia in these models comes from fixed investments on the demand side, mostly in power generation, while fossil fuel supply enters as a cost curve in cumulative extraction, and abatement in each period is treated as independent of past abatement (Grubb et al., 2021). This paper introduces inertia on the supply side and opens the black box of these models, by tracing lock-in to the specific investments that create it.

The third turns flows of emissions into stocks, either as emissions committed by existing infrastructure or as stranded assets. On the first side, the literature on carbon lock-in (Unruh, 2000; Seto et al., 2016) measures the emissions that existing infrastructure will release if it operates over its historical lifetime and at its historical utilization rate, whatever the policy in place. Following the accounting of Davis and Socolow (2014), Tong et al. (2019) show that existing energy infrastructure already commits most of the 1.5°C budget, and Erickson et al. (2015) and Trout et al. (2022) apply the same accounting to fossil fuel supply, the latter finding that fields already producing or under construction commit more emissions than a 1.5°C budget allows. These studies have no counterfactual and no early retirement: they cannot say which part of the committed stock exists because policy was delayed, nor which part would survive a carbon tax. Comparing a timely tax with a delayed one isolates the commitments the delay created, and counting only the production that remains profitable under the tax measures the part of them that policy can no longer remove, together with the year each commitment became irreversible and the tax needed to retire it.

On the other side, the literature on stranded assets has established the size of unburnable reserves (Meinshausen et al., 2009; McCollum et al., 2014) and their uneven distribution across countries (McGlade and Ekins, 2015; Welsby et al., 2021), which depends on extraction costs and carbon intensities (Coulomb et al., 2026). It has also traced the corresponding losses to their ultimate owners (Mercure et al., 2018; Semieniuk et al., 2022), and examined how owners respond when they expect the policy (Coulomb and Henriot, 2018; van der Ploeg and Rezai, 2020). In these studies, policy arrives at a given date, and stranding is computed on a supply curve with no memory of investment. This paper shows how a delay reallocates stranded assets through two channels, overproduction and lock-in, and how the stranding it requires builds up with each year of inaction.

Finally, measuring lock-in field by field makes it possible to evaluate specific supply-side restrictions rather than their principle, which contributes to the case for supply-side climate policy discussed in Section 5.3.

The rest of the paper is organised as follows. Section 2 presents the model of field development and production under a delayed carbon tax, defines the lock-in zone of a field and the factors that determine it, and defines the residual overshoot and the lock-in premium. Section 3 describes the data and the assumptions made on costs, prices, carbon intensities and fiscal regimes. Section 4 reports the overshoot caused by the delay, the emissions committed by new projects, the lock-in premium and the residual overshoot, and their sensitivity to expectations, assumptions and the design of the tax. Section 5 locates the locked-in fields, shows how lock-in changes the incidence of the delay across countries and firms, and evaluates targeted policies. Section 6 concludes.

2 A simplified model of investment and lock-in under delayed carbon taxation

This section presents a simple model of investment in oil fields when a carbon tax is implemented late. It serves two purposes: defining the objects measured in the empirical part, and showing which field characteristics determine their size. The *lock-in zone* of a field is the range of carbon taxes that would have prevented its development but cannot stop its production once the development cost is sunk. *Committed emissions* at a given tax are the future emissions of existing fields that survive this tax, and *overcommitted emissions* are the committed emissions of fields that a timely tax would have deterred. The *lock-in premium* is the additional tax required after the delay to offset overcommitted emissions, on top of the increase needed to compensate for overproduction during the delay. I use a constant tax to keep the model analytically tractable, and show in Section 2.6 that the main results carry over to an increasing tax. Closed-form expressions, which require no discounting and a parametric cohort of fields, are in Appendix A.1.

Production profiles are fixed by geology and by the initial design of the field, so that the tax affects when a field starts and stops producing, but not how much it produces in a given year. Okullo and Reynès (2016) model supply in the same way: production is bounded by installed capacity and by a geological limit on the share of reserves extracted each year, and capacity only expands through investment. Anderson et al. (2018) show that production from existing wells hardly responds to prices, while drilling does. With fixed profiles and

development dates taken from the data, producers who expect the tax cannot accelerate extraction before it is implemented, which rules out the green paradox and makes the total cost of the delay a lower bound. Expecting the tax can also deter investment during the delay, which reduces lock-in, as shown in Section 2.5.

2.1 Setup

A carbon tax τ is levied on producers. Under the timely policy it is in place from 2016; under the delayed policy it is implemented at date Δ . From Δ onwards, both policies apply the same tax.

A field is characterised by a development cost C , paid when the field is developed; a production flow $\bar{q}$; a carbon intensity θ , so that it emits $\theta\bar{q}$ per period; and a selling price p , constant over time.³ Its operating cost per barrel increases with its age a , counted from the start of production,

$$c(a) = c_0 + \gamma a, \quad \gamma > 0. \quad (1)$$

Operating costs rise as a field gets depleted because reservoir pressure declines and more water is produced with the oil, which requires more energy to lift each barrel (Chermak and Patrick, 1995; Leighty and Lin, 2012), and the production breakevens computed from the data increase accordingly across production stages (Section 3). Capital intensity $\kappa \equiv C/\bar{q}$ is the development cost per barrel of annual capacity. Firms discount profits at rate δ , while emissions are not discounted.

2.2 Two breakevens and the lock-in zone

Once developed, a field of age a earns $p - c_0 - \gamma a - \theta\tau$ per barrel. It keeps producing as long as this margin is positive, that is as long as the tax does not exceed its *production breakeven* $\bar{\tau}_P(a)$, and it is phased out at age $a^*(\tau)$:

$$\bar{\tau}_P(a) \equiv \frac{p - c_0 - \gamma a}{\theta}, \quad a^*(\tau) = \frac{p - c_0 - \theta\tau}{\gamma}. \quad (2)$$

The production breakeven only depends on operating costs, since the development cost is sunk once the field produces. It is highest at start-up, $\bar{\tau}_P(0) = (p - c_0)/\theta$, and falls by γ/θ each year.

³The selling price is the common reference price minus a discount that depends on the characteristics of the oil (Section 3.3).

Before development, investors decide to invest in a field if its net present value is positive:

$$-C + \int_0^{a^*(\tau)} e^{-\delta a} \bar{q} (p - c_0 - \theta\tau - \gamma a) da \geq 0. \quad (3)$$

Equivalently, the discounted profit per barrel of capacity must cover the capital intensity of the field,

$$\bar{v}_\delta(\tau) \equiv \int_0^{a^*(\tau)} e^{-\delta a} (p - c_0 - \theta\tau - \gamma a) da \geq \kappa. \quad (4)$$

The discounted profit $\bar{v}_\delta(\tau)$ decreases with the tax, so the condition holds if and only if the tax does not exceed the *development breakeven* $\bar{\tau}_D$, defined by

$$\bar{v}_\delta(\bar{\tau}_D) = \kappa. \quad (5)$$

$\bar{\tau}_D$ is the model counterpart of the maximum viable tax computed field by field in Section 3. Since $\kappa > 0$, $\bar{\tau}_D < \bar{\tau}_P(0)$: the taxes between the two breakevens would have prevented the development of the field but do not stop its production once it exists. This interval is the field's lock-in zone. I describe it by its depth, its width and its height, which Figure 1 illustrates. Panel (a) plots the two breakevens against capital intensity for a field of a given age at implementation, and panel (b) plots them against the age at implementation for a field of a given capital intensity. In both panels, region II is the set of taxes at which the field is locked in.

Depth. The depth of the zone is the distance between the two breakevens at start-up, $D \equiv \bar{\tau}_P(0) - \bar{\tau}_D$. Expressed per barrel, $\theta D = p - c_0 - \theta\bar{\tau}_D$ is the margin at start-up at which the field just recovers its development cost, and from (4) it solves

$$\int_0^{\theta D/\gamma} e^{-\delta a} (\theta D - \gamma a) da = \kappa. \quad (6)$$

The depth reflects how a field's costs split between fixed and variable costs. The production breakeven only has to cover variable costs, while the development breakeven also has to cover the fixed development cost, so the larger the fixed cost per barrel of capacity, the further apart the two breakevens. In panel (a), the development breakeven falls as capital intensity rises and region II opens up; in panel (b), a more capital-intensive field has a lower horizontal line $\bar{\tau}_D$ and a taller region II. Expressed per barrel, the depth depends on capital intensity, on the slope of the operating cost path and on the discount rate, but not on carbon intensity or on the level of operating costs. Without discounting, $\theta D = \sqrt{2\gamma\kappa}$. Discounting lowers the value of developing a field without changing the decision to keep it producing, so it lowers $\bar{\tau}_D$ and deepens the zone: the undiscounted depth is a lower bound (Appendix A.1).

Width. A field only locks in emissions if it still produces when the tax arrives. In panel (b), its production breakeven falls with age with slope γ/θ , and the zone closes where it crosses the development breakeven, at age $a^*(\bar{\tau}_D) = \theta D/\gamma$. A field developed during the delay can therefore carry overcommitted emissions only if it is young enough at implementation, $a_\Delta < a^*(\bar{\tau}_D)$. The zone is wider for fields whose operating costs rise slowly, since a flatter production breakeven crosses the development breakeven later, and for capital-intensive fields, since their development breakeven is lower.

Height. The tax that meets the carbon budget must also fall inside the zone. The height of the zone is set by its top, $\bar{\tau}_P(0) = (p - c_0)/\theta$. A lower operating cost at start-up raises both breakevens by the same amount and moves the whole zone up without changing its depth; a lower carbon intensity raises both breakevens and also deepens the zone in tax units. In panel (b), this shifts both lines upwards. Oil sands illustrate why height matters: their capital intensity is close to that of deepwater fields, but their high operating costs and carbon intensity place a deep zone below the taxes the budget requires. Deepwater fields have a deep zone that sits high, while shale wells have a shallow zone that sits low.

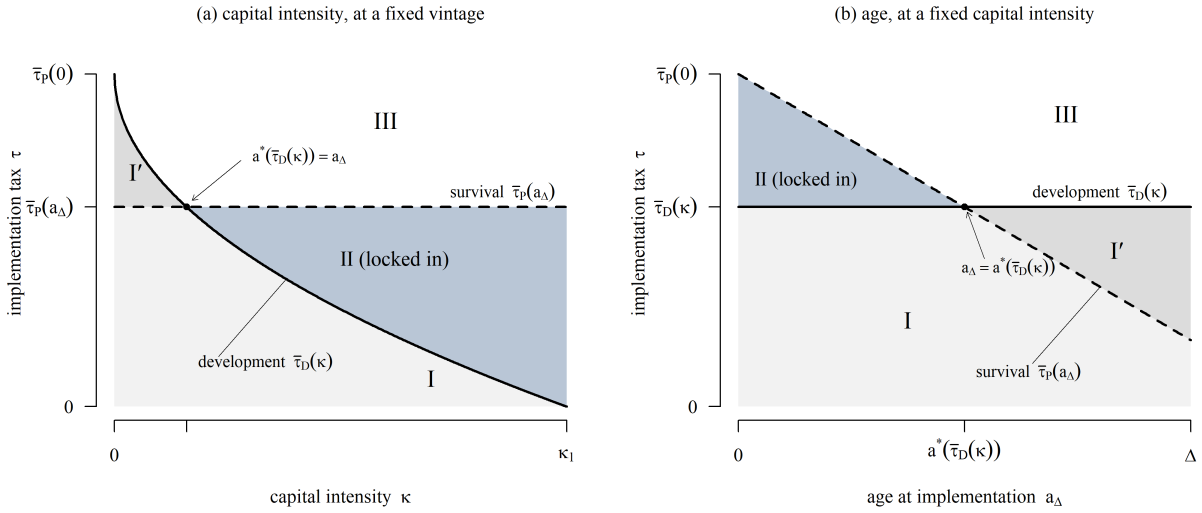


Figure 1: Which fields lock in.

Notes: The implementation tax is on the vertical axis. The development boundary is the development breakeven $\bar{\tau}_D$, the survival boundary the production breakeven at implementation $\bar{\tau}_P(a_\Delta)$, and only region II carries emissions after implementation. (a) At a given age at implementation, the lock-in zone deepens with capital intensity, and the two boundaries cross where $a_\Delta = a^*(\bar{\tau}_D)$. (b) At a given capital intensity, the survival boundary falls with age and the zone closes at $a_\Delta = a^*(\bar{\tau}_D)$. Drawn without discounting, with $p - c_0 = 10$, $\theta = \gamma = 1$, $\Delta = 8$, $a_\Delta = 4$ in (a) and $\kappa = 8$ in (b).

2.3 Which fields lock in, and the overshoot they cause

Fields developed during the delay differ in their costs and in their age at implementation: a field developed at date $s \in [0, \Delta)$ is $a_\Delta = \Delta - s$ years old when the tax arrives. Its contribution to the overshoot, compared with the timely policy, depends on two questions. Would a timely tax have allowed its development? From (4), it would not if $\kappa > \bar{v}_\delta(\tau)$, that is if its discounted profit per barrel of capacity under the tax does not cover its capital intensity, or equivalently if $\tau > \bar{\tau}_D$. Fields developed under both policies are called *intensive*, and fields developed only because of the delay are called *delay-induced*. Is the field still producing at implementation? It is if $a_\Delta < a^*(\tau)$. The two questions give four types of fields, which correspond to the four regions of Figure 1. Table 1 reports their excess emissions compared with the timely policy, per field and in units of $\theta\bar{q}$.

Type	Condition	released during the delay	overcommitted after Δ
I	$\kappa \leq \bar{v}_\delta(\tau), a_\Delta < a^*(\tau)$	0	0
I'	$\kappa \leq \bar{v}_\delta(\tau), a_\Delta \geq a^*(\tau)$	$a_\Delta - a^*$	0
II	$\kappa > \bar{v}_\delta(\tau), a_\Delta < a^*(\tau)$	a_Δ	$a^* - a_\Delta$
III	$\kappa > \bar{v}_\delta(\tau), a_\Delta \geq a^*(\tau)$	a_Δ	0

Table 1: Excess emissions of each type of field relative to the timely policy.

Notes: Per field, in units of $\theta\bar{q}$. Types I and I' are developed under both policies (intensive), Types II and III only because of the delay (delay-induced).

Only Type II fields emit after implementation what a timely policy would have avoided. They are capital-intensive and cheap to operate, like deepwater projects. Type III fields are also delay-induced, but their operating costs are too high for them to survive the tax at implementation. Some are cheap to develop, like shale wells, and others are not, like oil sands. Type I' fields would have been developed under both policies, but the timely tax would have stopped them before Δ , because they are cheap to develop but costly to operate. The table also applies to fields already producing in 2016, whose development cost is sunk and which can be treated as having $\kappa = 0$. They are of Type I' if the timely tax would have stopped them before Δ , and of Type I otherwise. Type I fields, typically the onshore conventional fields of the Middle East, are cheap to operate and cheap to develop or already developed, and they produce under both policies.

Summing over all fields splits the overshoot into three parts: the overshoot at the intensive margin, released before Δ by Type I' fields, the overshoot at the extensive margin

released before Δ by Type II and III fields, and the *lock-in wedge*

$$W(\tau) \equiv \sum_{\text{Type II at } \tau} \theta \bar{q} (a^*(\tau) - a_\Delta), \quad (7)$$

the overcommitted emissions of the cohort at tax τ . The wedge is defined for any tax, independently of the budget. It is zero when there is no tax, since no field is then delay-induced, and it is zero again at taxes high enough to stop every field at implementation. In between, a higher tax brings more fields into their lock-in zone, but shortens the remaining life of those already in it, so the wedge first increases and then decreases with the tax. Appendix A.1 gives its closed form for a parametric cohort.

2.4 The residual overshoot and the lock-in premium

Let B_Δ denote the carbon budget left at implementation, that is the budget minus the emissions released during the delay. The tax that meets this budget depends on the supply curve used to compute it.

The naive planner. The naive planner described in the introduction assigns each field a single cost per barrel, which includes its development cost. For this planner, a field developed during the delay stops producing as soon as the tax exceeds its development breakeven. Its lock-in zone disappears: in Table 1, Type II fields become Type III fields, and in Figure 1, region II merges with region III. The naive planner therefore expects no overcommitted emissions after Δ .⁴ Let $E(\tau)$ be the emissions of the cohort after implementation, that is the sum of $\theta \bar{q}(a^*(\tau) - a_\Delta)$ over all fields still producing at Δ , and $E^N(\tau)$ the same emissions as computed by the naive planner, which leaves out Type II fields. Then

$$E(\tau) - E^N(\tau) = W(\tau). \quad (8)$$

E depends on neither the discount rate nor the distribution of capital intensity, since emissions after implementation only depend on the fields that exist. The tax τ^* that meets the budget, defined by $E(\tau^*) = B_\Delta$, is therefore independent of both.

⁴I only apply the single cost per barrel to the decision of delay-induced fields to produce after Δ . Applying it to the producing life of every field would shorten these lives in the planner's calculation, lower its estimate of emissions further and widen the gap with actual emissions, so this choice is conservative.

The residual overshoot and the premium. Let τ^N be the tax that meets the budget according to the naive planner, $E^N(\tau^N) = B_\Delta$, which I call the naive tax. Implementing τ^N leaves a *residual overshoot* of the budget

$$\Omega^N \equiv E(\tau^N) - B_\Delta = W(\tau^N). \quad (9)$$

Meeting the budget requires increasing the tax by the *lock-in premium*

$$\pi \equiv \tau^* - \tau^N. \quad (10)$$

The wedge is a function of the tax, and the residual overshoot is its value at τ^N . The difference matters in the data, because the wedge takes different values at the timely tax, at τ^N and at τ^* , and only its value at τ^N measures what the naive planner leaves unabated. The residual overshoot and the premium measure the same error, in tonnes and in dollars. To a first approximation,

$$\pi \approx \frac{\Omega^N}{|E'(\tau^N)|}, \quad (11)$$

where $|E'|$ is the reduction in emissions after implementation obtained with one more dollar of tax. For a given residual overshoot, the premium is small when emissions respond strongly to the tax and large when they respond weakly. Section 4.3 reports both numbers and the slope that relates them.

Two properties of the premium hold in general. Appendix A.1 gives a closed form, without discounting and with capital intensity uniformly distributed, in which they are explicit. First, $\pi > 0$ if the delay led to the development of any field the tax τ^* would have deterred. Second, the premium increases with the discount rate. A higher δ lowers $\bar{v}_\delta(\tau)$, so that more fields are delay-induced and E^N falls while E is unchanged: τ^N decreases, τ^* does not change, and both W and π increase. The undiscounted case is therefore a lower bound, and the sensitivity analysis of Section 4.4 shows the same ordering with discount rates of 5% and 15%.

Who pays for the delay. Locked-in fields hardly respond to the premium. When their zones are deep, their production breakevens lie well above τ^N , so an additional dollar of tax does not stop them, and the budget is met by stranding production on fields that a timely policy would have left in place. Overproduction shifts stranding in the same way. Emissions released during the delay reduce the budget left at Δ , so that even with the right supply curve, the tax after the delay must be higher than the timely one, and this increase strands production that a timely policy would not have stranded. The delay thus shifts stranding

through two channels: overproduction, which raises the tax to τ^N , and lock-in, which raises it further to τ^* . Section 5 measures both.

2.5 Anticipation

So far, investors have been assumed not to anticipate the tax during the delay. Suppose instead that they screen projects with an anticipated tax $\tau^A \in [0, \tau]$, which can combine a probability of implementation, an expected implementation date and an expected level. The baseline corresponds to $\tau^A = 0$ and full anticipation to $\tau^A = \tau$. The only change concerns which fields are developed during the delay: a project is developed if $\kappa \leq \bar{v}_\delta(\tau^A)$, that is if $\tau^A \leq \bar{\tau}_D$. The delay-induced fields are then those whose development breakeven lies between the anticipated and the actual tax, $\tau^A \leq \bar{\tau}_D < \tau$. The four types, the decomposition of the overshoot, the wedge and the premium are defined as before.

Anticipation removes from the cohort the fields whose lock-in zone reaches lowest, that is whose development breakeven $\bar{\tau}_D = \bar{\tau}_P(0) - D$ is below τ^A . Both the depth and the height of the zone matter here. A field with a deep zone can still be developed under anticipation if its zone sits high enough, as for a deepwater project that is cheap to operate and relatively clean, while a field with a shallower zone can be deterred if its zone sits low. The premium remains positive as long as some fields developed during the delay have a development breakeven between τ^A and τ^* , and disappears when investors screen with the tax the budget turns out to require. The anticipation scenarios of Section 4.4 follow this pattern: the premium falls as anticipation increases and only disappears at full anticipation.

2.6 Increasing tax and field heterogeneity

The empirical analysis departs from the simple model in four ways. Each of them affects the results in a direction the model predicts.

An increasing tax. In the empirical analysis, the tax increases at rate r . Under a constant tax, the tax only enters the development decision through the margin at start-up, which is why the depth of the zone does not depend on the tax. With an increasing tax, it also enters through the path of the tax after start-up, and the development breakeven depends on the date of development, $\bar{\tau}_D(a_\Delta) = \bar{\tau}_D^0 e^{ra_\Delta}$, where $\bar{\tau}_D^0$ is the development breakeven of a field developed at Δ . Fields developed early in the delay would have faced a lower tax under the timely policy, and are therefore more likely to have been developed anyway. Figure 2

shows the resulting partition. The production breakeven, the four types and the identity $W = E - E^N$ are unchanged, but the closed forms are lost. Appendix A.2 gives the details.

An increasing tax affects lock-in in two opposite ways. It raises the development breakeven of young fields, so more of them become delay-induced, and it shortens the life of every locked-in field after implementation. After Δ , the zone now closes from both sides: the production breakeven falls by γ/θ each year while the tax rises by about $r\tau e^{r\Delta}$. At the increase rate used here, the two effects roughly offset each other, so that lock-in and the premium are of similar size under a constant and an increasing tax. The total cost of the delay differs much more. A tax that meets the same budget while rising over time starts at a lower level, so a timely policy retires fewer fields before Δ , and the overproduction added by the delay before implementation is smaller. Section 4.4 quantifies both, and Appendix A.2 illustrates the mechanism on a synthetic cohort.

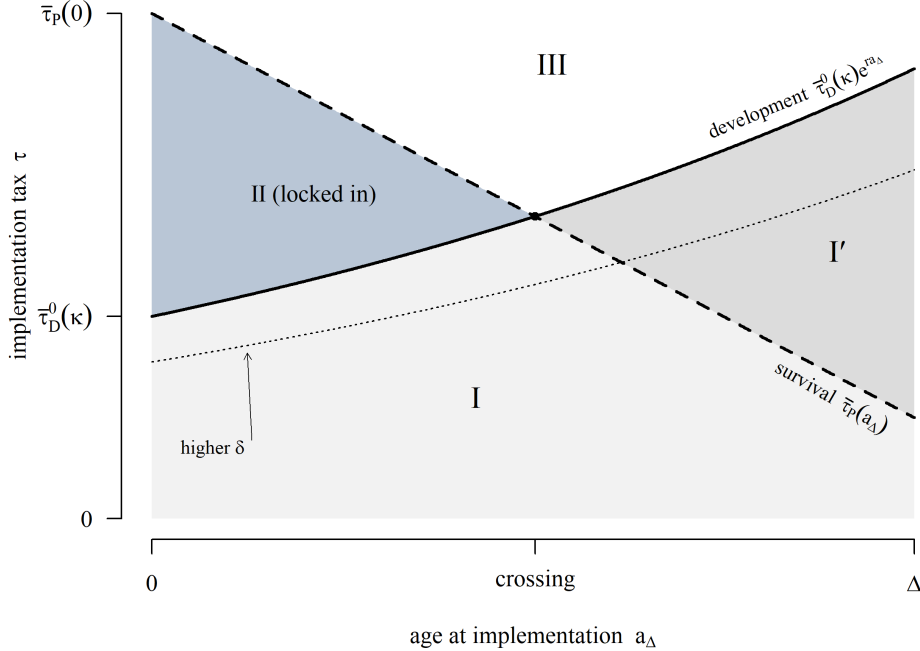


Figure 2: The taxonomy under an increasing tax, at a given capital intensity.

Notes: The survival boundary $\bar{\tau}_P(a_\Delta) = (p - c_0 - \gamma a_\Delta)/\theta$ is the same as in Figure 1. The development boundary is $\bar{\tau}_D(a_\Delta) = \bar{\tau}_D^0 e^{r a_\Delta}$, increasing with age because fields developed early in the delay would have faced a lower tax under a timely policy. Region II, the set of locked-in fields, lies between the two boundaries and closes where they cross. The dotted line shows the development boundary at a higher discount rate: δ shifts it down and enlarges region II, r tilts it, and neither moves the survival boundary. Drawn with $p - c_0 = 10$, $\theta = \gamma = 1$, $\Delta = 8$, $r = 0.10$ (exaggerated for legibility; the empirical rate is 2%), $\bar{\tau}_D^0 = 4$.

Heterogeneous production, cost paths and development periods. Three other features of the data affect the results in one direction each. First, the overcommitted emissions

of a locked-in field, $\theta\bar{q}(a^* - a_\Delta)$, increase with $\bar{q}$, whereas whether a field is locked in does not depend on $\bar{q}$. The size of fields therefore changes the amount of lock-in but not which fields carry it, which is why the concentration documented in Section 5.1 reflects the size of a few projects. Second, capital-intensive offshore projects also have lower operating costs and flatter cost paths than shale. A lower c_0 raises the zone and a lower γ widens it, so heterogeneity in operating costs reinforces the concentration of lock-in in deepwater fields. Third, developing a field takes several years, and some fields are only partly built when the tax arrives. Appendix A.3 adds a development period: a partly built field is completed if its production value covers the development cost it still has to pay, and abandoned otherwise. Since a development period can only add fields to the locked-in set, the model without it gives a lower bound.

The empirical analysis includes the increasing tax and these three features.

3 Data and methods

3.1 Oil field data and sample

The analysis relies on proprietary asset-level data from Rystad Energy. Rystad’s UCube database covers the vast majority of oil and gas production in existing fields worldwide since 1900, together with fields expected to be discovered and developed up to 2100, over 100,000 assets in total across oil, gas, LNG and biofuels. The analysis is conducted at the asset level, the smallest geographic scale available in the database, which typically corresponds to a field or to a portion of a field. For each asset, the data contain annual production by oil type and costs by category (operating expenses, split into production, transportation and SG&A opex; capital expenses for facilities and wells; exploration expenses; and government take), together with administrative, geographic and geological characteristics. Production and costs are observed up to 2025 and forecast by Rystad thereafter. Appendix B reproduces Rystad’s definition of each cost item.

The estimation sample keeps oil-producing assets discovered by 2025 with positive remaining oil reserves in 2016.⁵ This leaves 25,336 assets, of which 16,746 were already producing in 2016. Reserves in this paper are oil resources as defined by Rystad: all economically recoverable oil over the lifetime of the asset, which is typically larger than 2P reserves be-

⁵Gas assets are excluded. Assets that Rystad models as not yet discovered are excluded from the main sample because their production, costs and reserves are entirely modelled; the sensitivity analysis of Section 4.4 adds them back, with their exploration expenses treated as development costs.

cause it includes reserve growth. Oil is any type of crude (bitumen, extra heavy, heavy, regular, light or sour), together with condensate and natural gas liquids, which are counted as oil because (i) they are liquids and sometimes mixed with crude oil and (ii) their price is set mainly by the oil price.

Table 2 describes the sample. Panel A splits it by vintage: fields already producing in 2016, on which the delay acts only through overproduction; the delay cohort, the 3,136 fields on which some development takes place between 2016 and 2029, and which are the only fields a delay can lock in; fields approved in 2030 or later; and fields never developed on the baseline path (without carbon tax). Panel B splits the delay cohort by supply segment, and Panel C splits the full sample by region. For each group, the table reports the medians of capital intensity κ (development capex per barrel of peak annual capacity), of the development and production breakevens $\bar{\tau}_D$ and $\bar{\tau}_P(0)$, of the production breakeven in 2030, $\bar{\tau}_P(a_\Delta)$, when the delayed tax arrives, and of the depth of the lock-in zone $\bar{\tau}_P(0) - \bar{\tau}_D$. Medians are weighted by 2016 reserves, so that they describe the typical barrel rather than the typical field.

	Fields	Reserves 2016 (Gbbbl)	Emissions post-2016 (GtCO ₂ eq)	Emissions post-2030 (GtCO ₂ eq)	κ (\$/bbl)	$\bar{\tau}_D$ (\$/t)	$\bar{\tau}_P(0)$ (\$/t)	$\bar{\tau}_P(a_\Delta)$ (\$/t)	$\bar{\tau}_P(0) - \bar{\tau}_D$ (\$/t)
<i>Panel A. All fields, by vintage</i>									
Producing by 2016	16,746	1,072.9	479.4	286.6	40.8	–	215.1	83.8	–
Delay cohort	3,136	225.4	103.9	77.2	89.0	68.6	117.3	99.1	32.8
Approved 2030 or later	2,259	208.3	89.0	89.0	71.8	54.5	79.2	–	23.7
Never developed	3,195	48.8	–	–	10.5	–23.5	–3.7	–	88.0
Total	25,336	1,555.4	695.6	468.2	49.6	57.0	146.9	84.8	25.6
<i>Panel B. Delay cohort, by supply segment</i>									
Onshore conventional	1,427	77.4	35.9	26.7	73.4	80.4	121.3	109.3	28.9
Offshore shelf	544	56.2	24.2	18.5	92.8	58.6	124.2	117.0	46.8
Offshore midwater	206	12.8	6.0	4.3	128.7	79.9	116.2	100.2	28.8
Offshore deepwater	203	44.1	20.5	13.9	120.6	63.1	110.4	93.9	29.9
Shale and tight oil	708	27.7	13.1	10.6	2.7	34.0	45.7	36.3	7.4
Oil sands	48	7.2	4.1	3.2	102.1	34.5	72.4	53.7	23.6
<i>Panel C. All fields, by region</i>									
Middle East	1,540	711.7	295.9	225.0	35.9	59.4	230.3	104.2	19.9
North America	9,670	366.4	175.3	113.7	5.8	17.5	49.9	30.6	21.0
Russia	2,967	147.2	66.8	41.1	89.0	68.7	157.7	81.5	30.0
Asia	3,730	115.7	53.8	30.4	101.5	52.0	167.3	70.4	45.5
South America	2,393	102.5	49.1	30.9	107.6	49.0	106.0	80.7	36.5
Africa	2,116	79.2	39.5	20.8	107.5	51.3	163.2	88.7	35.2
Europe	2,242	27.9	13.0	5.3	108.7	39.1	160.1	85.5	51.7
Oceania	678	4.9	2.1	1.2	76.4	79.9	152.7	104.2	26.5

Table 2: Summary statistics.

Notes: Medians are weighted by 2016 reserves and are not additive: the depth column is computed field by field. Emissions are the lifecycle emissions of Rystad’s baseline production, without a carbon tax, after 2016 and after 2030. $\bar{\tau}_P(a_\Delta)$ is the production breakeven in 2030, when the delayed tax is implemented. Taxes are 2016-anchored levels of the increasing path, in \$/tCO₂eq. The delay cohort is the set of fields on which some development takes place during the delay (start-up after 2016 and approval by 2029).

Three features of the table matter for what follows. First, the delay cohort is more than twice as capital-intensive as the fields already producing in 2016, with a median κ of \$89 per barrel of capacity against \$41, and its lock-in zone is correspondingly deeper, \$33 per tonne against a full-sample median of \$26. Second, within the cohort, capital intensity separates shale from the rest: at \$3 per barrel of capacity, shale has a zone only \$7 deep, while the capital-intensive segments, at \$73–129, have zones of \$24–47. Third, the regional panel reads the same way. North America, where shale and oil sands are concentrated, has a median production breakeven of \$50, so most of its output is retired by any tax in the range studied here, while South America and the Middle East have medians of \$106 and \$230. The regional medians describe where a region’s production sits relative to the tax; where its lock-in sits is discussed in Section 4.1.

Capital intensity and depth. The model predicts that the depth of a field’s lock-in zone per barrel depends on its capital intensity and on the steepness of its operating cost path, but not on its carbon intensity or on the tax (equation (6)). Field by field, the ordering of Panel B holds within segments as well as across them (Appendix Figure 24). Measured per barrel, the depth of the lock-in zone rises with capital intensity with an elasticity of 0.55, against 0.5 in the model without discounting and more at any positive discount rate, so that a field ten times more capital-intensive than another has a lock-in zone about three and a half times deeper. Part of this relation is mechanical, since development capital enters both capital intensity and the full-cycle breakeven that defines depth. The slope should be read as a description of how depth scales with capital intensity.

3.2 Field-level carbon intensities

Rystad’s UCube data contain no carbon intensity. I estimate a lifecycle carbon intensity for each asset by combining upstream, midstream and downstream emissions. Appendix C gives the full procedure. For upstream emissions, I use the input data of Masnadi et al. (2018) for 737 oil fields, matched to 1,136 Rystad assets, and run the Oil Production Greenhouse Gas Emissions Estimator (OPGEE). The main source of upstream heterogeneity is flaring, which is not reported at the asset level. I use flared volumes derived from the satellite observations of NOAA’s VIIRS instruments over 2012–2024 (Elvidge et al., 2016), matched to Rystad assets following Ghosh et al. (2025), and predict flaring-to-oil ratios for assets whose production had not started by 2025. A regression on flaring, steam-to-oil and gas-to-oil ratios, offshore status and oil type then assigns an upstream intensity to every remaining asset, including those not yet developed. For midstream and downstream emissions, I use

the Petroleum Refinery Life-Cycle Inventory Model (PRELIM) and the Oil Products Emissions Module (OPEM), matching 493 publicly available crude assays to 392 Rystad crude streams, which cover 64.4% of 2024 production, and predicting intensities for unmatched assets from API gravity, sulphur content and predicted refinery configuration. Summing the three stages gives one lifecycle intensity per asset. In 2024, the production-weighted average is 466 kgCO₂eq/bbl, with a weighted standard deviation of 112. Carbon intensity does not enter the per-barrel depth of the lock-in zone: it sets the height of the zone, $\bar{\tau}_P(0) = (p - c_0)/\theta$, and converts barrels into emissions, and the sensitivity analysis replaces the asset-level estimates with a common value to show how much of the result depends on them.

3.3 Costs, prices and discounting

Development and production costs. Following Rystad’s split of costs, only expenses attributed to oil operations are kept. The capital expenses for drilling wells and building facilities before production starts and in its first years are the development cost of the asset, the empirical counterpart of C . Following Rystad’s own definition, all capital expenses incurred up to two years after start-up are counted as development. For modelling purposes, development expenses incurred after start-up are moved to the year before start-up and discounted accordingly, which leaves the net present value of the project unchanged. Production costs include all operating expenses (production, transportation and SG&A opex) and all other capital expenses, that is facilities and wells added more than two years after start-up. Shale and tight oil assets follow a different rule: only facility capex incurred up to two years after start-up counts as development, and all well capex is counted as a production cost. A shale asset keeps drilling new wells for as long as it produces, and each well produces most of its oil within a year or two, so drilling behaves like a recurring cost of production, more than a long-lasting investment.

The two-year rule matters for lock-in, because it determines how much of a field’s capital counts as sunk once production starts. On most projects, a large part of the development cost is spent in the first years of production, on the wells drilled to reach plateau, and this capital is committed by the time a tax can reach the field. Section 4.4 tests a stricter rule, which counts only capital spent before start-up as development. It moves these two years of capital expenditure into the operating cost and makes every lock-in zone shallower, therefore giving a lower bound on lock-in. Under this more restrictive rule, results remain remarkably close to the main specification.

Prices. The reference Brent price used in the empirical analysis is constant at \$80/bbl. What this paper measures is driven by investment decisions taken during the delay, so the relevant price is the long-run price investors use to screen those projects, not the price of any particular year. The price level matters for the results, since it sets the pool of assets that are viable without carbon tax. Changing it moves the tax that meets the budget more than it moves the overshoot, because both the timely and the delayed policy are re-calibrated on the same budget, and because few projects need more than \$80/bbl to be developed. Two arguments support this level. The first is the behaviour of the oil market. Its stability is maintained mainly by the three core countries of OPEC, Saudi Arabia, Kuwait and the United Arab Emirates (Okullo and Reynès, 2016; Pierru et al., 2018), and the IMF reports fiscal breakeven prices, the price at which an exporting country balances its budget, of \$80/bbl in the recent past and \$84/bbl in projection for Saudi Arabia, and lower for the other two (International Monetary Fund, 2024). Given the influence these countries exert on the price, their fiscal breakeven is a reasonable anchor for a long-run price. The second is that \$80/bbl keeps the investment margin studied here active. At much lower prices, few new projects are viable with or without a carbon tax, and the question of what a delay lets through does not arise. One may object that the introduction of a worldwide carbon tax would itself change long-run price expectations. The price does not respond to the tax in the model, so the tax levels reported in the paper should be interpreted as the levels borne by producers. Section 4.4 reports a variant in which part of the tax is passed on to consumers. Beyond this, the countries that anchor the price are also those a carbon tax affects least, since their fields have both low costs and low carbon intensities. If producing countries collect the carbon tax levied on their own production, their oil revenue, the rent plus the tax receipts, falls only with the production the tax strands, which is small for these low-cost fields; they lose from the delay (Section 5.2) more than from the tax itself.

Price volatility is not modelled either, because the investments studied here are long-lived and screened on long-run price expectations. OPEC’s use of its spare capacity reduces price volatility and supports investors’ confidence in a stable market (Pierru et al., 2018; Fattouh, 2007). Berntsen et al. (2018) find that volatility has no significant effect on oil and gas investment in Norway once the expected price level is accounted for, and Tang et al. (2017) argue that volatility changes the timing of investment rather than the decision to invest. The sensitivity analysis uses Brent prices of \$60 and \$100/bbl, which move the residual overshoot and the premium in opposite directions (Section 4.4).

Asset-specific price discounts. Reference prices such as Brent describe one particular crude stream, and each asset sells its oil at Rystad’s long-term discount to that reference,

which depends on crude quality. Every oil type sells at a discount on average, which reflects the high quality of the crudes constituting the Brent stream. Discounts are also very heterogeneous: heavier oils, which are more difficult to refine into the same products, carry larger discounts, from about 3% for light crudes to 29% for bitumen, and NGL sells at roughly half the price of oil. The reserve-weighted average discount is 12.9%, so that with a reference price of \$80/bbl the average selling price in the model is closer to \$70. Appendix Table 8 reports the discounts by oil type, and the sensitivity analysis replaces them with a common selling price across assets.

Discounting. Net present values are computed at a 10% discount rate. This is the most common rate used in the oil and gas industry to screen projects, the rate at which the industry reports its breakeven prices, and the one Rystad uses (Bailey et al., 2000; Jahn et al., 2008; Wood Mackenzie, 2013). It is also the rate at which listed producers must discount the future cash flows of their proved reserves in their financial statements, the so-called standardized measure or PV-10 (?). The model of Section 2.2 says in which direction this choice matters. A higher rate lowers the value of developing a field without changing the value of keeping it in production, so it deepens and widens every lock-in zone and raises both the premium and the residual overshoot. The sensitivity analysis reports results at 5% and 15%.

3.4 From the model to the data

The empirical model is a sequence of yearly decisions taken for each asset independently of the others: whether to invest in the development of an undeveloped asset, and whether to keep an active asset in production.

Investment decision. An asset is developed if the expected net present value of the project is non-negative, given what investors anticipate about the carbon tax τ :

$$\sum_{t=T_1}^T \frac{P(1-d_i) q_{it} - c_{it}}{(1+\delta)^{t-T_0}} - C_{iT_0} - \sum_{t=T_\tau}^T \frac{\tau \theta_i q_{it} (1+r)^{t-2016}}{(1+\delta)^{t-T_0}} \geq 0, \quad (12)$$

where $T_0 = T_1 - 1$ is the year before start-up, T_1 and T the first and last years of production, T_τ the anticipated year of implementation, $P(1-d_i)$ the reference price net of the asset's discount, C_{iT_0} the development cost expensed in T_0 , c_{it} total production costs in year t , θ_i carbon intensity and r the increase rate of the tax. Investors choose T to maximise the

value of the project. Rearranging (12) gives the maximum anticipated tax $\bar{\tau}_i$ consistent with a non-negative net present value, the empirical counterpart of the development breakeven $\bar{\tau}_D(\kappa)$ of equation (5).

Development often takes several years, and can stop and resume, as it did on many projects after the 2020 price collapse. The maximum viable tax is therefore computed at each development stage, with C_{iT_0} replaced by the development expenses that remain to be spent, and the asset is developed only if the anticipated tax is below this threshold at every stage. The threshold rises as the remaining cost falls, so an asset that clears it when development starts keeps clearing it as long as expectations do not change. When the tax is implemented, an asset under development may no longer clear the threshold at its remaining development cost. Development then stops, the capital already spent is lost and the infrastructure is abandoned. This remaining-cost threshold is the empirical counterpart of the completion boundary of Appendix A.3. For a given tax, the first year in which the remaining-cost threshold exceeds the tax is the year of investment responsible for the asset’s lock-in: from that year on, the asset would be completed even if the tax was implemented immediately.

Production decision. Once developed, an asset produces in a given year if the price net of the tax covers its production cost. Production costs are computed as breakeven prices by stage of production, and rise over the life of the asset: I compute the last year of production that minimises the breakeven price, which closes the first stage, and repeat the operation on the remaining years to identify the following stages. For assets developed after 2015, the average number of stages is 5.8 and the average stage lasts 5.2 years, the first stage usually being the longest. The breakeven of the current stage is the empirical counterpart of $\bar{\tau}_P(a)$ in equation (2): it rises with age because operating costs rise and because capital expenses per barrel rise as production declines. An asset is phased out in the first year in which the tax exceeds this breakeven, and does not come back once phased out.

Investment and production paths. The path of investment and production of each asset is the one observed up to 2025 and forecast by Rystad thereafter. Production rates depend mainly on the initial investment and cannot be changed at will afterwards, because of geological and technological constraints; changing the path would mean changing development expenses and all subsequent costs. During the delay, the tax is not, or only partially, anticipated, which makes the business-as-usual path forecast by Rystad the relevant counterfactual. A field’s production profile is therefore taken as given from its start-up year, and a later development date shifts that profile in time rather than reshaping it. The development

date still matters once the tax is implemented. Under a tax rising over time, a field that starts producing later meets a higher tax at each age and is phased out younger. This is why the year in which an asset is developed matters for what it still emits after 2030.

3.5 Scenarios, budget and parameters

The analysis compares a scenario in which the tax is implemented in 2016, the timely policy, with one in which it is delayed to 2030 without being anticipated by investors, which is equivalent to business as usual until 2030. From its implementation, the tax rises at 2% per year, the social discount rate. Under a cumulative carbon budget, the cost-effective tax rises at the discount rate (Dietz and Venmans, 2019), while Lemoine and Rudik (2017) obtain a slower-rising path when warming lags emissions, which a cumulative budget abstracts from. A social discount rate below market rates can be justified on ethical grounds (Stern, 2007) or by uncertainty about growth and interest rates, which lowers the certainty-equivalent rate at long horizons (Weitzman, 1998; Gollier, 2012). This literature points to rates between 1% and 2% in the distant future, while Nordhaus (2007) argues for a market-based rate of about 3%. I adopt a rate of 2%, between these approaches, and the sensitivity analysis uses rates from 1% to 6%. All tax levels in the paper are expressed at their 2016-anchored level, whatever the implementation date, so that a level of x on any axis or in any table corresponds to $1.32x$ in 2030.

Anticipation scenarios. I explore alternative anticipation scenarios in which investors screen projects either with a fraction of the eventual tax (a wrong level), or with the correct tax arriving at different dates (in 2030, or five years after the asset’s start-up). Appendix B.3 defines them, Section 2.5 maps them onto a single anticipated tax τ^A , and Section 4.4 reports the results.

The budget is the carbon dioxide that Welsby et al. (2021) allocate to oil, 747 Gbbl over 2018–2100 extended to 2016–2100 with the observed production of 2016 and 2017, converted with the average downstream carbon intensity of their model (IPCC, 2006). It covers combustion only, while the emissions of each field here also include extraction and refining, so it is a tight budget for lifecycle emissions. Section 4.4 study the results’ sensitivity to variations in the budget. The resulting budget is 350.3 GtCO₂eq from 2016. Under a timely policy, the tax that meets it starts at \$57.2/tCO₂eq in 2016. Under the delay, the budget remaining at implementation, B_Δ , is the initial budget net of the 205.4 GtCO₂eq released between 2016 and 2029 on the business-as-usual path, that is 144.9 GtCO₂eq.⁶

⁶Note that B_Δ can vary when assessing anticipation scenarios and early ban policies.

Leakage towards other fossil fuels is not modelled, and the results should be read as if similar policies applied to other fuels at the same time. The sensitivity analysis varies the budget between 75% and 125% of its central value.

An asset’s excess emissions under the delay are counted on the intensive margin if the asset produces under the timely policy or was already producing in 2016, and on the extensive margin, as a delay-induced field, if it does not produce under the timely policy. Since expectations affect only investment decisions, the anticipation scenarios differ only at the extensive margin. Table 3 summarises the parameters.

Brent price	\$80/bbl, constant; asset-specific discounts (12.9% on average)
Discount rate δ	10%
Tax increase r	2% per year; levels reported 2016-anchored
Oil carbon budget	350.3 GtCO ₂ eq from 2016 (Welsby et al., 2021)
Delay	2016–2029, tax in force from 2030
Anticipation (main case)	none
Development cost	capex up to two years after start-up
Carbon intensity	lifecycle, GWP-100, asset-level (Appendix C)
Sample	25,336 assets with oil reserves in 2016, discovered by 2025

Table 3: Parameters of the main specification.

3.6 Fiscal regimes

Government take is excluded from costs, except for taxes reported as opex by companies, which producers treat as operating costs unrelated to profitability. Profit taxes change the size of an asset’s net profitability but not its sign, and bonuses are paid before development decisions are made. The key assumption concerns royalties, which are levied on gross revenue and can therefore make a project unprofitable. I assume that, in the presence of a worldwide carbon tax, governments renegotiate royalty rates so as to keep viable assets in production and keep collecting a share of the rent, without subsidising assets that are no longer viable. Two cases bound this assumption. On the one hand, governments could absorb the whole carbon tax by lowering royalties.⁷ On the other hand, contractual rigidities in existing production-sharing agreements and concession contracts may prevent renegotiation in the short term, causing the carbon tax to pile on top of the existing fiscal regime and further

⁷A government at its Ramsey optimum could in principle respond to a worldwide carbon tax by adjusting fiscal regimes so as to leave the equilibrium unchanged, substituting carbon tax revenue for royalty revenue. Fully restoring the pre-tax equilibrium would, however, in some cases require subsidising development or production.

constrain investment and production beyond what the model predicts. The assumption made in this paper implicitly lies between these two extremes: governments are assumed to absorb the carbon tax to the extent necessary to keep viable assets in production (potentially capturing a higher rent in the process) without subsidising uneconomical ones, so that assets not resilient to the carbon tax are either phased out earlier than in a no-tax world or simply remain undeveloped. The locked-in emissions measured in the paper therefore correspond to an intermediate case, with full absorption giving an upper bound and full rigidity a lower bound.⁸ The same assumption makes Rystad’s forecast paths, which embed current fiscal terms, the relevant counterfactual without a carbon tax.

4 Results

All results in this section rely on a tax rising at 2% per year from its 2016-anchored level, under the no-anticipation scenario, with a Brent price of \$80/bbl. The tax that meets the budget under a timely policy is $\tau_{2016}^* = \$57.2/\text{tCO}_2\text{eq}$.

4.1 The overshoot and where it comes from

Figure 3 reports, for each level of the tax, the overshoot of the carbon budget: the excess of cumulative emissions under the delay over cumulative emissions under the timely policy. The overshoot is split into the cells of the decomposition of Section 2.3: (i) emissions released before 2030 by fields that produce under both policies but that a timely tax would have retired earlier (the intensive margin), (ii) emissions released before 2030 by delay-induced fields and (iii) emissions released after 2030 by delay-induced fields that the tax arriving in 2030 cannot stop. The fourth cell, intensive emissions after 2030, is zero, as the model predicts. Appendix Figure 21 draws the same bands on top of cumulative emissions under the timely policy.

The overshoot rises with the tax, slowly below about \$25 and steadily above. At low levels, a tax removes little production, during the delay or after it, because oil fields earn large rents. At the tax that meets the budget, the delay overshoots the oil carbon budget by 66.7 GtCO₂eq, 19.1% of the budget and about four years of global oil emissions at their current rate. Of this overshoot, 41.2 GtCO₂eq are released before 2030 by fields on

⁸On the interaction between extraction taxes and rent capture, see [Daubanes and Lasserre \(2023\)](#), on the response of investment to oil taxation, [Ahlvik and Harding \(2025\)](#), [Ahlvik et al. \(2026\)](#) and [Stefanski et al. \(2025\)](#), and on compensation for halted upstream projects [Paletta et al. \(2025\)](#).

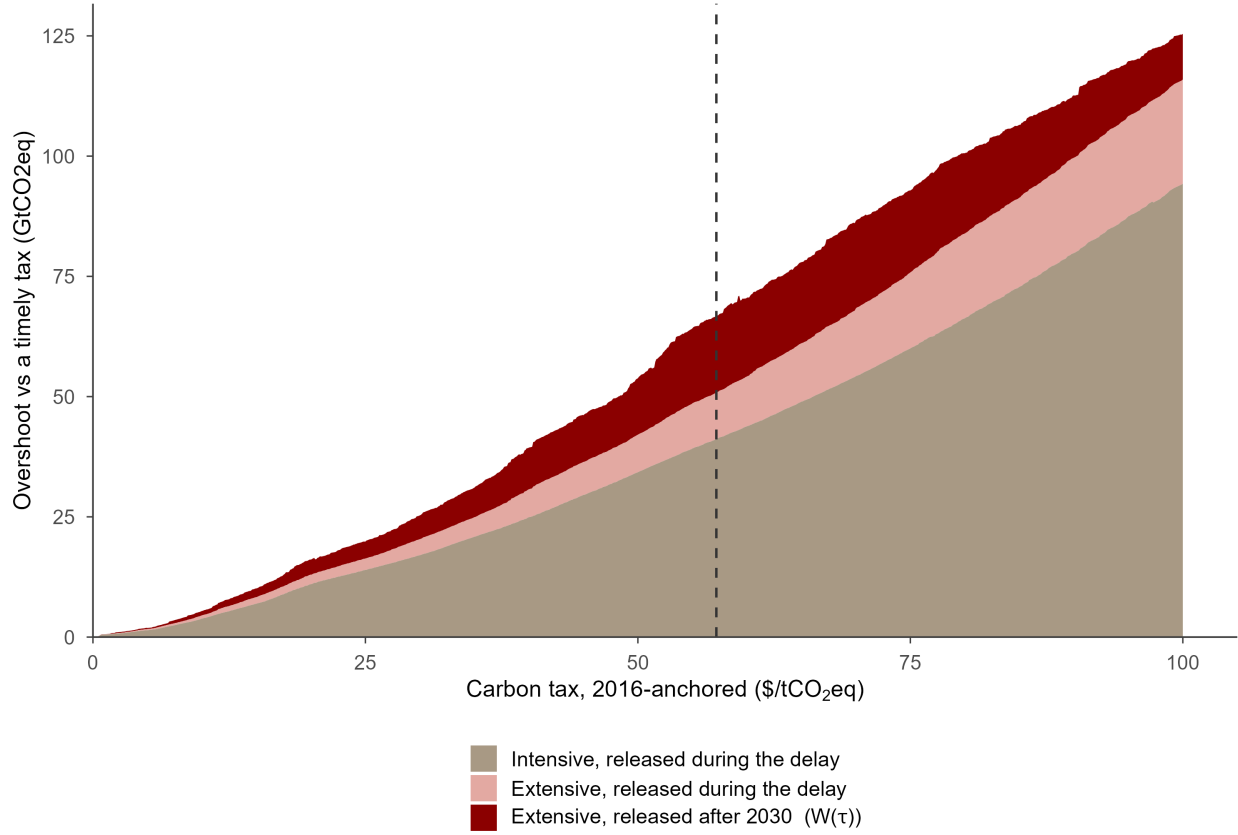


Figure 3: Carbon budget overshoot caused by the delay, by margin and date of emission.

Notes: The horizontal axis is the 2016-anchored level of the tax rising at 2% per year; the dashed line is the tax that meets the budget under a timely policy, \$57.2. Intensive emissions after 2030, the fourth cell of the decomposition, are zero and have no band. The top band read at \$57.2 is the lock-in wedge at the timely tax, $W(\tau_{2016}^*) = 15.9$ GtCO₂eq.

the intensive margin, while the extensive margin represents 9.6 GtCO₂eq before 2030 and 15.9 GtCO₂eq after. Delay-induced fields, those that exist because of the delay, therefore account for 38% of the overshoot, and their emissions after 2030 alone for 24%. Three quarters of the overshoot is already in the atmosphere when the tax arrives, in 2030. The rest of the paper focuses on the top band, which a policy implemented in 2030 can still act on.

Supply segments. The three bands come from different kinds of fields. Figure 4 repeats the decomposition by supply segment, on top of the emissions each segment would release under the timely policy. Emissions under the timely policy explain a large part of the pattern. A timely tax at \$57.2 would phase out or deter most oil sands and shale projects, while about 60% of deepwater emissions and 70% of onshore conventional and shelf emissions would remain. Segments that would almost disappear under a timely tax can only overproduce

during the delay; segments that would largely survive it can also lock in. Table 2 gives the median breakevens that place each segment's lock-in zone.

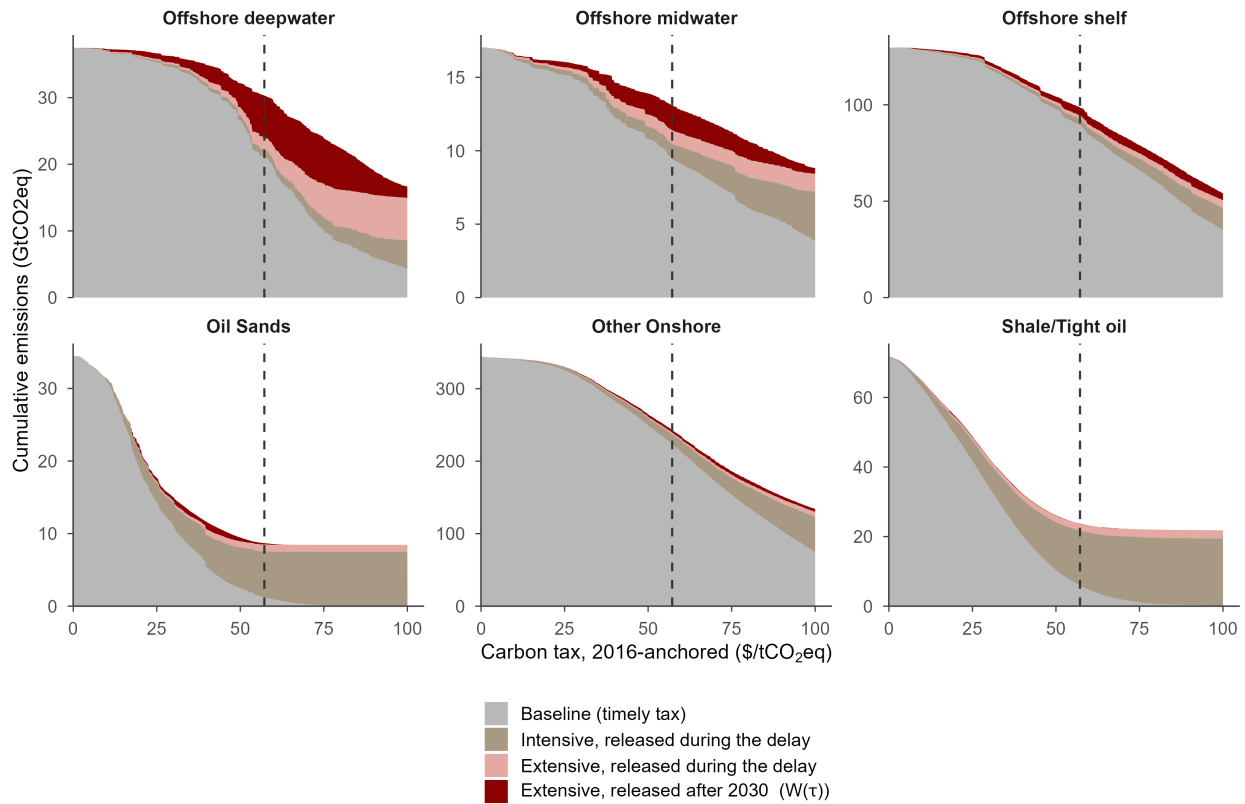


Figure 4: Overshoot caused by the delay, by supply segment.

Notes: Cumulative emissions under the timely policy (grey) and overshoot caused by the delay by margin and date of emission, tax rising at 2% per year. Panels have different vertical scales. The dashed line is the tax that meets the budget under a timely policy. Appendix Figure 22 shows the overshoot bands alone.

Shale and tight oil is the clearest case of a lock-in zone the tax passes over. The median shale barrel in the cohort costs \$3 per barrel of capacity to develop, so its zone is only \$7 deep, and it sits low, with a median production breakeven tax of \$46 at start-up and \$36 by 2030, well below the tax in force after 2030. Almost all of the segment's production is removed by a timely tax, and its overshoot is almost entirely on the intensive margin. Shale fields run untaxed during the delay and are shut down as soon as the tax arrives, whether or not the tax would have deterred their development. At the timely tax, shale carries 15.8 GtCO₂eq of the intensive overshoot, 38% of it and the largest contribution of any segment, and almost none of the emissions released after 2030.

Oil sands reach the same outcome by a different route, and it shows that the depth of a lock-in zone and its position both matter. Oil sands projects are as capital-intensive as offshore ones, at \$102 per barrel of capacity, and their zone is correspondingly deep, \$24,

but it sits low. The median production breakeven is \$72 at start-up, and has already fallen to \$54 by 2030, below even the timely tax, so the segment’s post-2030 band closes almost entirely at the timely tax level. The reason is the segment’s high operating costs and carbon intensity. Oil sands barrels are the most carbon-intensive in the sample, and since the upper edge of the zone is $\bar{\tau}_P(0) = (p - c_0)/\theta$, both push the whole zone down in tax terms.

Offshore fields meet both conditions. The median deepwater barrel has a development breakeven of \$63 and a production breakeven of \$110 at start-up, and the median shelf barrel \$59 and \$124. A tax in the \$67–70 range falls inside both zones, and their production breakevens are still \$94 and \$117 in 2030, so a delay-induced offshore field survives the tax and keeps producing for years. Offshore fields also have low operating costs and are rarely retired early, so their intensive band stays small, while their post-2030 band keeps growing beyond the tax that meets the budget. Offshore fields carry 76% of the 15.9 GtCO₂eq released after 2030 by delay-induced fields, and deepwater fields alone 39%.

Onshore conventional is the most resilient to the tax. Because it concentrates the largest reserves, its overshoot is the largest of all segments in absolute terms, 19.3 GtCO₂eq, and comes mostly from fields already producing in 2016. The median development breakeven of the segment’s delay cohort, \$80, is above the tax, so most of these projects would also have been built under a timely policy.

Regions. The segment pattern has a geography, which drives the incidence results of Section 5. Appendix Figure 23 repeats the decomposition by region. North America, where shale and oil sands are concentrated, alone accounts for 67% of the intensive overshoot, and very little of its overshoot survives 2030. The median North American barrel has a production breakeven of \$50, so its output is retired by a tax in the \$67–70 range whatever its development history. South America is the mirror image. Its overshoot is dominated by the post-2030 band, which grows steeply between \$40 and \$60 as the deepwater developments off Brazil and Guyana enter their lock-in zones, and the region carries 5.4 GtCO₂eq of the 15.9 released after 2030, 34% of the total and the largest regional share. The Middle East is dominated by low-cost fields already in production, with a median production breakeven of \$230. Its overshoot at the timely tax is small, 4.0 GtCO₂eq, but 2.7 GtCO₂eq of it is released after 2030, from the minority of its cohort projects that a timely tax would have deterred. The other regions are affected on both margins. Asia’s overshoot at the timely tax is 9.4 GtCO₂eq, mostly on the intensive margin, Russia’s 3.4 GtCO₂eq and Europe’s 2.0 GtCO₂eq. Africa’s overshoot is 4.4 GtCO₂eq, of which 44% is released after 2030 by delay-induced fields; Section 5 shows that this is enough to reverse the sign of the delay’s effect for Africa as a whole.

4.2 Committed and overcommitted emissions

Figure 3 counts emissions in the year they are released. Lock-in is better described by the year they are committed, that is the year of the investment after which, at a given tax, the full development of the asset can no longer be deterred. Figure 5 reports, by supply segment and period of the delay, the emissions committed by fields developed during the delay. It splits them between emissions that the timely tax of \$57.2 would also have committed and overcommitted emissions, which are due to the delay itself.

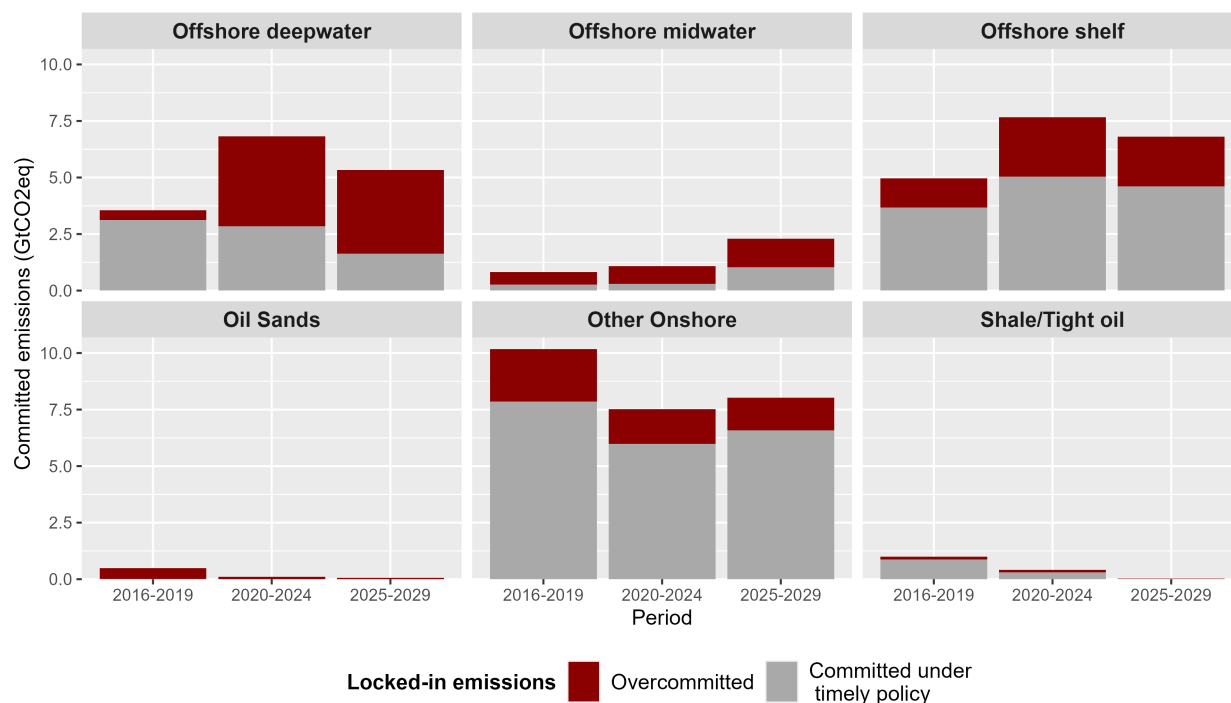


Figure 5: Committed and overcommitted emissions, by supply segment and period.

Notes: Tax rising at 2% per year from \$57.2/tCO₂eq. Each bar is the lifetime emissions committed by investments in new oil assets of the segment during the period. The grey part is committed under the timely policy as well; the red part is overcommitted, that is committed only because of the delay. Emissions are attributed to the year from which the full development of the asset can no longer be deterred at the tax shown.

Two features stand out. First, overcommitted emissions are a large share of what new projects commit: 34% of the emissions committed by new oil projects over 2016–2029 would not have been committed under a timely policy. Second, the share is highest where the model predicts, in the capital-intensive offshore segments. Over half of the emissions committed by deepwater projects are overcommitted, and 62% of those committed by midwater projects. Onshore conventional projects commit the most in absolute terms, but only a fifth of their commitments are due to the delay. Shale commits little at the timely tax, because most of its production does not survive that tax, and oil sands commit almost nothing at all, since

almost all of their production is phased out once the tax arrives. The time profile matters for policy. Overcommitted deepwater emissions rise from 0.4 GtCO₂eq in 2016–2019 to 4.0 GtCO₂eq in 2020–2024 and remain at 3.7 GtCO₂eq in 2025–2029, so that a large share of the lock-in measured in this paper is decided in the years just before implementation. Section 4.3 returns to this when describing how the residual overshoot builds up.

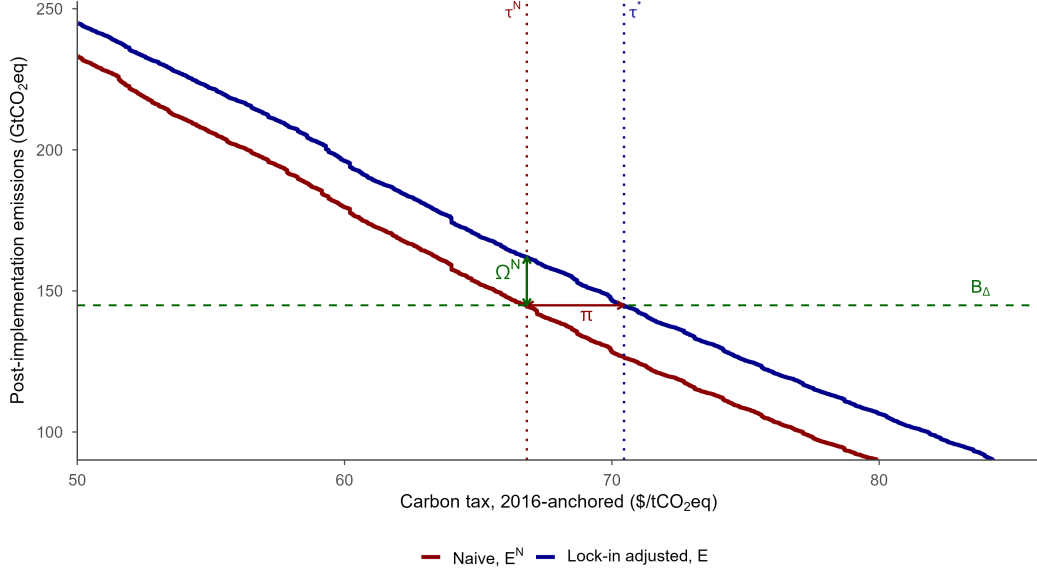
4.3 The lock-in premium and what the naive planner leaves behind

Overproduction during the delay raises the tax needed to meet the budget. The 205.4 GtCO₂eq released between 2016 and 2029, against 154.6 under the timely policy, leave 144.9 GtCO₂eq for the rest of the century. Investment during the delay does something more: it locks in additional reserves resilient to high taxes, so that a given tax removes fewer emissions than it would have before the delay. The tax has to rise for both reasons, and a naive planner using a single full-cycle cost per asset only sees the first.

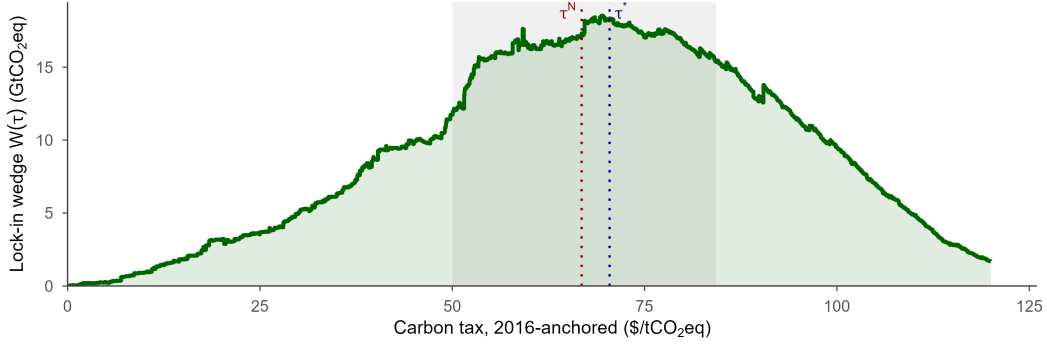
Panel (a) of Figure 6 reports post-implementation emissions under the two calibrations of Section 2.4. E^N is computed from full-cycle breakevens, as if the delay-induced fields could still be deterred, and E from their operating breakevens as of 2030. Both are compared with the remaining budget $B_\Delta = 144.9$ GtCO₂eq, drawn as a horizontal line. The line is flat because no tax is levied before 2030, so the choice of the 2030 tax does not change what has already been emitted. The tax at which E^N meets the budget, the naive tax, is $\tau_{2030}^N = \$66.8$. The tax at which E meets it is $\tau_{2030}^* = \$70.5$. The premium is the difference between the two, $\pi = \tau_{2030}^* - \tau_{2030}^N = \$3.6/\text{tCO}_2\text{eq}$, 5.4% of the naive tax.⁹ Both curves fall in small steps because emissions are counted field by field. They are close to parallel over the whole range shown, 12 to 18 GtCO₂eq apart, so the premium does not depend on reading the two curves at a particular point.

Panel (b) draws the wedge itself as a function of the tax, and it has the hump shape the model predicts. It is zero at a zero tax, where no field is delay-induced. It rises as the tax draws the capital-intensive fields of the cohort into their lock-in zone, with steps where large fields enter, then forms a plateau between about \$55 and \$80, and finally declines, as taxes that high phase out even locked-in fields after 2030. Both the naive tax and the lock-in-adjusted tax sit on this plateau, close to its highest point. This is not a choice made in the analysis, since both taxes are set by the remaining budget, and it means that a moderate

⁹The premium is computed before rounding the two taxes: $70.45 - 66.82 = 3.64$. In 2030 dollars, the premium amounts to \$4.8, on a tax of \$93.0 against \$88.2.



(a) Post-implementation emissions under the full-cycle and lock-in-adjusted calibrations



(b) Lock-in wedge $W(\tau) = E(\tau) - E^N(\tau)$

Figure 6: Full-cycle and lock-in-adjusted calibration of the 2030 tax.

Notes: (a) Post-implementation emissions computed from full-cycle (E^N) and operating (E) breakevens of the delay-induced fields, against the remaining budget $B_\Delta = 144.9$ GtCO₂eq; the dotted lines mark $\tau^N = 66.8$ and $\tau^* = 70.5$. The premium π is the horizontal distance between the two curves at the budget, the residual overshoot Ω^N the vertical distance at τ^N . (b) The lock-in wedge $W(\tau) = E(\tau) - E^N(\tau)$ as a function of the tax; the dotted lines mark the naive tax, at which the wedge gives Ω^N , and the lock-in-adjusted tax, and the shaded band is the range of taxes shown in panel (a). Taxes are 2016-anchored levels of the tax rising at 2% per year.

error in the naive tax would change the residual overshoot very little. The plateau may look relatively narrow in tax terms, but it is wide in terms of emissions, because each dollar of tax moves about 4.8 GtCO₂eq near the naive tax: moving the budget by 10% in either direction leaves the residual overshoot within 1 GtCO₂eq of its central value (Section 4.4).

The residual overshoot. A tax set on full-cycle costs and implemented from 2030 would still overshoot the budget by $\Omega^N = 17.1$ GtCO₂eq, 11.8% of the budget remaining in 2030, and about one year of global oil emissions. Ignoring lock-in therefore leads to setting

the carbon tax too low, even when the budget itself is correctly identified.¹⁰

The size of the premium needs to be read against the tax it adds to. Because the oil price does not respond to the tax in the model, the tax studied here is the part borne by producers, which is why the tax that meets the budget in 2016 is only \$57.2. And because it applies to the lifecycle emissions of all the oil produced worldwide, each dollar of it moves a large quantity of emissions. Near the naive tax, post-implementation emissions fall by about 4.8 GtCO₂eq per dollar. A premium of a few dollars therefore corresponds to a residual overshoot of about one year of global oil emissions, and equation (11) gives the conversion from one to the other.

How the residual overshoot builds up with the delay. Table 4 reports, for each implementation year, the remaining budget, the lock-in-adjusted tax, the premium, and the residual overshoot left by the naive tax, for delays starting in 2016 and in 2025. The cost of ignoring lock-in grows roughly linearly until 2026 and more slowly afterwards. It reaches 8.2 GtCO₂eq after six years and 17.1 GtCO₂eq after fourteen. Two forces are at work. Each additional year of delay adds new fields that will produce for their whole life after implementation, but it also ages every field already built, so that commitments made earlier are partly emitted during the delay rather than after it. As the delay lengthens, the second channel slows the build-up. The right-hand columns show the same build-up from a later start: ignoring only the five years of lock-in of the 2025–2029 period would still leave 10.2 GtCO₂eq above the budget, 7.1% of what remains, and require a premium of \$1.9. Most of the lock-in measured here is decided in the last years before implementation, mainly by deepwater projects, as Figure 5 shows.

What the premium measures. The premium rises with the length of the delay in Table 4, because a longer delay lets more projects be screened with a tax that is too low. The length of the delay is not the whole story, however. The premium measures how far the supply curve has moved below the full-cycle calibration, which depends both on how many projects were developed during the delay and on how far the tax their investors used fell below the tax the budget turns out to require (Section 2.5). Delay-induced fields are, by definition, projects that would not have been worth developing had investors expected the tax: they cover their operating costs after 2030, but will not necessarily recover their

¹⁰The relevant number is read at $\tau^N = 66.8$ because the full-cycle calibration solves $E^N(\tau^N) = B_\Delta$. This differs from the 15.9 GtCO₂eq of Figure 3, which is the wedge at the timely tax, set in 2016. The two measures count locked-in fields at different tax levels. A higher tax draws more fields into their lock-in zone and shortens the lives of those already in it, and between \$57.2 and \$66.8 the first effect dominates.

Year	Budget (GtCO ₂ eq)	Adjusted tax (\$/tCO ₂ eq)	From 2016		From 2025	
			Premium (\$/tCO ₂ eq)	Residual overshoot (GtCO ₂ eq)	Premium (\$/tCO ₂ eq)	Residual overshoot (GtCO ₂ eq)
2016	350.3	57.2	—	—	—	—
2018	320.9	58.3	0.5	2.8 (0.9%)	—	—
2020	291.4	59.4	0.7	5.2 (1.8%)	—	—
2022	264.0	60.6	0.9	8.2 (3.1%)	—	—
2024	235.1	62.8	2.1	11.1 (4.7%)	—	—
2026	205.8	64.8	2.4	13.8 (6.7%)	0.4	1.9 (0.9%)
2028	175.6	67.5	3.2	15.3 (8.7%)	1.3	5.8 (3.3%)
2030	144.9	70.5	3.6	17.1 (11.8%)	1.9	10.2 (7.1%)

Table 4: Lock-in premium and residual overshoot by year of implementation.

Notes: Tax rising at 2% per year. Taxes are 2016-anchored levels. The remaining budget is the initial budget net of emissions released before implementation on the business-as-usual path. Overshoot percentages are relative to the remaining budget. “From 2016” and “From 2025” indicate the year from which the delay is measured, the state of the industry in that year being taken as given.

development cost. Section 4.4 reports the results with a variety of anticipation scenarios by investors, in which the gap between their expectations and the actual tax needed narrows.

4.4 Sensitivity

Figure 7 reports the residual overshoot and the lock-in premium across six groups of assumptions: expectations, costs and the computation of net present values, carbon intensities, the carbon budget, the increase rate of the tax, and pass-through. Outside the anticipation group, the main specification is not an extreme case: several specifications, such as constant marginal costs, a 15% discount rate, a GWP-20 conversion and a 1% tax increase rate, give a residual overshoot above 19 GtCO₂eq, and outside the anticipation and tax increase groups the premium ranges from \$2.5 to \$4.6. Within the anticipation group, the main specification is by construction the upper bound for the residual overshoot, but the scenarios closest to observed investor behaviour give results of the same order.

Expectations. Anticipation changes which fields are developed during the delay, and nothing else. Investors who expect part of the tax stop developing the projects that would not be worth building even at that anticipated tax, and these are the fields with the lowest development breakevens, typically the most capital-intensive ones, which carry the deepest lock-in (Section 2.5). The anticipation scenarios of Figure 7 span two ways of being wrong about the tax. Investors who expect the correct tax but only from 2030 stop developing the late projects that would have carried the largest post-2030 commitments, and the residual

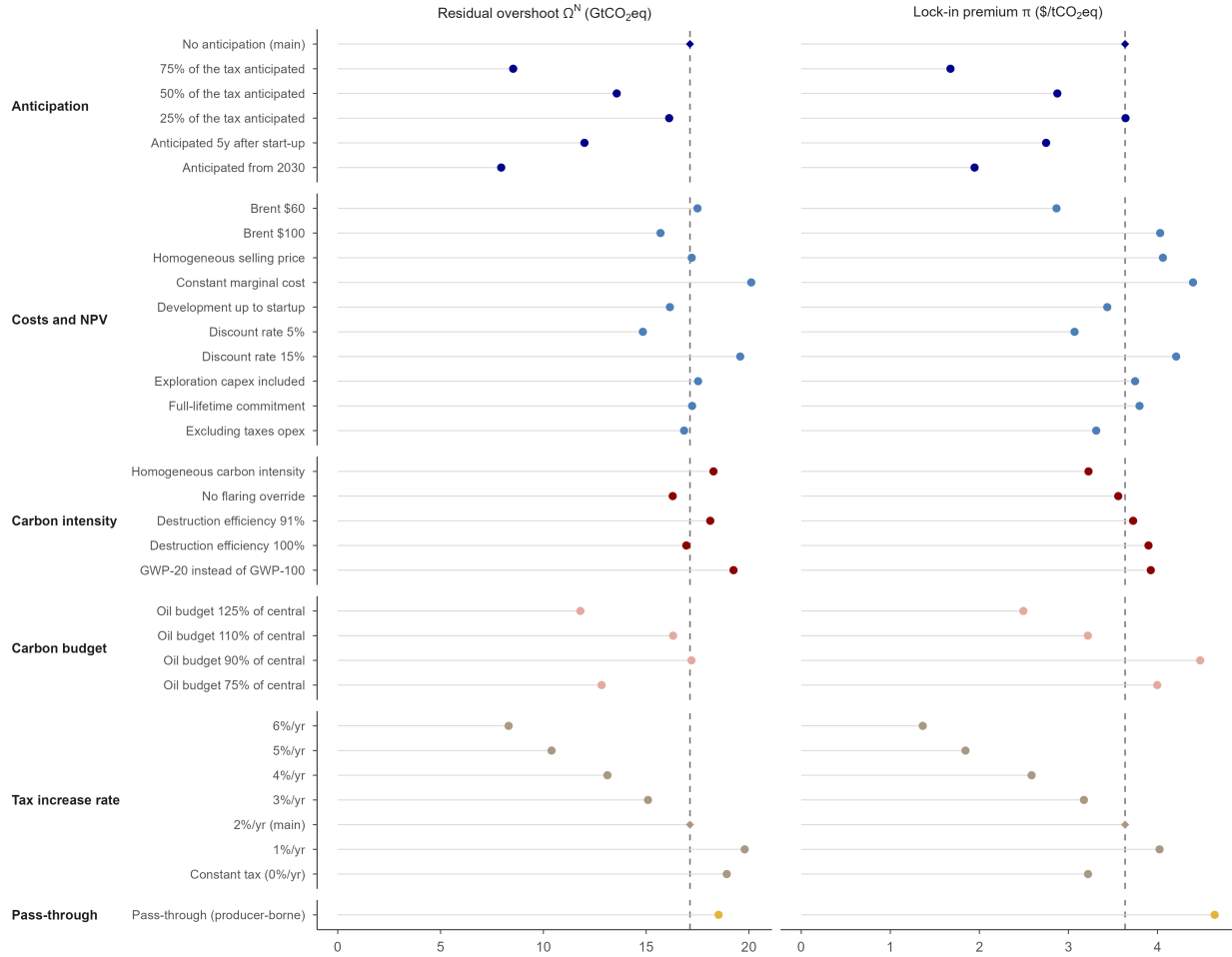


Figure 7: Sensitivity of the residual overshoot and the lock-in premium, by group of assumptions.

Notes: Each row is one specification; the main specification is marked by the diamond and the dashed line. Taxes are 2016-anchored levels of the tax rising at 2% per year. The pass-through row reports the premium per dollar borne by producers at implementation, which makes it comparable with the other rows.

overshoot falls to 7.9 GtCO₂eq (and the premium to \$1.9). If investors screen their projects as if the correct tax were implemented five years after the projects' start-up, the residual overshoot amounts to 12.0 GtCO₂eq and the premium to \$2.7. Investors who expect the tax immediately but underestimate its level leave 8.5 GtCO₂eq and \$1.7 when they expect three quarters of it, and 13.6 GtCO₂eq and \$2.9 when they expect half of it.

Which of these scenarios describes firms best? Oil companies report internal carbon prices, mostly between \$23 and \$60 (Ahluwalia, 2017), but these are shadow prices: they describe the tax in a policy scenario, not the probability that the scenario occurs (Fried et al., 2022). The internal carbon fees that do embed this probability are much lower, with a modal value of \$10/tCO₂ among the few firms that disclose one (Fried et al., 2022). An

anticipated tax of this level, about 15 to 17% of the timely tax, leaves the residual overshoot at 16.5 GtCO₂eq and the premium at \$3.5.

Figure 8 generalises the share-of-the-tax scenarios to any share of the tax anticipated by investors. The residual overshoot and the premium fall only slowly as long as investors expect less than about half of the tax. The residual overshoot is still 15.3 GtCO₂eq when they expect 30% of it and 13.6 GtCO₂eq when they expect half, and the premium is still \$2.9 at half. The fields removed by weak expectations are the most capital-intensive ones, but there are few of them, because at these tax levels most projects remain worth developing. The decline is faster above half: the residual overshoot falls to 8.5 GtCO₂eq at three quarters and 2.8 GtCO₂eq at 90%, and both measures reach zero at full anticipation, where investors screen with the tax the budget requires. Reducing the cost of the delay substantially would thus have required investors to expect a tax close to the one the budget requires, that is a commitment to the tax that was both announced and credible.

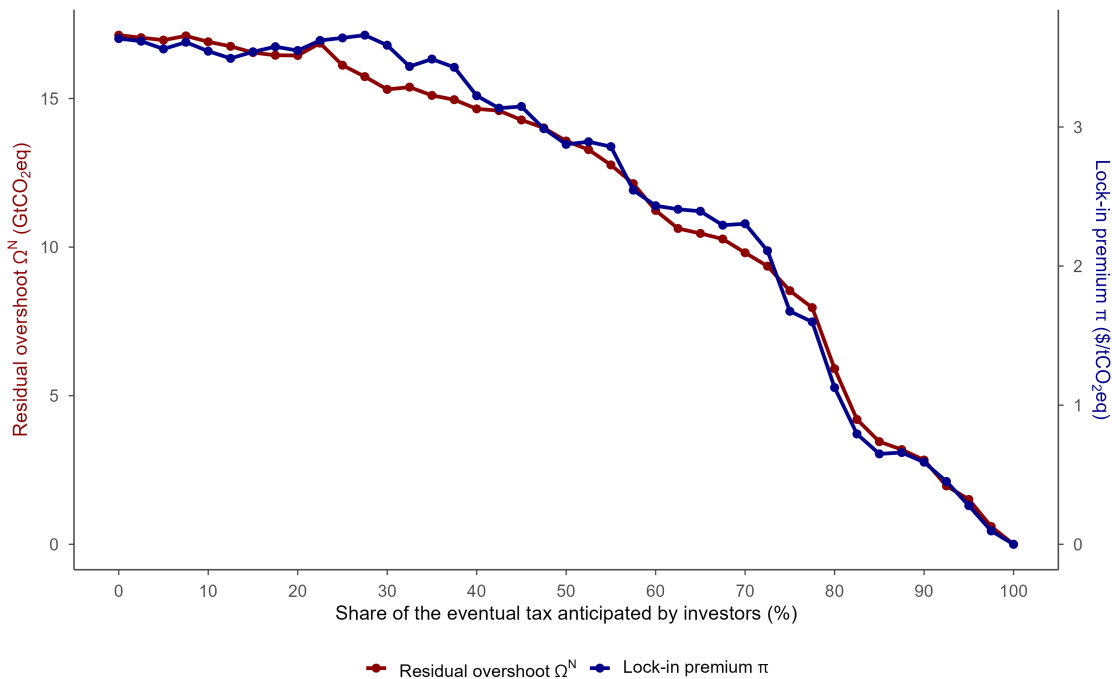


Figure 8: Residual overshoot and lock-in premium, by share of the tax anticipated.

Notes: Share of the eventual tax anticipated by investors on the horizontal axis. Both are zero at full anticipation, where investors screen projects with the tax the budget requires (Section 2.5).

Costs, carbon intensities and budget. The cost assumptions move the results in the direction predicted by the model. A higher discount rate deepens every lock-in zone. The residual overshoot is 14.8 GtCO₂eq at 5% and 19.6 GtCO₂eq at 15%, and the premium

\$3.1 and \$4.2. Constant marginal costs, which remove the rising cost path that phases out locked-in fields over time, raise the residual overshoot to 20.1 GtCO₂eq. Counting as development only the capital spent up to start-up makes every lock-in zone shallower and gives a lower bound on lock-in, with a residual overshoot of 16.2 GtCO₂eq and a premium of \$3.4 (Section 3.3): the results change little with where the line between development and production capital is drawn. Adding the fields not yet discovered, with their exploration costs treated as development costs, changes little, with a residual overshoot of 17.5 GtCO₂eq, because most of them are not developed before 2030 and are stopped by the tax. The Brent price is one of the few assumptions that move the two measures in opposite directions. At \$60, the residual overshoot rises to 17.5 GtCO₂eq and the premium falls to \$2.9; at \$100, the residual overshoot falls to 15.7 GtCO₂eq and the premium rises to \$4.0. A higher price raises every production breakeven and makes emissions respond less to the tax near τ^N , so the same quantity requires a larger premium. Assumptions on carbon intensity move the residual overshoot between 16.3 and 19.2 GtCO₂eq. A common carbon intensity across assets raises it to 18.3 GtCO₂eq, which indicates that locked-in fields are on average cleaner than the rest of the sample. Varying the budget traces the hump of the wedge from another angle: the residual overshoot is 11.8 GtCO₂eq at 125% of the central budget, 16.3 GtCO₂eq at 110%, 17.2 GtCO₂eq at 90% and 12.8 GtCO₂eq at 75%. It changes little around the central value. It falls with a much larger budget, because the lower tax deters fewer fields in the timely counterfactual, and with a much smaller one, because the higher tax starts to shut down locked-in fields themselves.

Instrument design. The constant tax that meets the same budget is \$99.4/tCO₂eq, and the delay then overshoots the budget by 34%, against 19% under the increasing tax. Appendix D.4 reports the full set of constant-tax results. This comparison is itself a result: between a constant tax and a tax rising at 2%, the total cost of the delay changes a lot, while its lock-in component changes little. Under the constant tax, the residual overshoot is 18.9 GtCO₂eq and the premium \$3.2, against 17.1 GtCO₂eq and \$3.6 under the increasing tax, whereas the total overshoot is almost twice as large. Across increase rates from 0 to 6%, the residual overshoot varies between 8.3 and 19.8 GtCO₂eq. As Section 2.6 explains, the tax increase rate acts on lock-in in two opposite directions: it moves the development boundary and makes more young fields delay-induced, and it shortens the post-implementation life of every locked-in field. The first effect dominates between 0 and 1%, where the residual overshoot rises, and the second above. The shape of the tax matters mostly for the intensive margin, which is released before implementation and which no tax set in 2030 can recover.

Pass-through. The tax studied so far is entirely borne by producers. I relax this assumption using exogenous pass-through rates from [Heal and Schlenker \(2019\)](#), who solve for the oil price path under a carbon tax using the same field-level data. For a tax rising at 2% per year, consumers initially bear 40% of the tax through a higher oil price and producers 60%, and the share borne by producers then rises until they bear the whole tax. I assume that this share rises at a constant rate and reaches 100% sixty years after the tax is introduced.¹¹ Only the share borne by producers deters development and retires production, which changes the results in two ways. First, the gross tax required to meet the budget is higher, and so is the gross premium. Second, because the pass-through path starts when the tax is introduced, a tax delayed to 2030 is borne by producers at a lower share in 2030 than a timely tax would have been at the same date. This is a second channel through which the delay raises the tax required, and it is specific to this variant. Expressed per dollar borne by producers at implementation, which makes it comparable with the other rows of Figure 7, the premium is \$4.6 against \$3.6 in the main case, and the residual overshoot is 18.5 GtCO₂eq against 17.1 GtCO₂eq. The gross premium, which also includes the second channel, is \$7.7.

5 Incidence and targeted policy

This section asks where the locked-in fields are, who bears the cost of the premium, and what can be done to alleviate the cost of lock-in. The three questions are linked. Because locked-in fields barely respond to a higher tax (Section 5.1), the premium is paid by other fields, adding a second level of redistribution across fields after the one created by the overproduction during the delay, and this changes who gains and who loses from the delay (Section 5.2). Because the locked-in fields are few and similar, targeted policies can partially achieve what the delayed tax cannot, provided they target the right projects early enough (Section 5.3).

5.1 Where the lock-in sits

Overcommitted emissions are concentrated in a small number of projects, mostly offshore. At the naive tax, the emissions committed beyond 2030 by delay-induced fields come from 694 assets, and 41 of them carry half of the total (Appendix Table 18). Figure 9 shows

¹¹Values are drawn from Figure A4 in [Heal and Schlenker \(2019\)](#). A producer share of 60% reaching 100% in sixty years corresponds to a growth rate of 0.855% per year, which is the rate used in the analysis.

where these projects are, pooled by province.¹² Offshore projects account for 75% of these emissions, and deepwater projects alone for 40%. The coasts of Brazil and Guyana and the Gulf of Mexico account for 34% of the total, Brazil alone for 20%, and South America for 31%. The Middle East accounts for 22%, mostly through a few large offshore and onshore redevelopments approved during the delay, the largest single asset of the sample being the expansion of the Marjan field in Saudi Arabia. What these projects have in common is high capital intensity combined with operating costs low enough for the tax to fall inside their zone, as Section 2.2 predicts. The timing is also concentrated, since 55% of these emissions are committed by investments made between 2025 and 2029.



Figure 9: Locked-in projects by province.

Notes: Circle size is the post-2030 emissions of delay-induced fields that survive the naive tax implemented in 2030. A province is a level of aggregation in Rystad's data between the asset and the country. No-anticipation scenario, tax rising at 2% per year.

The locked-in stock in these assets is also deep. Figure 10 reports, for each level of the 2030 tax, the quantity of emissions locked in at the naive tax that would still be released, split by supply segment; Appendix Figure 25 gives the same curve for all delay-induced fields together. Raising the tax to the lock-in-adjusted level of \$70.5 leaves 92% of them in place. Even an \$80 tax, which would undershoot the remaining budget in 2030 by 27%, leaves 70% in place. The few locked-in shale and oil sands projects, by contrast, are almost all phased out by a tax of \$80, because their lock-in zones, although sometimes deep, sit low. For most

¹²Province is a level of aggregation available in Rystad smaller than countries, but comprising multiple assets. The median number of assets per dot on the graph is 2, the average 5.1, and the maximum 40 (Western Siberia, Russia).

locked-in projects, the carbon tax arrives too late. An extra dollar of tax removes little of their production, and meeting the budget requires stranding fields elsewhere.

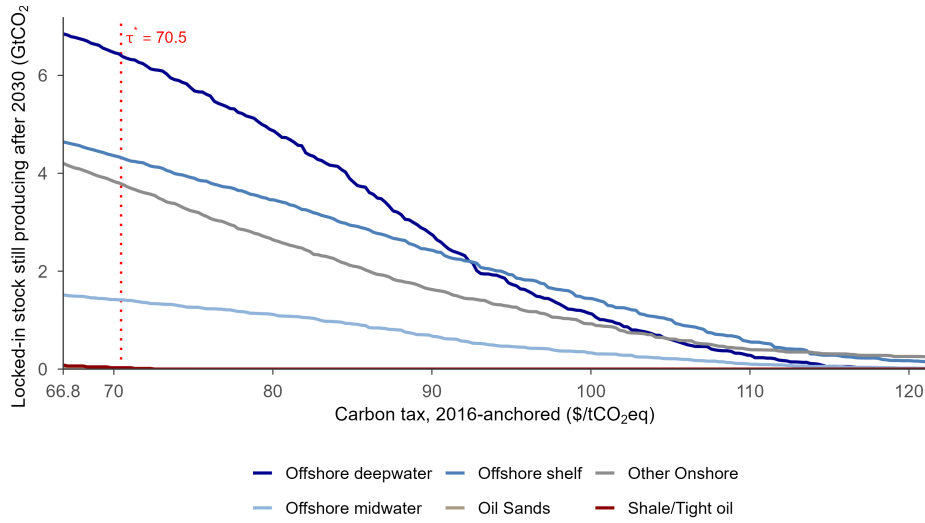


Figure 10: Depth of lock-in, by supply segment.

Notes: Post-2030 emissions locked in at the naive tax that are still released when the tax implemented in 2030 is raised to the level on the horizontal axis, by supply segment, in GtCO₂eq. The horizontal axis starts at the naive tax, $\tau^N = \$66.8$, and the dotted line marks the lock-in-adjusted tax, $\tau^* = \$70.5$. Taxes are 2016-anchored levels of the tax rising at 2% per year.

5.2 Who gains from the delay

The delay changes each producer's cumulative emissions, relative to the timely policy, through two channels. The overproduction channel runs through the production the delay allows fields to sell untaxed before 2030, and through the production lost after 2030 when the tax has to rise from τ_{2016}^* to τ_{2030}^N to compensate for it. The lock-in channel runs through the fields that exist because of the delay. These fields gain their post-2030 locked-in production, and other fields lose the production stranded when the tax has to rise further, from τ_{2030}^N to τ_{2030}^* , to compensate for it. The 50.8 GtCO₂eq excess emissions before 2030 are offset by 50.9 GtCO₂eq from assets stranded after 2030 by the first tax increase, and the 15.8 GtCO₂eq that delay-induced fields emit after 2030¹³ are offset by 15.6 GtCO₂eq stranded by the second.¹⁴ Table 5 reports the decomposition by continent, in GtCO₂eq and as a share of each continent's own 2016 reserves, and Appendix Table 19 reports it for the largest producing countries.

¹³These 15.8 GtCO₂eq correspond to the 17.1 GtCO₂eq residual overshoot, minus 1.3 GtCO₂eq removed from the delay-induced assets participating in the residual overshoot by the increase in tax from \$66.8 to \$70.5.

¹⁴While both channels should balance, small differences between gains and losses can exist, due to steps in the supply curve caused by large assets.

	Overproduction (GtCO ₂ eq)	Lock-in (GtCO ₂ eq)	Net (GtCO ₂ eq)	Net (% of 2016 reserves)
North America	28.38	1.47	29.85	16.6
South America	1.29	3.98	5.26	10.5
Asia	3.34	1.26	4.60	7.6
Europe	1.10	0.46	1.56	11.2
Africa	-0.39	1.19	0.80	1.6
Oceania	0.03	0.04	0.07	3.2
Russia	-5.41	-2.13	-7.54	-11.8
Middle East	-28.46	-6.09	-34.55	-10.8

Table 5: Incidence of the delay by continent.

Notes: Change in cumulative emissions over 2016–2100 relative to a timely tax, no-anticipation scenario, tax rising at 2% per year. The overproduction column nets the production sold untaxed before 2030 against the production stranded when the tax rises from τ_{2016}^* to τ_{2030}^N ; the lock-in column nets the post-2030 production of delay-induced fields against the production stranded when the tax rises from τ_{2030}^N to τ_{2030}^* . The two sum to the net effect. The last column is the net change in production (in barrels) as a share of the continent’s own 2016 reserves; since carbon intensities differ across fields, it can differ in sign from the change in emissions for small net effects.

Three facts stand out. First, the usual view of the delay’s incidence, which counts only overproduction, has North America gaining the most and the Middle East paying the most. North American fields, mostly shale and oil sands, sell 31 GtCO₂eq more before 2030 than a timely tax would have allowed and gain 28 GtCO₂eq on net, while the low-cost fields of the Middle East are the ones a higher tax strands afterwards, 28 GtCO₂eq of them on net. Russia is on the same side as the Middle East, with a net loss of 5 GtCO₂eq. Second, lock-in changes the picture in the middle of the distribution. It accounts for 5% of the net effect for North America, where few projects would have been deterred by a timely tax, but for three quarters of the net effect for South America. It adds to the losses of Russia and the Middle East rather than offsetting them. The production stranded by the premium falls on low-cost, long-lived fields, 9.7 GtCO₂eq in the Middle East and 3.0 GtCO₂eq in Russia, 81% of the total. These two regions therefore pay for the lock-in of other regions on top of paying for the overproduction of the delay.

Third, for some producers the lock-in channel reverses the sign of the total. Africa loses 0.4 GtCO₂eq through overproduction and gains 1.2 GtCO₂eq through the fields it develops because of the delay, which makes it a net winner, at 0.8 GtCO₂eq. The same reversal holds for Brazil, which loses 0.4 GtCO₂eq through overproduction, gains 3.0 GtCO₂eq through its deepwater fields, and ends a net winner at 2.7 GtCO₂eq, and on a smaller scale for Angola, Namibia, Senegal and Vietnam. Producers whose lock-in gain falls short of their overproduction loss, such as Kazakhstan, Qatar and Azerbaijan, remain net losers: lock-in makes them smaller losers, not winners. Guyana, whose oil production only started in 2019,

does not reverse, because it also gains through overproduction.

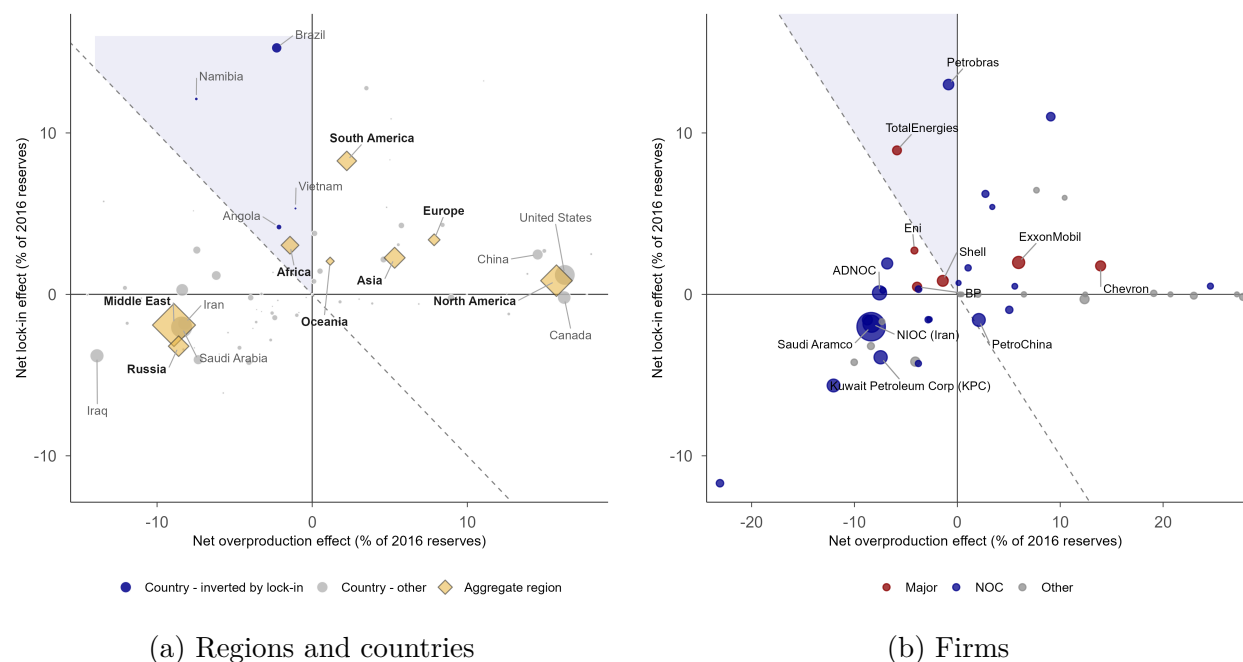


Figure 11: Incidence of the delay as a share of 2016 reserves.

Notes: The horizontal axis is the net effect through overproduction, the vertical axis the net effect through lock-in. (a) Aggregate regions (diamonds) and countries (circles, area proportional to 2016 reserves); countries with less than 500 Mbbl of 2016 reserves are excluded. (b) The 50 largest firms by 2016 equity reserves; production is attributed to equity holders at their share in each year, and shares after 2026 are held at their 2026 level. EOG Resources, Midland Oil Company and Cenovus Energy lie outside the frame. In both panels, the dashed line is where the net effect is zero, and the shaded area contains the producers that lose through overproduction and gain more through lock-in, whose net sign is reversed by lock-in.

Figure 11 shows the decomposition as a scatter, normalised by 2016 reserves so that producers of very different size can be compared. The horizontal position is the usual view of the delay's incidence, and the vertical position is what this view misses. Real winners from the delay, accounting for both overproduction and lock-in, lie above the dashed line, where the net effect is positive. In the rest of the upper-left quadrant, producers see their losses diminished by the lock-in, without being net winners. As highlighted in Tables 5 and 19, the US, China, and most of Europe and Asia win on both counts, while Russia and Middle Eastern countries see lock-in add a loss of two to four percent of reserves to a loss of eight to fourteen percent through overproduction.

The shaded area highlights producers that the usual view counts as losers, because a higher tax strands their existing fields, but whose investment during the delay is large enough to more than compensate: Brazil, with a gain of 15% of its reserves through lock-in against a loss of 2% through overproduction; Namibia and Senegal, whose oil industry mostly dates from after 2016; Angola; and Africa as a whole, where these countries outweigh losers on both counts such as Algeria. South America as a region lies just to the right of the vertical

axis, with lock-in accounting for most of its gain. The reversals are the same whether the effect is measured in barrels, as in the figure, or in emissions, as in Table 5.

Firms. The same decomposition can be run by company, attributing each year’s production to equity holders according to their share in that year (Appendix Table 20 and panel (b) of Figure 11). As expected, national oil companies as a group lose 26 GtCO₂eq through overproduction, Saudi Aramco alone 10.6 GtCO₂eq, and lose again through lock-in. Independent producers, mostly North American and focused on shale and oil sands, gain 15 GtCO₂eq through overproduction and little through lock-in. Majors are split in two groups: American majors win on both counts while European majors lose from overproduction and gain from lock-in. The firms the usual view misclassifies are those whose investment during the delay is in deep offshore projects. Petrobras loses 0.1 GtCO₂eq through overproduction and gains 2.0 GtCO₂eq through lock-in, 13% of its 2016 equity reserves, and TotalEnergies loses 0.5 GtCO₂eq and gains 1.0 GtCO₂eq. For these firms, the assets that make the delay profitable are those a targeted restriction would have to reach.

5.3 Targeted development restrictions

A worldwide carbon tax on oil producers as envisioned in the paper seems unlikely in the near future. At the international level, the climate regime does not favour supply-side policies. As [van Asselt \(2021\)](#) puts it, the “United Nations Framework Convention on Climate Change (UNFCCC) territorial accounting for greenhouse gas emissions emphasizes fossil fuel consumption over supply, so countries are incentivized to pursue demand-side policies to fulfil international agreements”. Targeted action could nonetheless alleviate a substantial share of the cost arising from infrastructure lock-in. The literature on supply-side policy makes the general case for such instruments, on grounds of abatement certainty ([Erickson et al., 2018](#)), public support ([Green and Denniss, 2018](#); [Green et al., 2024](#)) and commitment value ([Harstad and Holtmark, 2026](#)). Measuring lock-in field by field makes it possible to evaluate specific restrictions rather than their principle.

This subsection studies a variety of targeted policies aiming at alleviating the cost arising from lock-in. All instruments, with the exception of targeted early taxes, act on development only instead of production. A ban prevents the approval of new projects in a given perimeter from a given year. Projects already approved are left alone, so the exercise never cancels capital that is already partly spent. Table 6 reports, for each instrument, the naive tax, the lock-in-adjusted tax, the premium and the residual overshoot, each instrument being evaluated at its own budget-consistent tax, and Appendix D.9 defines each perimeter and

each instrument precisely. The remaining budget also changes, because a project that is not developed does not emit during the delay either, and this second effect is reported in the last column: it is what a restriction buys before implementation, as opposed to what it buys on lock-in.

	τ^N (\$/tCO ₂ eq)	τ^* (\$/tCO ₂ eq)	π (\$/tCO ₂ eq)	Ω^N (GtCO ₂ eq)	Remaining budget (GtCO ₂ eq)
No targeted policy (delay only)	66.8	70.5	3.6	17.1	144.9
<i>Ban on new approvals from 2016</i>					
Deepwater	64.9	67.4	2.5	10.4 (−40%)	149.4
Shale	66.2	70.0	3.8	17.0 (−1%)	147.3
The 50 largest locked-in projects (upper bound)	66.2	68.1	1.9	7.5 (−56%)	147.2
The 20 largest locked-in projects (upper bound)	66.4	69.1	2.7	11.9 (−31%)	145.9
Deepwater, OECD-headquartered firms	65.6	68.7	3.1	13.7 (−20%)	147.5
Deepwater, European firms	66.4	69.9	3.5	15.4 (−10%)	145.8
<i>Ban on new approvals from 2025</i>					
All new development	59.8	63.5	3.7	11.6 (−32%)	145.3
Deepwater	66.3	69.9	3.7	15.3 (−11%)	145.0
Shale	66.8	70.4	3.6	17.1 (−0%)	144.9
<i>Other instruments</i>					
Carbon tax on deepwater from 2016	66.1	68.4	2.4	10.3 (−40%)	150.1
Carbon tax on shale from 2016	66.3	70.0	3.8	16.9 (−1%)	165.4
Carbon tax on OECD deepwater from 2016	66.4	69.3	2.9	13.3 (−22%)	147.4
Development capital cost +50%, deepwater	66.0	70.0	4.0	18.1 (+6%)	145.2
Development capital cost +50%, all projects	63.4	68.2	4.8	19.7 (+15%)	147.2

Table 6: Targeted development restrictions and alternative instruments.

Notes: Each instrument is evaluated at its own budget-consistent tax. Taxes are 2016-anchored levels of the tax rising at 2% per year, in \$/tCO₂eq. The change in the residual overshoot relative to the delay without restriction is in parentheses. The remaining budget rises under a restriction because less is emitted before 2030. Premiums are computed before rounding the two taxes, so they can differ by 0.1 from the difference of the rounded values. Appendix Figure 26 reports the full set of instruments.

Three lessons come out of the table, and each of them follows from a property of lock-in established earlier. First, what a restriction achieves depends on the depth of the lock-in zones it removes. A ban on new deepwater approvals from 2016 cuts the residual overshoot by 40%, from 17.1 to 10.4 GtCO₂eq, and the premium by a third. A ban on new shale approvals cuts the residual overshoot by 1% only and slightly raises the premium, to \$3.8. The shale ban, the closest instrument here to the politically salient ban on fracking, still has an effect. It enlarges the budget remaining in 2030 by 2.4 GtCO₂eq, because shale projects that are not developed do not produce during the delay either.¹⁵ But it leaves lock-in untouched, since shale lock-in zones are shallow and sit below the tax. The two ban targets pursue different objectives. Policies aimed at shale buy budget while policies aimed at capital-intensive offshore projects lower the cost of lock-in.

¹⁵Since a large share of shale assets are already producing by 2016, the reduction in delay-era overproduction remains limited. However, shale production within an asset is maintained or increased through continuous drilling of new wells. A policy banning the drilling of any new well would hence affect shale overproduction much more.

The second lesson is that the concentration documented in Section 5.1 makes very narrow restrictions effective. Banning the approval of the 50 assets that carry the most locked-in emissions cuts the residual overshoot by 56%, more than a ban on all new deepwater development, and brings the premium down to \$1.9. Banning the 20 largest cuts it by 31%. This is the practical consequence of the result that 41 assets carry half of the locked-in stock. The ranking uses what the delay eventually built, which a planner in 2016 could not know, so these two rows are an upper bound on what targeting by size can achieve. The deepwater ban, which requires no such foresight, is the implementable benchmark.

The third lesson is that a restriction does not have to be worldwide, but it does have to reach the firms that build capital-intensive projects. A ban applying to assets at least half owned and operated by firms headquartered in an OECD country cuts the residual overshoot by 20%, half of what the worldwide deepwater ban achieves, and the same ban restricted to European firms cuts it by 10%. A coalition can reach the deepwater development its own firms own and operate, which is a much larger share than the development taking place on its territory. More than 70% of the deepwater projects approved during the delay, and 58% of the reserves they hold, are at least half owned and operated by firms headquartered in an OECD country, so a restriction decided in a handful of jurisdictions reaches projects on three continents. Ahlvik et al. (2026) estimate that 47–73% of the emission reductions from an OECD-only tax on oil production are offset by non-OECD production in the medium term, through the oil price. The model excludes this channel, so the ownership-based rows of Table 6 are upper bounds. Yet, the offset should be smaller than in their setting, where non-OECD production is untaxed, because the restricted projects produce mostly after 2030, when every producer faces the tax.

Bans. Applied from 2025 rather than 2016, the deepwater ban cuts the residual overshoot by 11% instead of 40%, and a ban on all new approvals from 2025, which is as broad an instrument as one can write down, cuts it by 32%. Most of the lock-in of the delay is already approved by 2025, consistent with the concentration of deepwater commitments in the 2020–2024 and 2025–2029 periods (Figure 5). Approval precedes commitment: a project is approved before most of its capital is spent, and its emissions become committed only once enough of that capital is sunk for it to be completed under the tax (Section 3.4). Acting later than that means cancelling projects that are already approved, and possibly already under construction, with the compensation issues this raises (Paletta et al., 2025), rather than deterring them.

Alternative instruments. Two instruments are worth comparing with the bans. Early action through a tax targeting deepwater projects alone from 2016, at the level the budget requires, achieves the same reduction in the residual overshoot as the deepwater ban, 40%. It also enlarges the remaining budget in 2030, 5.2 against 4.5 GtCO₂eq, since it also retires active deepwater assets during the delay. The same instrument applied to shale from 2016 leaves lock-in where it is but enlarges the remaining budget by 20.5 GtCO₂eq, by far the largest budget effect of any instrument in the table: it confirms that segment-specific early taxation is a policy for the intensive margin when it targets shale, and for lock-in when it targets offshore projects. This contrast also points to a limit of the model that favours deepwater restrictions over early taxes on shale. A tax applied early to shale, or to OECD production more generally, removes a large volume of barrels while the rest of the world is still untaxed, and the price response this would trigger, which the model excludes, would bring part of that production back elsewhere. A restriction on deepwater development is much less exposed to that objection. It barely changes production during the delay, since the projects it stops are not producing yet, and what it removes materialises after 2030, when the tax is in force everywhere, which leaves less room for a price-driven response by other producers. The same argument applies to the ownership-based restrictions of Table 6: an early tax levied by OECD countries on their own deepwater assets cuts the residual overshoot by 22%, close to the ban, but it is the ban that avoids the spillover. Raising the cost of development capital, by contrast, does little against lock-in and can make it worse. Raising the development cost of deepwater projects by 50% raises the residual overshoot by 6% and the premium to \$4.0, and the same shock applied to all new projects raises the residual overshoot by 15% and the premium to \$4.8. Raising development costs lowers a project's development breakeven without changing its operating cost, which deepens the lock-in zone of every asset. The naive planner reads the shock as deterring more assets that are in reality already built by 2030, which is why the naive tax falls, from \$66.8 to \$63.4 when the shock applies to all projects, while the emissions these projects still deliver after 2030 rise through the lower tax.

Incidence. Restrictions also change the incidence described in Section 5.2, and they change both sides of it at once. By removing part of the lock-in, they remove the gain that this lock-in gives to the producers who hold it. Because the premium falls, they also reduce the production stranded elsewhere to make up for it. The deepwater ban from 2016 cuts the post-2030 emissions of delay-induced fields from 15.8 to 9.8 GtCO₂eq worldwide, and the ban on the 50 largest projects cuts them to 7.1 GtCO₂eq. By the accounting of Section 5.2, the stranding that compensates for them falls by the same amounts, benefiting mainly the Middle East and Russia. South America and Africa, which gain 4.8 and 2.0 GtCO₂eq gross through

lock-in without any restriction, keep only 1.0 and 1.1 GtCO₂eq of it under the deepwater ban. The producers that gain from lock-in are therefore those that would oppose a restriction on it, and the producers that pay for it through a higher tax are those that would gain from it. Part of what a restriction removes also comes back elsewhere. Under the deepwater ban the budget constraint is relaxed and the tax the budget requires after 2030 falls from \$70.5 to \$67.4, so delay-induced fields in regions with little deepwater exposure produce for longer after 2030, which raises their post-2030 emissions by 4% in Asia and 17% in Russia. This leakage is internal to the model, distinct from the price-driven supply leakage the model excludes, and it goes in the same direction. An early tax on shale works on the other channel and therefore has a different incidence. It removes most of the overproduction gain of North America during the delay, which enlarges the remaining budget by 20.5 GtCO₂eq and lowers the tax the budget then requires, so the stranding borne after 2030 by the Middle East and Russia falls while lock-in is left untouched.

Scope for smaller initiatives. Convincing large producing countries to adopt such bans may prove difficult, since governments that depend on oil revenue are the least likely to adopt any constraint on production (Lujala et al., 2022), but the concentration documented above changes what smaller initiatives can achieve. Most bans adopted so far have come from public campaigns against specific projects (Piggot, 2018; Carter and McKenzie, 2020). With 41 assets carrying half of the post-2030 locked-in emissions, campaigns against specific projects can matter, and Table 6 puts a number on what they can achieve: stopping the approval of the 50 largest of them would have cut the residual overshoot by more than half. The Willow project in Alaska is an example. The project was widely debated in the United States, and was finally approved by the Biden administration in 2023. It embeds 235 MtCO₂eq of locked-in emissions that survive the naive tax, 1.4% of the total. Large offshore developments can weigh even more: the eight delay-induced phases of the Buzios deepwater project in Brazil¹⁶ carry 1.4 GtCO₂eq, 8% of the total. Many such projects are owned and operated by firms headquartered in OECD countries, as shown above, and their development depends on technologies these firms master. OECD countries could therefore restrict these specific projects and encourage the redirection of the corresponding capital, which is large: the investment made during the delay in fields that a timely tax would have deterred amounts to \$739 billion, discounted to 2025, and it is spread across regions, with about a fifth each in North America, Africa, Asia and South America.

¹⁶The Buzios project is split into 13 assets in Rystad data, each corresponding to a phase of the project. Four are already developed by 2016.

Equity. The two channels of the delay point to different producers, which allows efficiency and equity to be addressed together (Pye et al., 2020). Overproduction during the delay is dominated by North American shale and oil sands, where early action by developed countries is possible within a few jurisdictions (mainly Texas, North Dakota and Alberta) and would be justified on distributive grounds (Lenferna, 2018). Much of the deep lock-in sits in offshore projects owned and operated by majors headquartered in developed countries, which is where a restriction can be decided, while the incidence results show where it would be felt. For Brazil, Angola, Namibia and Senegal, the locked-in projects are what turns the delay into a gain, and a restriction on them is one these countries would experience as a cost. How large this cost is depends on what happens to the capital a restriction frees. The restriction does not remove the capital itself, and part of it could be invested in the same countries in other projects, so that the share that stays there determines how much of their gain from the delay these producers actually lose. Whether the remaining cost should be compensated depends, as Paletta et al. (2025) show, on contractual provisions and on whether undeveloped reserves count as assets at all.

6 Conclusion

This paper measures what delaying carbon pricing costs in the oil sector once the lock-in effects from fields developed during the delay are taken into account. During the delay, firms develop fields that a timely tax would have deterred, and once their development cost is sunk, a tax high enough to have prevented them is no longer high enough to stop them. Models of oil supply that give each field a single cost per barrel cannot account for this, because a field keeps the same position on their supply curve whether or not its capital has already been spent. Separating the development and production decisions of 25,336 oil fields brings this lock-in back into the cost of the delay. Delaying a 1.5°C-aligned carbon tax from 2016 to 2030 overshoots the oil carbon budget by 19%, and 16 GtCO₂eq of this overshoot is released after 2030 by fields that exist because of the delay. A tax set in 2030 on full-cycle costs leaves 17 GtCO₂eq above the budget, 12% of what remains, and closing this gap requires a lock-in premium of \$3.6/tCO₂eq.

Lock-in concentrates in fields that are expensive to develop and cheap to produce from, which in the data means deepwater offshore projects, while the overproduction during the delay comes mostly from fields that are cheap to develop, or already developed in 2016, and expensive to run, shale and oil sands above all. Because locked-in fields are resilient to a higher tax, the premium is paid by low-cost fields elsewhere, mostly in the Middle East and

Russia. For producers whose recent investment are into deep offshore projects, such as Brazil or Africa as a whole, lock-in turns what the usual view counts as a loss from the delay into a gain.

Most of this lock-in is decided in the last years before implementation, and announcing the tax does little against it if investors do not find it credible: when they screen their projects with half of the tax the budget requires, four fifths of the residual overshoot remain. Because lock-in is concentrated in a few hundred projects, and half of it in 41, restrictions on specific projects can do part of what the delayed tax cannot. A ban on new deepwater approvals from 2016 cuts the residual overshoot by 40% and the premium by a third, while a ban on new shale projects leaves both almost unchanged. When a delay in carbon pricing is expected, early restrictions on capital-intensive projects can therefore complement the tax that eventually comes.

These estimates have three limitations, the first two of which lead to understating the cost of the delay. First, the analysis is confined to oil, so the results should be read as if comparable policies applied to gas and coal at the same time; to the extent that a tax on oil alone shifts demand towards other fossil fuels, total emissions are understated. Second, the model excludes the green paradox, the response of producers who anticipate the tax by bringing production and investment forward, so the total cost of the delay computed here is likely a lower bound. Third, the paths of development and production are those forecast by Rystad at a constant oil price, which does not respond to the tax. If the price rose as the tax reduces supply, part of the tax would be offset for producers, and the post-2030 supply curves under both calibrations would become flatter, as they do at a higher price level or with partial pass-through in Section 4.4¹⁷. A given gap between them would then require a larger premium, while the residual overshoot itself moves by less than 1.5 GtCO₂eq in either direction in these variants. The same separation between development and operating costs applies wherever capital is sunk before climate policy arrives, from LNG terminals to gas power plants, and the lock-in premium could be measured there in the same way.

¹⁷The model also leaves aside the optimal fiscal response of resource-owning governments to an external carbon price (Section 3.6), and the political economy that produces delays in the first place.

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A Model extensions

A.1 Closed forms for a parametric cohort

This appendix collects the closed-form expressions behind the statements of Section 2. They rely on two simplifications that the body does not use: no discounting where indicated, and a parametric cohort of fields. None of the definitions in the body depend on them.

The discounted profit per barrel. Evaluating the integral in (4) with $a^* = a^*(\tau)$,

$$\bar{v}_\delta(\tau) = \frac{p - c_0 - \theta\tau}{\delta} (1 - e^{-\delta a^*}) - \frac{\gamma}{\delta^2} \left[1 - e^{-\delta a^*} (1 + \delta a^*) \right], \quad (13)$$

using $\int_0^{a^*} e^{-\delta a} da = (1 - e^{-\delta a^*})/\delta$ and $\int_0^{a^*} a e^{-\delta a} da = [1 - e^{-\delta a^*} (1 + \delta a^*)]/\delta^2$. By Leibniz, $\partial \bar{v}_\delta / \partial \tau = -\theta \int_0^{a^*} e^{-\delta a} da = -\theta(1 - e^{-\delta a^*})/\delta < 0$, the boundary term vanishing because the margin is zero at $a = a^*$. Since a^* does not depend on δ , $\partial \bar{v}_\delta / \partial \delta = -\int_0^{a^*} a e^{-\delta a} (p - c_0 - \theta\tau - \gamma a) da < 0$, the integrand being positive on $(0, a^*)$. As $\delta \rightarrow 0$, $\bar{v}_\delta(\tau) \rightarrow (p - c_0 - \theta\tau)^2 / 2\gamma$.

Depth without discounting. With $\bar{v}_0(\tau) = (p - c_0 - \theta\tau)^2 / 2\gamma$, the development breakeven, the depth and the width of the zone are explicit:

$$\theta \bar{\tau}_D = p - c_0 - \sqrt{2\gamma\kappa}, \quad \theta D = \sqrt{2\gamma\kappa}, \quad a^*(\bar{\tau}_D) = \sqrt{2\kappa/\gamma}, \quad (\delta = 0). \quad (14)$$

Because $\bar{v}_\delta$ decreases with δ , a higher discount rate lowers $\bar{\tau}_D$ and deepens the zone, so the square root is a lower bound on the depth of lock-in at any positive discount rate. The log-log slope of depth per barrel on capital intensity is 1/2 at $\delta = 0$ and above it at any positive rate. This is the reference for the slope of 0.55 reported in Section 3.1, which, as noted there, is a check on the functional form rather than a test of the model.

The cohort. Let a mass N of fields be developed during the delay, with capital intensity distributed with c.d.f. F_κ on $[0, \kappa_1]$, independent of the age at implementation a_Δ , which is uniform on $(0, \Delta]$. All other characteristics, θ , $\bar{q}$, γ and $p - c_0$, are common across fields. The most capital-intensive field of the cohort is pinned down by the model. A field is developed during the delay only if it is worth developing at the tax investors expect to face, which in the baseline is no tax at all, so the highest capital intensity in the cohort is the one at which

development just breaks even without a tax:

$$\kappa_1 = \bar{v}_\delta(0), \quad \text{equivalently} \quad a^*(\bar{\tau}_D(\kappa_1)) = \frac{p - c_0}{\gamma} = a^*(0), \quad (15)$$

with $\kappa_1 = (p - c_0)^2/2\gamma$ at $\delta = 0$. The lock-in zone of this field stays open over its whole untaxed life. Under anticipation, the zero tax is replaced by τ^A . Only one feature of F_κ appears in what follows, the share of delay-induced fields

$$G(\tau) \equiv \Pr [\kappa > \bar{v}_\delta(\tau)] = 1 - F_\kappa(\bar{v}_\delta(\tau)), \quad (16)$$

the fraction of the cohort a timely tax τ would have deterred. Since fields only differ in κ and a_Δ , the development question depends on κ alone and the survival question on a_Δ alone. Every row of Table 1 therefore aggregates into $G(\tau)$ or $1 - G(\tau)$ times an integral over ages, and the expressions below hold for any F_κ . With κ uniform and $\delta = 0$, $G(\tau) = 1 - (1 - \tau/\bar{\tau}_P(0))^2$.

The wedge. Take a tax at which $a^*(\tau) \leq \Delta$, so that the oldest fields of the cohort have already been retired when the tax arrives. A delay-induced field of age a_Δ then produces for $(a^* - a_\Delta)^+$ years after implementation, and integrating over the uniform age gives

$$W(\tau) = \frac{N\theta\bar{q}}{2\Delta} G(\tau) a^*(\tau)^2. \quad (17)$$

At lower taxes, when all fields of the cohort are still producing at implementation, $a^{*2}/2$ is replaced by $\Delta(a^* - \Delta/2)$. The wedge is zero at $\tau = 0$, where $G = 0$, and at $\tau = \bar{\tau}_P(0)$, where $a^* = 0$, and positive in between. With κ uniform, W is proportional to $u(1 - u)$ with $u \equiv (1 - \tau/\bar{\tau}_P(0))^2$, so it is single-peaked.

Actual and naive emissions. In the same range of taxes, the emissions of the cohort after implementation and the emissions computed by the naive planner are

$$E(\tau) = \frac{N\theta\bar{q}}{2\Delta} a^*(\tau)^2, \quad E^N(\tau) = \frac{N\theta\bar{q}}{2\Delta} a^*(\tau)^2 (1 - G(\tau)), \quad E(\tau) - E^N(\tau) = W(\tau), \quad (18)$$

which is (8) for the parametric cohort.

The premium in closed form. At $\delta = 0$, with κ uniform and $a^*(\tau^N) \leq \Delta$, solving $E^N(\tau^N) = E(\tau^*) = B_\Delta$ and eliminating B_Δ gives

$$\begin{aligned} \bar{\tau}_P(0) - \tau^N &= \sqrt{\bar{\tau}_P(0) (\bar{\tau}_P(0) - \tau^*)}, & \text{equivalently} \\ \frac{\pi}{\bar{\tau}_P(0)} &= \sqrt{x} - x, & x \equiv 1 - \frac{\tau^*}{\bar{\tau}_P(0)} \quad (\delta = 0). \end{aligned} \tag{19}$$

Three properties follow. The premium depends on the primitives only through $\bar{\tau}_P(0)$ and τ^* : two cohorts with different cost slopes, carbon intensities and capital-intensity distributions but the same production breakeven and the same budget-consistent tax have the same premium. It is strictly positive whenever $\tau^* > 0$, since $\bar{\tau}_P(0) - \tau^N$ is the geometric mean of $\bar{\tau}_P(0)$ and $\bar{\tau}_P(0) - \tau^*$ and lies strictly between them. Relative to the production breakeven, it is largest at $x = 1/4$, when the budget-consistent tax is three quarters of the production breakeven.

Anticipation. When investors screen projects with an anticipated tax τ^A , the cohort ends at $\kappa_1 = \bar{v}_\delta(\tau^A)$. With κ uniform and $\delta = 0$, G and a^* keep the same form once $\bar{\tau}_P(0)$ is replaced by $\bar{\tau}_P(0) - \tau^A$ and taxes are measured from τ^A , so (19) holds with these substitutions as long as $\tau^N \geq \tau^A$. The premium is zero when investors screen with τ^* or with a higher tax, since no field developed during the delay is then delay-induced at τ^* .

A.2 An increasing tax

This appendix gives the algebra behind Section 2.6. The tax increases at rate r from its 2016-anchored level τ . A field that is a_Δ years old at implementation earns, x years after implementation, a margin per barrel of

$$p - c_0 - \gamma(a_\Delta + x) - \theta\tau e^{r(\Delta+x)}. \tag{20}$$

Three objects of the constant-tax model are affected.

The survival boundary. The field is phased out when (20) reaches zero. The corresponding age has no closed form but is unique, since the margin decreases with x , and it is lower than under a constant tax because the tax keeps rising. To first order in r , the margin falls by $\gamma + r\theta\tau e^{r\Delta}$ per year after implementation: the operating cost rises by γ and the tax in force at implementation, $\tau e^{r\Delta}$, rises by $r\tau e^{r\Delta}$. This is the statement used in Section 2.6.

Whether a field survives implementation depends on its margin at $x = 0$ only, so the survival boundary is still $\bar{\tau}_P(a_\Delta)$ once the tax is measured at its implementation level $\tau e^{r\Delta}$.

The development boundary. Under a timely policy, a field developed at date s faces the tax $\tau e^{r(s+a)} = (\tau e^{rs}) e^{ra}$ at age a . Its discounted profit per barrel of capacity therefore depends on τ and s only through τe^{rs} , the tax in force at its start-up, and the development condition is a threshold on that product. Writing $\bar{\tau}_D^0$ for the development breakeven of the field if it were developed at $s = \Delta$, the development breakeven of a field that is $a_\Delta = \Delta - s$ years old at implementation is exactly $\bar{\tau}_D(a_\Delta) = \bar{\tau}_D^0 e^{ra_\Delta}$. Fields developed early in the delay would have faced a lower tax under a timely policy and are more likely to have been developed anyway. Because the tax keeps rising after start-up, $\bar{\tau}_D^0$ depends on the whole tax path and not only on the margin at start-up, so the depth of the zone now depends on the tax. Figure 2 shows the resulting partition.

Aggregation. The four types of Section 2.3 and the identity $W = E - E^N$ are unchanged, and intensive fields still carry no overcommitted emissions: a field that the timely tax would also have developed produces for the same period under both policies, whatever the tax path. What is lost is the factorisation of the overshoot into $G(\tau)$ times an integral over ages, because whether a field is delay-induced now depends on its age at implementation as well as on its capital intensity. The wedge and the premium are computed numerically over (κ, a_Δ) :

$$W(\tau) = \frac{N\theta\bar{q}}{\Delta} \int_0^\Delta \int_{\bar{v}_\delta(\tau, a_\Delta)}^{\kappa_1} \left(a^*(\tau, a_\Delta) - a_\Delta\right)^+ dF_\kappa(\kappa) da_\Delta, \quad (21)$$

where $\bar{v}_\delta(\tau, a_\Delta)$ is the discounted profit per barrel of capacity of a field of age a_Δ at implementation under the timely tax path, and $a^*(\tau, a_\Delta)$ is its phase-out age. E , E^N , τ^N , τ^* and π follow as in Section 2.4.

Numerical check. Equation (21) and the code used for the empirical analysis have been checked against each other on a synthetic cohort of 3,000 fields that satisfy the model exactly: flat production, an operating cost rising linearly with age, development capital spent in the year before start-up, a common carbon intensity and start-up margin, capital intensity uniformly distributed, and start-up years spread over the delay. Run through the empirical code at taxes from 0 to 100 and at increase rates of 0 and 2%, the cohort reproduces the analytic development decisions, the phase-out year of every field, given by the root of (20), and the wedge of (21), all exactly.

Increasing tax and the premium: an illustration. The same synthetic cohort, with no discounting and a budget set at 60% of its untaxed emissions, shows how the tax increase rate affects the objects of Section 2.4. Table 7 reports them for a constant tax and for a tax rising at 2% per year. A rising tax that meets the same budget starts lower, so the premium falls in 2016-anchored terms, from 19.1 to 12.0, and in dollars of the implementation year, from 19.1 to 15.8. As a share of the tax, however, it does not change: π/τ^* is 0.44 in both cases, and π/τ^N is 0.80 and 0.78. The residual overshoot falls by 15%. These magnitudes describe a cohort developed entirely during the delay, with no field already in production, and are not comparable with those of Section 4.

	Constant tax	Tax rising at 2%
Naive tax τ^N	23.9	15.3
Lock-in-adjusted tax τ^*	43.0	27.3
Premium π , 2016-anchored	19.1	12.0
Premium π , at implementation	19.1	15.8
Premium as a share of τ^*	0.44	0.44
Residual overshoot Ω^N	4,560	3,901

Table 7: Increasing tax and the lock-in premium in a synthetic cohort.

Notes: Naive and lock-in-adjusted taxes, premium and residual overshoot for a constant tax and a tax rising at 2% per year. 3,000 fields satisfying the model exactly, no discounting, budget set at 60% of untaxed emissions. Taxes and emissions are in the units of the synthetic cohort.

A.3 A development period

In Section 2 development is instantaneous. In the data it takes several years, and a field may be partly built when the tax arrives. Let development take B years, with development spending spread evenly over them, so that a field whose development began at date $s > \Delta - B$ has paid a share $(\Delta - s)/B$ of its development cost at implementation, and a field whose development began earlier has paid all of it. Production starts at $s + B$, so a completed field is $a_\Delta = \Delta - s - B$ years old at implementation.

Nothing changes for a field completed before implementation: its lock-in zone at implementation is $(\bar{\tau}_D, \bar{\tau}_P(a_\Delta)]$ and Section 2.2 applies with the shifted age. A field under construction at implementation is completed if its discounted profit per barrel of capacity covers the development cost that remains to be paid, rather than the full development cost:

$$\bar{v}_\delta(\tau) \geq \kappa \left(1 - \frac{\Delta - s}{B}\right), \quad \text{equivalently} \quad \tau \leq \tau_L(s), \quad \bar{v}_\delta(\tau_L(s)) = \kappa \left(1 - \frac{\Delta - s}{B}\right). \quad (22)$$

The completion boundary $\tau_L(s)$ falls from $\bar{\tau}_P(0)$ at $s = \Delta - B$, when nothing remains to be paid and the field is completed at any tax that leaves it a positive margin, to $\bar{\tau}_D$ at $s = \Delta$, when nothing has been paid and the full development condition applies. Without discounting, $\theta \tau_L(s) = p - c_0 - \sqrt{2\gamma\kappa(s - \Delta + B)/B}$, so the boundary falls steeply as soon as part of the development cost remains to be paid, and flattens as s approaches Δ . The threshold on remaining development cost used in Section 3.4 is its empirical counterpart.

Figure 12 shows the partition in (s, τ) at a given capital intensity. Two types are added to the four of Section 2.3. Fields under construction at implementation whose completion satisfies (22) are completed and locked in (Type IV-a). They produce over their whole life after implementation, since they only start producing once the tax is in place. Fields under construction whose completion does not satisfy (22) are abandoned, with their capital stranded and no emissions (Type IV-b). A development period can only add fields to the locked-in set, since a partly built field is kept at any tax below $\tau_L(s) \geq \bar{\tau}_D$, so the case of instantaneous development in the body is a lower bound. Lock-in is largest for fields whose development began just before $\Delta - B$, which have just started producing when the tax arrives.

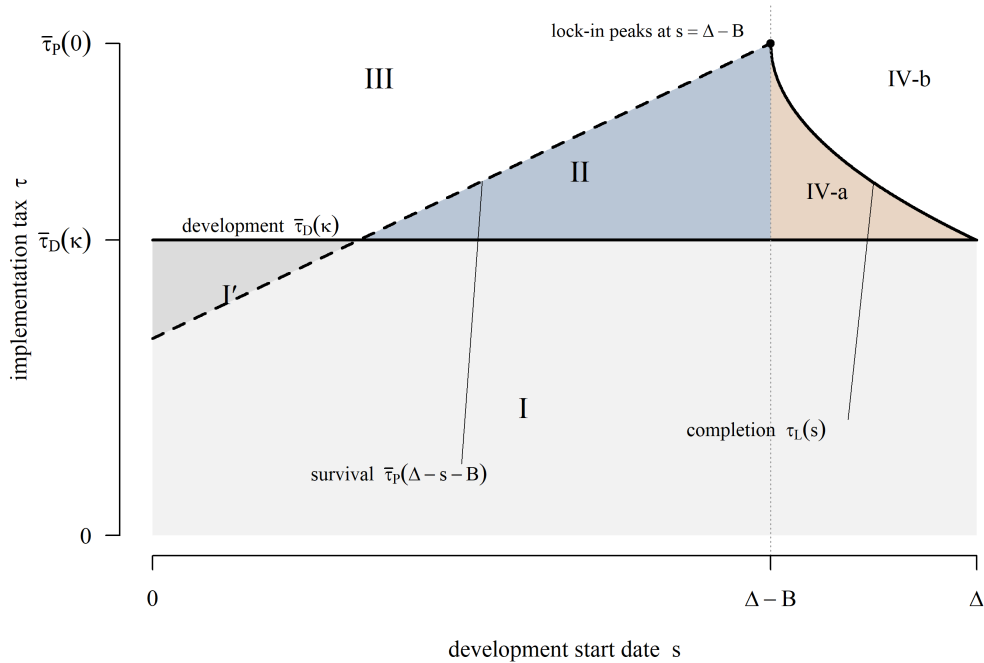


Figure 12: The taxonomy with a development period, at a given capital intensity.

Notes: The horizontal axis is the date s at which development starts; fields starting after $\Delta - B$ are under construction at implementation. The development boundary $\bar{\tau}_D$ is horizontal; the survival boundary $\bar{\tau}_P(\Delta - s - B)$ rises with s because later fields are younger at implementation; the completion boundary $\tau_L(s)$ falls from $\bar{\tau}_P(0)$ at $s = \Delta - B$ to $\bar{\tau}_D$ at $s = \Delta$. Region II is the locked-in set among completed fields, IV-a among partly built fields that are completed, IV-b the partly built fields abandoned at implementation. Lock-in is largest at $s = \Delta - B$.

B Data appendix

B.1 Rystad cost categories

Figure 13 reproduces Rystad’s definition of each cost item in the UCube database. Development costs in this paper are the facility and well capex incurred up to two years after production start-up; production costs are production, transportation and SG&A opex, taxes reported as opex, and all facility and well capex incurred later. For shale and tight oil assets, all well capex is counted as a production cost, and only facility capex incurred up to two years after start-up as development. Exploration capex and government take other than taxes reported as opex are excluded from the main specification.

B.2 Asset-specific price discounts

Table 8 reports the reserve-weighted average discount to Brent by oil type, which Section 3.3 discusses.

Oil type	Average discount	2016 reserves (Gbbl)
Regular	−5.4%	760
NGL	−50.6%	222
Light	−2.7%	203
Condensate	−5.9%	113
Sour (> 3%)	−4.6%	52
Bitumen	−29.0%	47
Heavy oil 15–19	−14.0%	34
Heavy oil 20–22	−10.4%	22
Extra heavy oil	−12.4%	21
Synthetic crude	−7.4%	19
Total	−12.9%	1,493
Excluding NGL	−6.3%	1,271

Table 8: Discounts relative to the Brent price by oil type.

Notes: Reserve-weighted averages of the discounts observed and forecast by Rystad over 2016–2100, estimation sample. Discount data are missing for a small part of the sample’s production, so the reserves in the table (1,493 Gbbl) fall slightly short of the total of Table 2 (1,555 Gbbl).

Economics Category	The Economics Category splits the Economics variable (=Revenue) into higher granularity than Economics Group, e.g. capital expenses is subdivided into facility and well expenses. The sum of all items equals the revenue. Use Economics Category to analyze breakdown of cost models or specific cost elements.
Free Cash Flow	FCF is the incurred cash flow available to investors and creditors, thus largely used in DCF valuation analysis. $FCF = \text{revenue} - \text{capex} - \text{opex} - \text{gvt take}$. FCF can be compared with cash from operations and investments activity in the cash flow statement from companies (only upstream). The present value of (all discounted) FCF equals NPV (Net Present Value also called Gross Asset Value) that can be compared with Enterprise Value (=Market cap+Debt for upstream activities), or compared with asset transaction costs. The discount factor in UCube is 10% flat for all assets.
Government Profit Oil	Gvt PO is the PSA equivalent to petroleum taxes, but paid in kind. Gvt PO reduces the company's entitlement production and is thus treated as a royalty effect in company reports.
Royalty effects	Royalty effects is the sum of all gross taxes, including royalty and oil and export duties.
Income Tax	Income tax is a sum of all profit based taxes, ie corporate taxes, where the tax rate is equal to the country's corporate tax rate, and special petroleum taxes. The tax base is the net sales revenue minus opex and depreciation. For petroleum taxes there may be applied an additional deduction called uplift. In many cases, some profit based taxes can be deducted in the tax base for other profit taxes. The income taxes in the UCube assumes there is no interest expenses.
Bonuses	Signature bonus for the license at award, as officially reported.
Well Capex	Well capex is capitalized costs related to well construction, including drilling costs, rig lease, well completion, well stimulation, steel costs and materials.
Facility Capex	Development capex except well construction costs, includes costs to develop, install, maintain and modify surface installations and infrastructure. As reported by operators, field partners or officials, or modelled. Distribution of costs over time is largely modelled, one exception is Norway, where all historic costs are as reported in national budgets.
Exploration Capex	Costs incurred to find and prove hydrocarbons: seismic, wildcat and appraisal wells, general engineering costs, based on reports and budgets or modelled.
Abandonment cost	Abandonment cost represents the costs associated with shutting down and dismantling the surface and subsea facilities such as surface equipment, platforms, subsea structures and pipelines. Activities included is topside and jacket removal as well as pipeline cleaning and decommissioning, onshore disposal, surveys and monitoring, and project management costs. Note that the Abandonment cost only includes the remaining P&A work for wells, and not full cycle P&A work. Abandonment cost is mostly modelled and assumed to be incurred and expensed over couple of years after the production has ceased. The estimate is proportional to initial greenfield capex.
Production Opex	Represents operational expenses directly related to the production activity. The category includes materials, tools, maintenance, equipment lease costs and operation related salaries. Depreciation and other non-cash items are not included.
Transportation Opex	Represents the costs of bring the oil and gas from the production site/processing plant to the pricing point (only upstream transportation). The category includes transport fees and blending costs.
SG&A Opex	Represent operation expenses not directly associated with field operations. The category includes administrative staff costs, office leases, related benefits (stocks and stock option plans) and professional expenses (legal, consulting, insurance). Only E&P related SG&A are included.
Taxes Opex	Represent gross tax reported by companies under operational costs and applies for US and Brazil. For US the category represent ad valorem taxes (county based) and severance taxes (state based).

Figure 13: Rystad economic data, split items.

Notes: Descriptions are provided by Rystad alongside the UCube database.

B.3 Anticipation scenarios

The main specification assumes investors do not anticipate the tax during the delay. The alternative scenarios reported in Section 4.4 are defined as follows. In the *tax in 2030* scenario, investors anticipate the correct level of tax to be implemented in 2030, and thus predict the planner's behaviour exactly. In the *five years after start-up* scenario, investors anticipate the correct level of tax to be implemented five years after production starts in the asset, a simple way of representing investors who expect the tax only once the project is

running.. In the *share of the tax* scenarios, investors anticipate the tax to be implemented immediately after investing, on the totality of the asset’s lifetime production, but underestimate its level; the sensitivity panel reports 25, 50 and 75%, and Figure 8 a grid in steps of 2.5%. In every scenario the anticipated tax enters only the investment decision (12), through T_τ and τ ; production decisions after implementation are taken under the implemented tax. The limit case, screening with the implemented tax τ_{2030}^* , reproduces $\pi = 0$ exactly.

C Field-level carbon intensities

While Rystad’s databases provide detailed asset-level data on oil production, they lack comprehensive carbon intensity (CI) estimates. This appendix fills this gap by combining engineering-based models (OPGEE, PRELIM, OPEM), satellite-derived flaring data (VIIRS), and predictive regressions to estimate lifecycle CI for all assets, including those not yet in production. Carbon emissions related to oil can be split into 3 stages (Gordon et al., 2015). The upstream stage includes all emissions arising from extraction processes to the transport of crude oil to the refinery inlet. The midstream stage corresponds to refining processes and is heavily dependent on the refinery configuration. Finally, the downstream stage comprises emissions from the transport of oil products to their end users and from their final use. While downstream emissions usually represent around 80% of total emissions, the upstream stage features important variability from one field to the other.

For each of these stages, I rely on a state-of-the-art engineering model used by the Oil-Climate Index (Gordon et al., 2015), run on a sample of currently produced oils, to estimate their carbon intensity. I then fit a regression model for each stage to assign a carbon intensity to all other Rystad assets.

Figure 14 reports cumulative 2016 reserves ordered by lifecycle carbon intensity. Most reserves lie between 400 and 500 kgCO₂eq/bbl, with a short tail of cleaner assets and a longer tail of more carbon-intensive ones. Fields developed during the delay and fields already producing in 2016 have similar distributions.

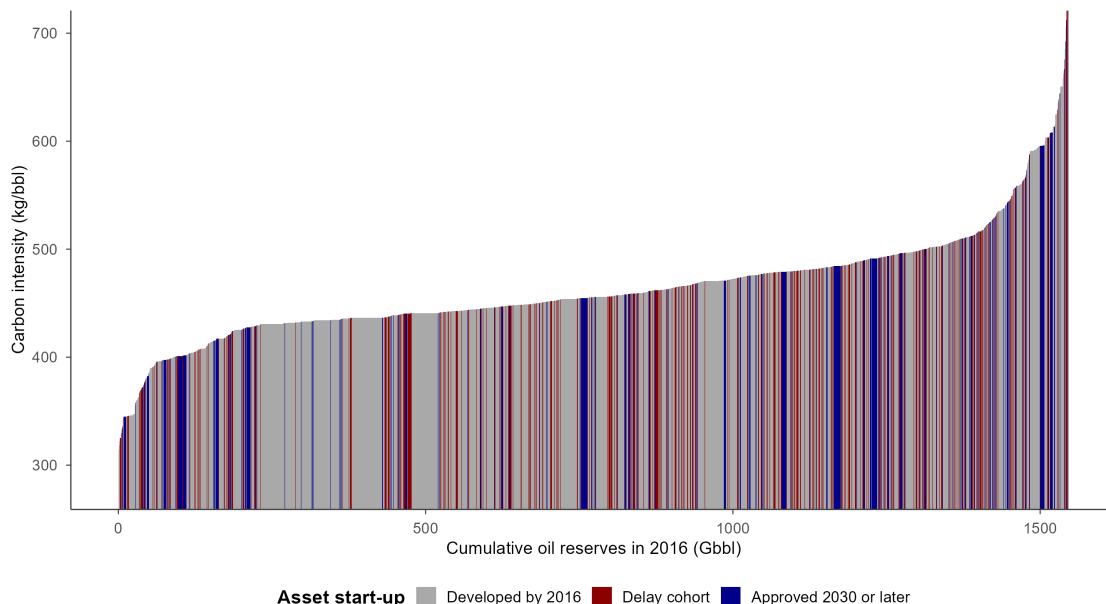


Figure 14: Cumulative 2016 oil reserves ordered by lifecycle carbon intensity, by development timing.

C.1 Upstream CI

C.1.1 Flaring data

Upstream emissions exhibit substantial heterogeneity across assets, and the major contributor to this heterogeneity is the flaring-to-oil ratio (FOR). Flaring corresponds to the combustion on-site of co-produced gas¹⁸, and represents more than 150 billion cubic meters worldwide in 2024 (World Bank, 2025). To compute precise estimates of upstream CI, I use satellite data on flares to derive field-specific flaring-to-oil ratios.

Satellite flare observations and imputed emissions. Rystad’s UCube database does not contain information on gas flared volumes at the asset level. This is mainly due to the fact that most oil and gas operators do not publicly report flared volumes. Moreover, under-reporting by field operators remains an important issue making the use of self-reported flaring less reliable.

To fill this gap, I use geocoded flaring volumes calculated by the Earth Observation Group

¹⁸Co-produced gas is often used to power equipment on site, or sold, if the asset is connected to the local gas market or has the equipment to liquefy and ship it. If not, excess gas is then either reinjected, to increase pressure in the reservoir, flared, or directly vented. Given the higher global warming potential of methane, flaring is preferable to venting, and venting is banned in most countries except for rare emergencies. Gas reinjection is often costly, making flaring an easier decision in many oil fields.

(EOG) at the Payne Institute for Public Policy, within the Colorado School of Mines. The EOG analyzes satellite data from the Visible Infrared Imaging Radiometer Suite (VIIRS), on board satellites launched in 2012 and 2017, which detects heat from gas flares at night. Flaring volumes are then estimated using the VIIRS Nightfire algorithm (VNF), as detailed by [Elvidge et al. \(2016\)](#) and further refined by [Zhizhin et al. \(2021\)](#). I use the 141,841 observations with positive flared volumes between 2012 and 2024 to assign them to specific assets from Rystad’s database.

Rystad assets geographic information. In order to assign flaring observations to Rystad assets, I need geographic information on these assets. I complement data from the UCube with two sets of geocoded data obtained from the same company, one in 2022 and the other in 2025.

First, geocoded boundaries were shared by Rystad in 2022 for 41,762 assets, among which 16,801 were generic (i.e. a 500m radius circle). Since the universe of Rystad assets has evolved between 2022 and 2025, 1,916 generic shapes were added for assets missing in the shapefile, based on their geocoded centroid.

Second, data were obtained on the location of wells completed between 2000 and 2025. This dataset contains the latitude and longitude of 1,733,698 producer, injector or wildcat wells, from 39,494 assets. In some regions, the location of wells corresponds to the asset centroid, meaning that Rystad probably does not have its precise location. In total, 857,900 unique locations exist in the dataset. Crucially, well locations are precisely defined for onshore US and Canada assets, where all asset boundaries are generic, making this dataset a useful complement to the first one.

Matching satellite observations to Rystad assets. The assignment of satellite observations to Rystad assets relies on both assets’ eligibility (the existence or not of surface wells, excluding for example subsea tiebacks and not yet completed wells) and geographic distances to existing wells and assets’ boundaries. For each observation, distances to all completed wells and assets in a 20km radius were computed. A score reflecting the probability of a detected flare to originate from each nearby asset is then computed. For observations with existing wells in a 2km radius, only information on wells was used to compute this score. For other observations, distances to both assets and wells were incorporated in the score. When scores were too close to decide between assets, flared volumes were assigned based on each eligible asset’s score and production capacity. [Ghosh et al. \(2025\)](#) provide more detailed information on the 5-stage matching procedure used to assign flaring observations to Rystad assets.

Overall, all observations with at least one eligible well or asset in a 20km radius are matched, leaving 7,261 observations unmatched (corresponding to 4.7% of total flared volumes between 2012 and 2024). Among matched observations, around two thirds were assigned to a single asset (70.4% of matched flared volumes) and the rest split between multiple assets.

Computing flaring-to-oil ratios (FOR). Once the matching procedure is done, total flared volumes over the 2012-2024 period are computed and divided by total oil production to obtain asset-specific FOR. For not-yet-active assets over this period, an average FOR is assigned. Two categories of assets producing over the period were also attributed averages: assets with at least 75% of years in the sample with positive production outages, mainly due to wars in Libya and Syria (109 assets), and assets whose production started less than 2 years before the end of the period (911 assets). The average FOR is computed at the lowest scale possible, for assets of similar size if possible¹⁹, and always by oil type. Averages are capped at 12,000 scf/bbl, corresponding to twice as much gas flared as oil produced, in barrels of oil equivalent. Table 9 reports the number and type of average FORs assigned to Rystad assets, and the corresponding oil reserves.

FOR origin	Number of assets	Share of oil reserves in 2016
From satellite matching	20,571	64.3%
Area $\times$ Oil Type $\times$ Field Size	5,133	4.2%
Province $\times$ Oil Type $\times$ Field Size	7,350	7.3%
Province $\times$ Oil Type	10,497	8.9%
Country $\times$ Oil Type $\times$ Field Size	6,662	3.3%
Country $\times$ Oil Type	4,250	4.8%
Region $\times$ Oil Type $\times$ Field Size	2,948	3.4%
Region $\times$ Oil Type	3,136	3.1%
World $\times$ Oil Type $\times$ Field Size	946	0.9%
World $\times$ Oil Type	2	0.0%

Table 9: Origin of FOR values.

Table 10 shows the binned distribution of computed FOR. Most assets display relatively low levels of flaring with around a third of them having a FOR equal to 0 (representing around 22% of oil reserves). In these assets, the 0 value can be explained by either complete use of co-produced gas or particularly low gas-to-oil ratio, but it can also be due to flared volumes so low (or so intermittent) that it falls below the satellite detection threshold. Among assets producing mainly oil (assets with gas-to-oil ratios below 10,200 scf/bbl), the

¹⁹Assets are split into 10 categories depending on their reserve size, with thresholds at 1, 3, 10, 30, 100, 300, 1000, 3000 and 10000 MMboe.

reserve-weighted average FOR is 104.4 scf/bbl, and the median is 13.4 scf/bbl.

FOR (kscf/bbl)	Assets count	Reserves (Gbbl)
0	11,651	345.6
0–0.1	10,820	748.7
0.1–1	10,974	435.4
1–10	1,069	21.0
>10	291	2.5

Table 10: Distribution of flaring-to-oil ratios.

C.1.2 OPGEE model

The Oil Production Greenhouse Gas Emissions Estimator (OPGEE) is an engineering-based model developed at Stanford University and used by the Oil-Climate Index ([Gordon et al., 2015](#))²⁰. It measures GHG emissions from extraction, processing and transport of crude oil in grams of CO₂ equivalent per megajoule of oil produced.

Data inputs and matching with Rystad assets. Running the model requires a wide range of inputs for each field, from extraction methods to gas composition. When information is missing, the model computes smart default values (deduced from other variables when possible). I recover publicly available input data from [Masnadi et al. \(2018\)](#) on 958 oil fields worldwide, and match it to Rystad assets manually. Among the 958 original fields, 737 were matched to 1,136 assets in the Rystad database. Most unmatched fields correspond to marginal heavily-depleted oil fields in the US. The difference between the number of OPGEE fields and matched Rystad assets comes from differing levels of aggregation. In many instances, a unique OPGEE field is split into multiple assets in Rystad data. On rare occasions, multiple OPGEE fields were matched to a unique Rystad asset.

After completing the matching procedure, I complement the 737-field dataset with information on 6 variables from Rystad data in order to improve the accuracy of my estimates. These variables are: production level, field age, gas-to-oil ratio, flaring-to-oil ratio, number of producer wells and number of injection wells. Since Rystad databases do not contain information on the quantity of gas reinjected, used on site, or vented, I assume the amount of gas produced to be equal to the sum of flared and sold gas in order to compute gas-to-oil ratios. Finally, extraction methods reported in [Masnadi et al. \(2018\)](#) input data were

²⁰Latest versions of the model are available at: <https://eao.stanford.edu/research-project/opgee-oil-production-greenhouse-gas-emissions-estimator>

adjusted if necessary to match information retrieved from Rystad²¹.

Computing CI. The resulting input dataset was fed into the OPGEE model to export results. Two distinct carbon emissions accounting methods are available within the model: energy-based allocation (EBA) and carbon product displacement (CPD). In the EBA approach CI are defined such that $CI_{EBA} = \frac{\text{TotalEmissions}_{\text{field}}}{\text{Prod}_{\text{oil}} + \text{Prod}_{\text{gas}}}$, with Prod_{oil} and Prod_{gas} in MJ. This means that total emissions from the oil field are computed, and allocated to oil and gas sold on the market according to their energy content. In the CPD approach, the definition becomes $CI_{CPD} = \frac{\text{TotalEmissions}_{\text{field}} - \text{Prod}_{\text{gas}} \cdot \overline{CI}_{\text{gas}}}{\text{Prod}_{\text{oil}}}$. Co-produced gas gives rise to carbon credits corresponding to the energy content of gas produced multiplied by $\overline{CI}_{\text{gas}}$, a worldwide average carbon intensity of gas production computed from the CA-GREET model. The latter approach can lead to negative oil CI, which is why the retained approach was the EBA one.

Figure 15 displays CI computed using the model. The production-weighted average carbon intensity is equal to 59 kgCO₂eq/bbl (indicated by the dashed red line on the figure). The variance among fields is considerable, with 10% and 90% weighted deciles ranging from 35 kgCO₂eq/bbl to 113 kgCO₂eq/bbl. Three key features influencing CI are indicated on the graph:

- **FOR:** as expected, large CI values often originate in large quantities of co-produced gas flared. The influence of FOR on CI largely depends on the assumptions made on destruction removal efficiency (DRE). This value measures the efficiency of flares, i.e. the share of gas effectively combusted and thus transformed into CO₂. Since the global warming potential of methane is 28 times larger than the one of carbon dioxide, assumptions on DRE greatly affect computed CI. I keep the default value of 96.1% retained by the model to compute baseline CI, and provide in sensitivity analysis results using values of 91.1% and 98% to cover the range of estimates found in the literature (Plant et al., 2022; Gvakharia et al., 2017; Zavala-Araiza et al., 2021).²²
- **Steam injection:** Heavier oils (API below 20) tend to be thicker than lighter ones, and often require the injection of steam in order to lower their viscosity and allow their extraction. This is captured by the steam-to-oil ratio (SOR), corresponding to the number of barrels of vaporized water injected to extract one barrel of oil. High

²¹This is for example the case for water reinjection. If Rystad data reports existing injection wells, at least water or gas injection must be indicated in OPGEE input data. When this was not the case, water reinjection was assumed to happen, since it is more common in oil fields than gas reinjection.

²²While playing a much smaller role than FOR, the quantity of co-produced gas (GOR) can still influence extraction emissions through fugitive emissions of methane related to its processing and transport on site.

SOR values imply larger quantities of liquids extracted, and therefore more energy consumed to extract oil.

- **Upgrading into synthetic oil:** some particularly viscous oils (bitumen or extra heavy) are extracted without first reducing their viscosity (for instance using steam injection). This is mainly the case for Canadian oil sands, which are often directly extracted in open air mines. However, they are often too viscous to flow through pipelines and be handled by existing refineries. These oils are therefore “upgraded” into synthetic crude on site. This requires using chemical treatments and/or distillation in order to break long hydrocarbon chains into shorter ones, thus reducing viscosity. These processes are often very energy intensive, translating into large CI for fields using them (Brandt and Farrell, 2007). Produced synthetic crude is however easier to refine, translating into relatively lower midstream CI²³.

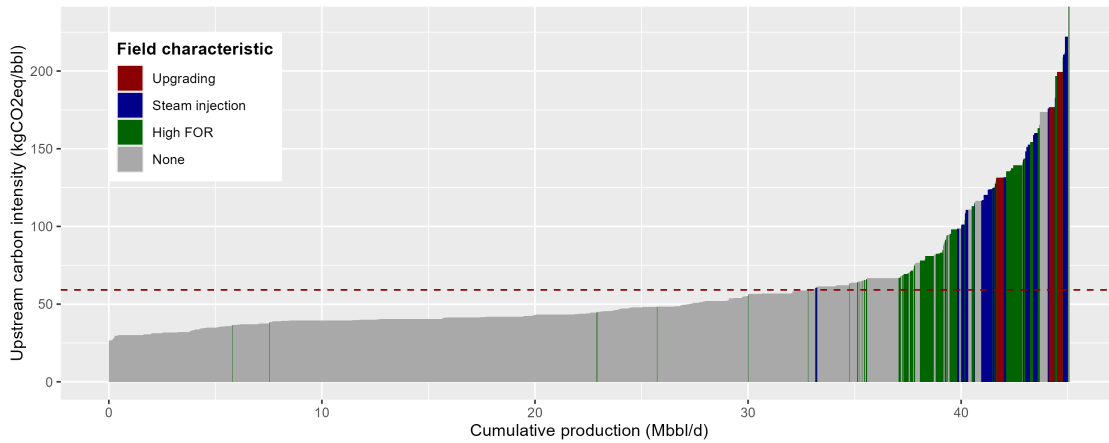


Figure 15: OPGEE-computed CI.

Notes: The figure depicts CI computed in the OPGEE model, with cumulative production in 2015 on the x-axis. The horizontal dashed red line indicates the production-weighted average.

C.1.3 Estimation model

Model and sample restrictions. Once the 737 OPGEE-computed CI are attributed to the corresponding 1,136 Rystad assets, I fit a regression model to explain across-field variations. Following the logic of the EBA approach, the dependent variable in the model thus consists of the field carbon intensity multiplied by its “energy denominator” $ED_f =$

²³Midstream CI for synthetic crudes are on average 30% lower when compared to other extra heavy or bitumen oils, and still 8% lower when compared to all types of oil.

$\frac{Prod_{oil,f} + Prod_{gas,f}}{Prod_{oil,f}}$, with production in energy units, so that $CI_f \times ED_f$ measures upstream emissions per barrel of oil. The retained model is:

$$CI_f^{OPGEE} * ED_f = \sum_{i=1}^{10} \beta_i^C ShareOilType_{i,f} + \beta_{11} Offshore_f + \beta_{12} SOR_f + \beta_{13} GOR_f + \beta_{14} FOR_f + \epsilon_f, \quad (23)$$

where f denotes a Rystad asset, $ShareOilType_f$ the share of each oil type in its production²⁴, $Offshore_f$ a dummy indicating if the field is offshore, SOR_f its steam-to-oil ratio (in bbl of water/bbl of oil), GOR_f its gas-to-oil ratio (in scf/bbl) and FOR_f its flaring-to-oil ratio (in scf/bbl).

Oil-type variables, together with the offshore dummy, capture various aspects of the extraction methods and facility characteristics on site. This is especially relevant for synthetic crude, where upgrading processes such as distillation or chemical treatments significantly increase CI. SOR, GOR and FOR have a more straightforward and linear impact on carbon emissions, and explain most of the remaining variance. SOR data used for the estimation is the one used as input in the OPGEE model since precise values are not available in Rystad data.

The sample used in the regressions was restricted to fields with water-to-oil ratio below 40 bbl of water/bbl of oil, and gas-to-oil ratio below 40,000 scf/bbl. The first restriction excludes fields almost fully depleted, characterized by exponentially increasing WOR²⁵. The second one is motivated by the fact that my focus is on oil producing fields. Oil fields are defined by Rystad as reservoirs with GOR below 10,200 scf/bbl. I exclude fields with GOR higher than four times this threshold in order to keep all fields producing non-negligible oil quantities. Altogether, these two restrictions exclude 20 Rystad assets, making the final sample 1,116 assets (associated with 719 OPGEE fields). The regressions are weighted so that each unique OPGEE-computed value carries the same weight. If a unique OPGEE value was matched to multiple Rystad assets, weights were computed based on oil reserves size and sum to 1.

Regression results. Table 11 shows coefficient estimates from the regression for two sets of CI: the first corresponds to my baseline estimates, using a 100-year horizon for global

²⁴An asset produces a unique type of crude oil, among 8 possible (synthetic, bitumen, extra heavy, heavy (15-19), heavy (20-22), regular, light and sour). It can however produce simultaneously condensate and/or natural gas liquids (NGL), which are sold as oil, similar to light crude.

²⁵I exclude these fields because Rystad data does not contain information on this ratio. The threshold of 40bbl/bbl is particularly high and only excludes 7 fields from the sample.

Variables	EBA - AR6 100y	EBA - AR6 20y
Oil type: Synthetic	157.096*** (6.514)	192.358*** (9.943)
Oil type: Bitumen	52.854*** (4.555)	78.442*** (6.952)
Oil type: Extra Heavy	48.122*** (4.169)	78.368*** (6.363)
Oil type: Heavy (15-19)	54.778*** (3.73)	96.841*** (5.693)
Oil type: Heavy (20-22)	58.861*** (3.561)	97.629*** (5.436)
Oil type: Regular	46.522*** (1.105)	79.708*** (1.687)
Oil type: Light	46.939*** (1.862)	77.204*** (2.842)
Oil type: Sour (> 3%)	43.308*** (5.14)	71.654*** (7.846)
Oil type: Condensate	43.535*** (4.235)	72.446*** (6.465)
Oil type: NGL	48.05*** (6.641)	83.603*** (10.137)
Offshore	-6.048*** (1.273)	-8.435*** (1.943)
SOR (bbl water/bbl oil)	26.317*** (0.983)	27.369*** (1.501)
GOR (kscf/bbl)	4.601*** (0.154)	8.7*** (0.236)
FOR (kscf/bbl)	83.908*** (0.533)	146.98*** (0.813)
Corrected R ²	0.970	0.977
Predicted R ²	0.936	0.943

Table 11: Regression results from upstream emissions estimation.

Notes: The table displays coefficients from the regressions using equation (23), with standard errors in parentheses. R² are corrected in order to account for the absence of intercept, as well as predicted R².

warming potential, and the second uses a 20-year horizon (associated with higher global warming potential for methane as compared to carbon dioxide, due to its shorter lifetime). R-squared values are close to 1, and predicted R-squared are also quite close to 1, indicating that the model's predictive power is relatively high.

As expected, synthetic crude production is associated with large CI, and coefficients for heavier crudes tend to be larger than the ones of lighter crudes. While coefficients associated with bitumen and extra heavy crudes are not significantly higher than others,

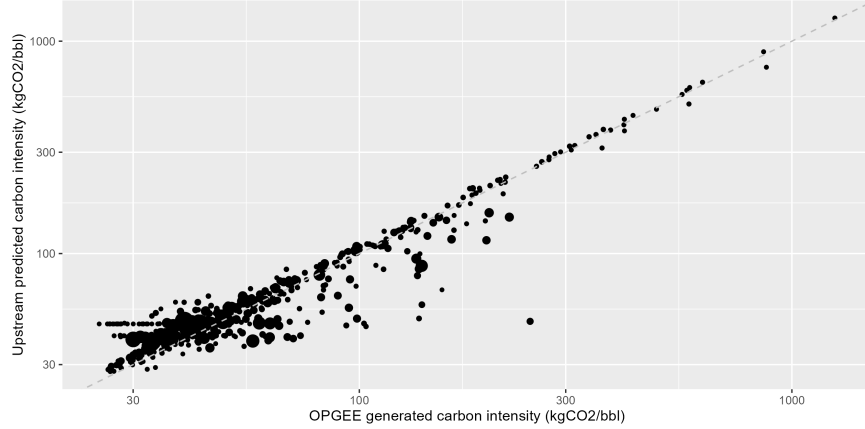


Figure 16: Predicted vs OPGEE-computed upstream CI.

Notes: The figure displays upstream CI computed using the OPGEE model on the x-axis and predicted using the model on the y-axis. Only assets matched with OPGEE fields are included. Both axes are transformed using the decimal logarithm. The size of each point reflects the log volume of oil reserves for each asset.

production of these two types often implies the use of steam injection, which means that most of the variation is captured by the SOR variable. The negative coefficient associated with the offshore dummy reflects the absence of land-use transformation, source of GHG emissions. The three remaining variables explain most of the variance. Steam injection greatly increases emissions (the average steam-to-oil ratio is 3 bbl of water/bbl of oil). While having a relatively low magnitude, GOR still plays an important role in explaining the variance in carbon intensity across fields. Fields producing gassy oils, even when selling their co-produced gas, are subject to higher quantities of fugitive emissions. Finally, FOR plays the most crucial role in explaining the variance across fields, as noted by [Masnadi et al. \(2018\)](#).

Regressions using 20-year horizon CI as the dependent variable display similar results. By construction, reducing the time horizon can only increase CI: while carbon dioxide emissions are not affected, methane emissions lead to higher global warming potential. Consequently, variables associated at least partially with methane emissions see their associated coefficients increase in large proportions. This is particularly visible when comparing the SOR, GOR and FOR variables. While steam injection does not entail methane emissions, gas production leads to fugitive methane emissions and flaring does not fully convert methane into carbon dioxide (depending on the retained DRE value as explained above). The large increase in the coefficients associated with GOR and FOR, while the SOR coefficient remains nearly unchanged, is thus consistent with this mechanism.

Sensitivity analysis. Figure 16 highlights the strong correlation between OPGEE-

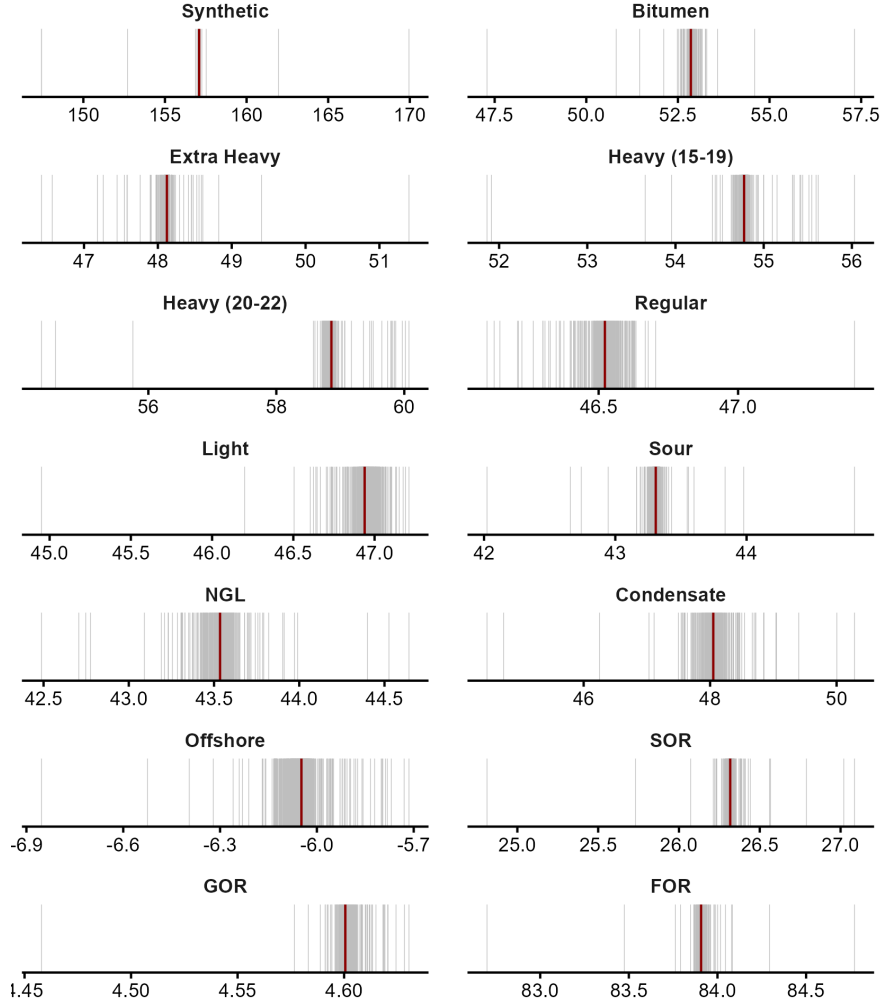


Figure 17: Upstream CI estimation: Estimates of the coefficients on the explanatory variables removing the OPGEE fields one-by-one.

Notes: The panels show the estimated coefficients on each of the explanatory variables in equation (23) removing observations one-by-one. Each grey line corresponds to a different estimation with the same number of observations: $N(OPGEE_{sample}) - 1$. The thick red lines show the original estimates obtained with the full sample of OPGEE publicly-available fields (OPGEE sample).

computed CI and CI predicted using the model for the same assets sample. To verify the influence of outliers, I re-estimate the coefficients of equation (23) excluding each observation one by one. Figure 17 displays estimated coefficients, with the baseline estimate indicated in red. While the coefficients of heavier oil types vary because few assets produce them, results indicate that my estimates are overall robust to changes in the sample.

Finally, I test the sensitivity of estimated coefficients to the exclusion of the most influential observations. I compute the Cook's distance of each observation, defined by $CD_i = \sum_{j=1}^N (\hat{y}_j - \hat{y}_{j(i)})^2 / ps^2$, where $\hat{y}_j$ and $\hat{y}_{j(i)}$ are the fitted response values from using

	Synthetic	Bitumen	Extra Heavy	Heavy 15–19	Heavy 20–22
0	157.1*** (6.51)	52.85*** (4.55)	48.12*** (4.17)	54.78*** (3.73)	58.86*** (3.56)
1	157.12*** (6.29)	53.14*** (4.40)	48.48*** (4.02)	55.42*** (3.60)	59.49*** (3.44)
10	133.06*** (6.51)	56.86*** (3.70)	51.35*** (3.37)	53.23*** (3.07)	50.40*** (2.99)
50	154.73*** (14.33)	51.56*** (2.88)	47.81*** (2.50)	49.14*** (2.40)	49.60*** (2.64)
	Regular	Light	Sour (>3%)	Condensate	NGL
0	46.52*** (1.11)	46.94*** (1.86)	43.31*** (5.14)	43.54*** (4.24)	48.05*** (6.64)
1	47.42*** (1.07)	46.87*** (1.80)	43.98*** (4.96)	44.40*** (4.09)	49.04*** (6.41)
10	47.62*** (0.90)	45.37*** (1.50)	44.17*** (4.12)	45.48*** (3.41)	53.37*** (5.37)
50	44.59*** (0.65)	41.86*** (1.07)	44.73*** (3.05)	43.83*** (2.51)	46.00*** (3.87)
	Offshore	SOR	GOR	FOR	Pred. R ²
0	-6.05*** (1.27)	26.32*** (0.98)	4.60*** (0.15)	83.91*** (0.53)	0.934
1	-6.85*** (1.23)	26.22*** (0.95)	4.62*** (0.15)	82.70*** (0.53)	0.931
10	-7.39*** (1.03)	24.97*** (0.81)	4.51*** (0.13)	82.67*** (0.54)	0.918
50	-5.21*** (0.74)	26.02*** (0.67)	4.54*** (0.09)	83.93*** (0.46)	0.910

Table 12: Upstream CI estimation: coefficient stability to influential observations (Cook’s distance).

Notes: Each panel shows re-estimated coefficients using equation (23) after excluding the 0, 1, 10, and 50 most influential observations. Cook’s distances were computed to determine the influence of each observation. The last panel indicates predicted R-squared for each of the re-estimated models.

the full sample and after excluding field i , respectively; s^2 is the mean squared error of the regression model, p the number of coefficients, and N the number of observations. Table 12 displays re-estimated coefficients excluding the 1, 10 and 50 most influential observations. Heavier oil types see their estimates vary slightly, due to the small number of assets producing them in my OPGEE sample, while SOR, GOR and FOR coefficients are particularly stable. Predicted R-squared stay high, even when removing the 50 most influential observations. Overall, re-estimated coefficients remain consistent with my original estimates using the full sample.

C.1.4 Predicting upstream carbon intensity for all Rystad assets

Using estimated coefficients, predicted upstream CI were computed for each Rystad asset. Data on reserves by oil type, offshore status, and GOR was directly available in Rystad databases. Asset-specific FOR were either derived from satellite observations or predicted for not-yet-producing assets, as described in Appendix C.1.1. Values for other needed variables were assigned as described below:

- SOR: on top of assets matched to OPGEE fields, assets using steam injection were identified and assigned a SOR value of 3, in line with both the default value in the

OPGEE model and the average SOR value in OPGEE fields data from [Masnadi et al. \(2018\)](#). All assets containing “Steam”, “Thermal” or “SAGD” in their name were identified as using steam injection. In addition, the list of Enhanced Oil Recovery projects made available by the EIA was used to manually identify remaining steam injecting assets ([McGlade et al., 2018](#)).

- ED_f : energy denominators for each asset were approximated by $ED_f = 1 + \frac{GOR_f - FOR_f}{6000}$, using a 6,000 MJ average energy content per barrel. This approximation is remarkably close to energy denominators derived from the OPGEE model, with a correlation coefficient of 0.975 within the OPGEE matched assets sample.

When a unique Rystad asset was matched with a unique OPGEE field, the carbon intensity computed using the OPGEE model was kept. For all remaining assets, predicted values prevail. Figure 18 shows the 2012-2024 production-weighted distribution of CI for both samples, with weighted means depicted by the vertical lines. Although averages are remarkably similar, the predictive model does not seem to fully replicate the variance across OPGEE-computed values. The general shape of the distribution remains coherent with OPGEE outputs.

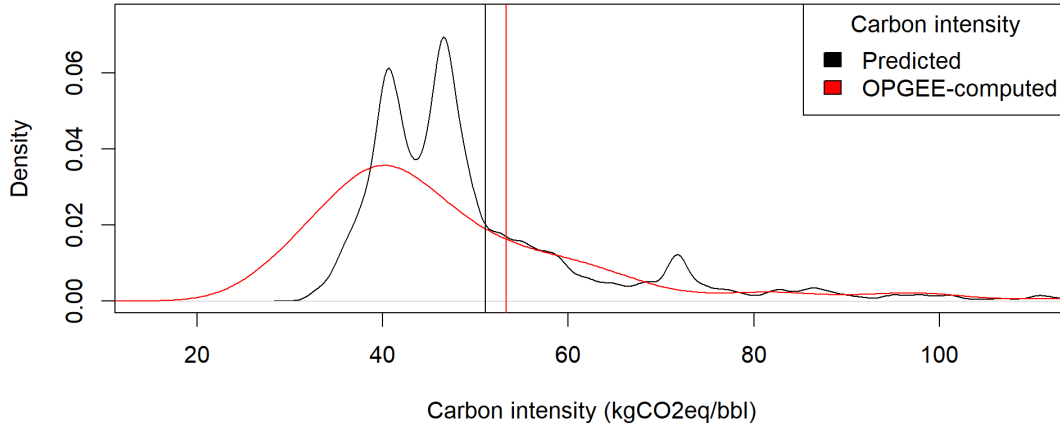


Figure 18: Distribution of CI: Predicted vs OPGEE-computed.

Notes: The figure shows in red the distribution of baseline CI (100-year horizon, EBA) for the OPGEE sample, i.e. assets matched with OPGEE fields and for which the OPGEE value was kept, and in black predicted CI for all remaining assets. Densities are weighted by oil production volumes over the 2012-2024 period. Vertical lines indicate weighted averages for both samples.

C.2 Midstream-downstream CI

Midstream emissions correspond to emissions from refining crude oil, while downstream emissions include emissions from both transport to end-users and final use (often combustion) of the different oil products obtained from refining. Refining crude oil yields multiple oil products depending on its characteristics and refinery configurations. Midstream and downstream emissions are typically computed at the crude stream level, which corresponds to the actual crude stream arriving at the refinery inlet. In some instances, a crude stream originates in a single oil field, while the majority of crude streams consist of the output of multiple fields blended together. For instance, the Brent crude used as reference in oil pricing is a blend of crude oils produced in 13 separate assets in the North Sea.

C.2.1 PRELIM and OPEM models

Midstream and downstream CI can be computed using the Petroleum Refinery Life Cycle Inventory Model (PRELIM) and Oil Products Emissions Module (OPEM) respectively²⁶. The versions used in this study to compute CI are PRELIM 1.6 and OPEM 1.1. The PRELIM model requires detailed information on crude assays, such as API gravity, sulphur content, and various boiling points. This data is typically not made available by producers, and publicly available data usually does not follow the same format and is therefore unusable in the models. The models' authors therefore included data on 657 unique crude assays, which I match to the Rystad assets they originate from.

Running the model. The PRELIM model allows for a large array of refinery configurations, designed to handle different types of oils and aimed at maximizing the production of specific oil products. Since I do not have sufficient information to match Rystad assets to the refineries treating their oil production, I follow the assumptions of the model on refinery configurations. Among the three main configurations used worldwide, the model selects the adapted configuration based on crude oil characteristics in terms of API gravity and sulphur content. All heavy crudes (API gravity below 22) are assigned the “deep conversion” configuration, entailing additional treatments aimed at reducing the length of carbon chains yielded. Light crudes with low sulphur levels (API gravity above 32 and sulphur content below 0.5%) are assigned the “hydroskimming” configuration, associated with fewer treatment stages and in turn lower emissions. All other crudes are refined using the “medium conversion” configuration. The importance of refinery configuration is illustrated

²⁶Information and most recent versions of the models can be found at: <https://ociplus.rmi.org/methodology>.

in Figure 19, depicting PRELIM-computed midstream CI. Intensive treatment of crude in the case of deep conversion is associated with higher emissions, while hydroskimming units yield much smaller emissions.

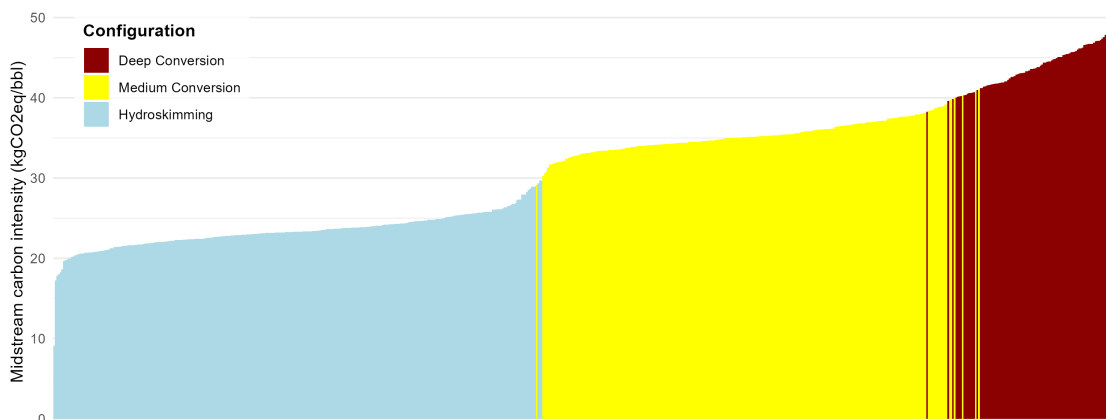


Figure 19: PRELIM-computed midstream CI.

Once midstream emissions are computed, PRELIM output sheets are imported to the OPEM model in order to estimate downstream CI of each crude stream. Emissions from transport and final combustion (for oil products used as fuel) are computed, giving me for a given barrel of crude oil its final products yield and associated emissions. Final combustion represents the largest share of emissions over the oil lifecycle, and displays smaller variability across crudes.

C.2.2 Matching PRELIM crude assays to Rystad assets

Crude streams available in the PRELIM model are matched to Rystad assets using data made available by Rystad on crude streams and associated assets. The matching is performed using fuzzy matching methods on crude names. After cleaning crude names from both sources, Jaro-Winkler distances were calculated and capped at 0.25. Matches were then manually verified, leaving me with 493 crude assays from PRELIM (from the original 657) matched to 392 crude streams from Rystad²⁷. Additionally, 25 unmatched PRELIM assays were manually matched to 41 Rystad assets with no specified crude streams. Overall, 518 PRELIM crude assays were linked to 8,386 Rystad assets, representing 64.4% of world production in 2024.

²⁷The difference in number comes from unique crude streams being split into different crude assays in PRELIM differing by the data source. For instance, the “Hibernia” crude in Canada is split into 3 assays, corresponding to information on this specific crude obtained from ExxonMobil, Statoil and Chevron.

A unique crude stream often consists of a blend of crude oils from multiple assets, but in certain cases (mainly in Canada), production in a unique asset is split between different crudes. For these assets, midstream and downstream CI from PRELIM/OPEM were averaged weighting by the share of production being allocated to each crude stream.

C.2.3 Estimation models

Model and sample restrictions. Using PRELIM output, I fit a regression model to explain the variance in midstream and downstream CI separately. In both cases, I use the two most influential variables, API gravity and sulphur content, which are also available at the asset level in Rystad data. Refinery configuration plays the most important role in explaining variations. I follow the assumptions made in the PRELIM model and assign “deep conversion”, “hydroskimming” and “medium conversion” based on the same rules on API gravity and sulphur content. Both variables (in log form for sulphur content) are subsequently introduced in polynomial form to account for as much of the residual variation as possible. Higher-order terms are iteratively dropped to avoid overfitting based on predicted R-squared. The final models used to explain midstream and downstream carbon emissions are as follows:

$$CI_f^{PRELIM} = \sum_1^3 \beta_i^C RefineryConfig_{i,f} + \beta_4 API_f + \beta_5 API_f^2 + \beta_6 API_f^3 + \beta_7 \log(Sulphur) + \beta_8 \log(Sulphur)^2 + \epsilon_f \quad (24)$$

$$CI_f^{OPEM} = \sum_1^3 \beta_i^C RefineryConfig_{i,f} + \beta_4 API_f + \beta_5 API_f^2 + \beta_6 API_f^3 + \beta_7 API_f^4 + \beta_8 \log(Sulphur) + \beta_9 \log(Sulphur)^2 + \beta_{10} \log(Sulphur)^3 + \epsilon_f \quad (25)$$

Regression results. Table 13 shows results from both estimations. All coefficients are significant and predicted R-squared are 0.95 and 0.88 respectively. As expected, refinery configuration explains a large share of variations across crudes, especially in the case of midstream emissions. Adding polynomial terms in API gravity and log sulphur content improves the variance explained marginally, as the relatively low within R-squared indicates. By contrast, API gravity and sulphur content play a much more important role in explaining the variance in downstream emissions.

Sensitivity analysis. Figure 20 compares midstream and downstream CI to the values predicted by the model. Midstream values appear grouped according to each refinery configuration, illustrating the crucial role played by the dummies in the regression results.

Variables	Midstream	Downstream
Configuration: Deep conversion	56.552*** (1.374)	530.269*** (17.341)
Configuration: Hydroskimming	40.093*** (1.608)	587.056*** (17.764)
Configuration: Medium conversion	51.361*** (1.604)	515.626*** (18.037)
API	-0.933*** (0.124)	-16.122*** (2.159)
API ²	0.016*** (0.003)	0.781*** (0.096)
API ³	-9.62e-05*** (2.74e-05)	-0.017*** (0.002)
API ⁴		1.11e-04*** (1.16e-05)
log(Sulphur)	-0.988*** (0.103)	-7.784*** (0.832)
log(Sulphur) ²	-0.135*** (0.025)	-2.404*** (0.474)
log(Sulphur) ³		-0.19*** (0.072)
Corrected R ²	0.948	0.887
Predicted R ²	0.947	0.883
Within R ²	0.300	0.875

Table 13: Regression results from midstream and downstream CI estimations.

Notes: The table displays coefficients from the regressions using equations (24) and (25), with standard errors in parentheses. R² are corrected in order to account for the absence of intercept, as well as predicted R². Within R² are computed using fixed-effects regressions to capture the effect of configuration dummies.

While the inclusion of API gravity and sulphur content in the model explains a large share of within-group variation for medium and deep conversion, it does not appear to be the case for crudes using the hydroskimming configuration (bottom-left of the midstream panel), explaining the low within R-squared value. Downstream CI are more evenly distributed across the graph, and predicted values remain close to OPEM-computed ones.

Contrary to upstream CI, midstream and downstream CI computed using PRELIM and OPEM models do not contain visible outliers. The risk of overfitting particular values is thus low. Tables 14 and 15 report re-estimated coefficients excluding most-influential observations for midstream and downstream CI respectively. In both cases, estimates remain significant and stable enough to exclude the risk of overfitting.

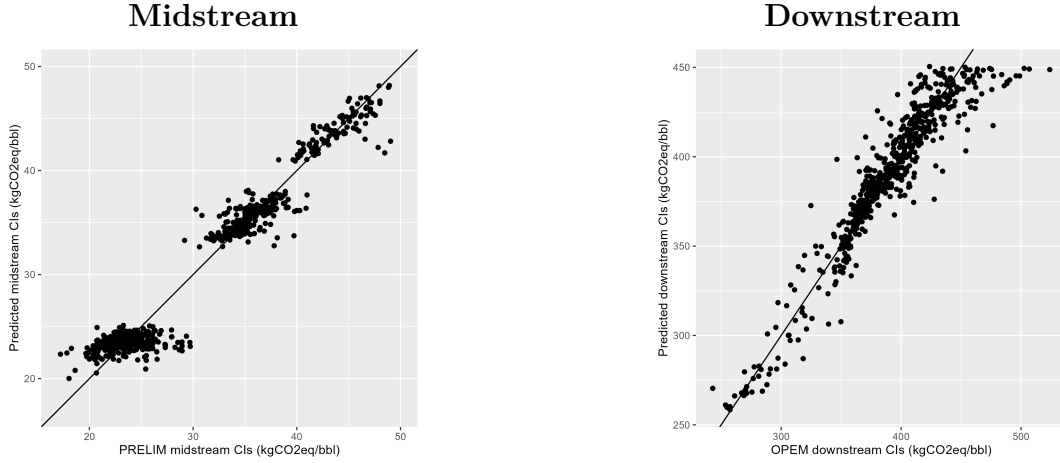


Figure 20: Comparison between PRELIM/OPEM CI and predicted ones.

	Deep Conversion	Hydroskimming	Medium Conversion	API
0	56.55*** (1.37)	40.09*** (1.61)	51.36*** (1.60)	-0.93*** (0.12)
1	56.38*** (1.30)	40.05*** (1.52)	51.22*** (1.51)	-0.91*** (0.12)
10	56.69*** (1.23)	40.58*** (1.44)	51.73*** (1.44)	-0.93*** (0.11)
50	56.56*** (1.05)	40.80*** (1.22)	51.91*** (1.21)	-0.89*** (0.10)
	API ²	API ³	log(Sulphur)	log(Sulphur) ²
0	0.0159***	-9.62e-05***	-0.99*** (0.10)	-0.13*** (0.03)
1	0.0152***	-9.03e-05***	-1.00*** (0.10)	-0.13*** (0.02)
10	0.0153***	-8.77e-05***	-1.10*** (0.09)	-0.18*** (0.02)
50	0.0126***	-5.57e-05***	-1.18*** (0.08)	-0.22*** (0.02)

Table 14: Midstream CI estimation: coefficient stability to influential observations (Cook's distance).

Notes: Each panel reports coefficients re-estimated after excluding the 0, 1, 10, and 50 most influential observations according to Cook's distance. Standard errors are reported in parentheses.

	Deep Conversion	Hydroskimming	Medium Conversion	API	API ²
0	530.27*** (17.34)	587.06*** (17.76)	515.63*** (18.04)	-16.12*** (2.16)	0.78*** (0.10)
1	543.82*** (18.09)	599.01*** (18.32)	527.93*** (18.62)	-18.39*** (2.33)	0.90*** (0.11)
10	544.71*** (16.65)	600.12*** (16.83)	528.63*** (17.12)	-18.46*** (2.15)	0.91*** (0.10)
50	555.93*** (14.05)	609.59*** (14.18)	540.02*** (14.41)	-19.59*** (1.83)	0.94*** (0.09)
	API ³	API ⁴	log(Sulphur)	log(Sulphur) ²	log(Sulphur) ³
0	-1.66e-02***	1.11e-04***	-7.78*** (0.83)	-2.40*** (0.47)	-0.19*** (0.07)
1	-1.92e-02***	1.30e-04***	-7.81*** (0.83)	-2.24*** (0.48)	-0.16*** (0.07)
10	-1.93e-02***	1.31e-04***	-7.60*** (0.76)	-2.40*** (0.45)	-0.15*** (0.07)
50	-1.95e-02***	1.30e-04***	-7.31*** (0.60)	-2.74*** (0.36)	-0.21*** (0.06)

Table 15: Downstream CI estimation: coefficient stability to influential observations (Cook's distance).

Notes: Each panel reports coefficients re-estimated after excluding the 0, 1, 10, and 50 most influential observations according to Cook's distance. Standard errors are reported in parentheses.

C.2.4 Predicting midstream/downstream CI

Rystad assets matched to PRELIM crudes are assigned midstream and downstream CIs computed using the PRELIM and OPEM models. These assets represent 59% of oil reserves in 2016. For remaining assets, I use the coefficients estimated in both regressions to predict their CI, at the crude level in the case of crude streams unmatched to PRELIM crude assays (17.3% of reserves) and at the asset level if the crude stream is unspecified (11.7%).

Several assets, representing 12% of world reserves, lack information on either API gravity or sulphur content. For these assets, reserve-weighted averages from similar oils were computed, according to the following order: reservoir, sub-basin, similar API within the same country, same main oil type within the same country, and finally same main oil type. The vast majority of assets are attributed values computed within the same reservoir or sub-basin. Since geographically close assets are more likely to be blended together before refining, averages computed from nearby assets are preferable. Table 16 sums up the methods and granularity levels of midstream and downstream CI predictions.

Method	Granularity level	Share of 2016 reserves
PRELIM/OPEM	Crude stream	58.9%
Predicted	Crude stream	17.3%
	Asset-specific	11.7%
Average	Reservoir	5.4%
	Sub-basin	5.7%
	API category $\times$ country	0.7%
	Oil type $\times$ country	0.2%
	Oil type	< 0.1%

Table 16: Data sources and aggregation levels for midstream and downstream carbon intensity values.

D Additional results

D.1 Overshoot against emissions under the timely policy

Figure 21 draws the bands of Figure 3 on top of cumulative emissions under the timely policy. Emissions under the timely policy fall from 635 GtCO₂eq at a zero tax to the 350.3 GtCO₂eq budget at \$57.2, so that at the timely tax the overshoot adds 19% to what the budget allows. The top band, the part a tax set in 2030 can still act on, is a thin layer

on top of a much larger stock, which is why a small error in the calibration of the 2030 tax matters.

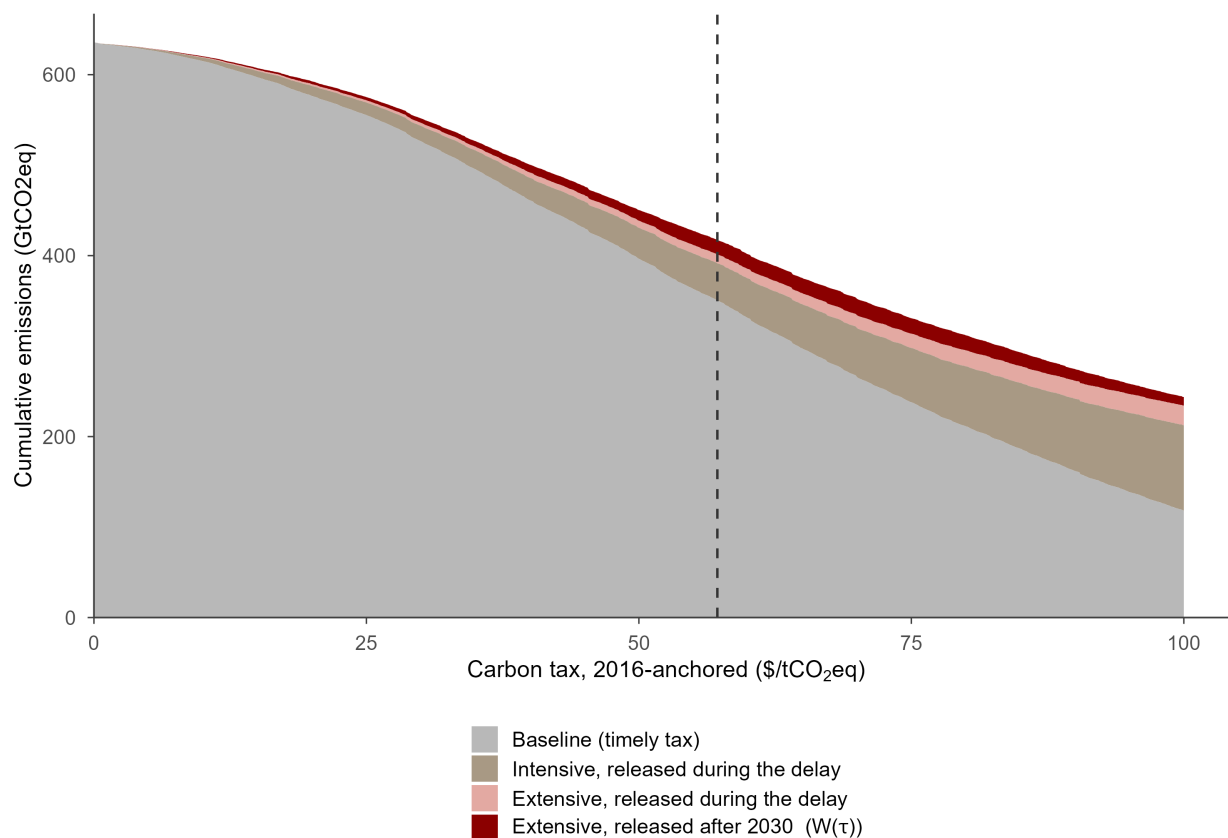


Figure 21: Overshoot caused by the delay and emissions under a timely policy.

Notes: Cumulative emissions under the timely policy (grey) and overshoot caused by the delay, by margin and date of emission. The dashed line is the tax that meets the budget under a timely policy. Taxes are 2016-anchored levels of the tax rising at 2% per year.

D.2 Overshoot by supply segment and region

Figure 22 shows the overshoot bands of Figure 4 without the emissions under the timely policy, which makes the post-2030 band of the onshore and shelf segments visible. Figure 23 shows the same decomposition by region, which Section 4.1 discusses.

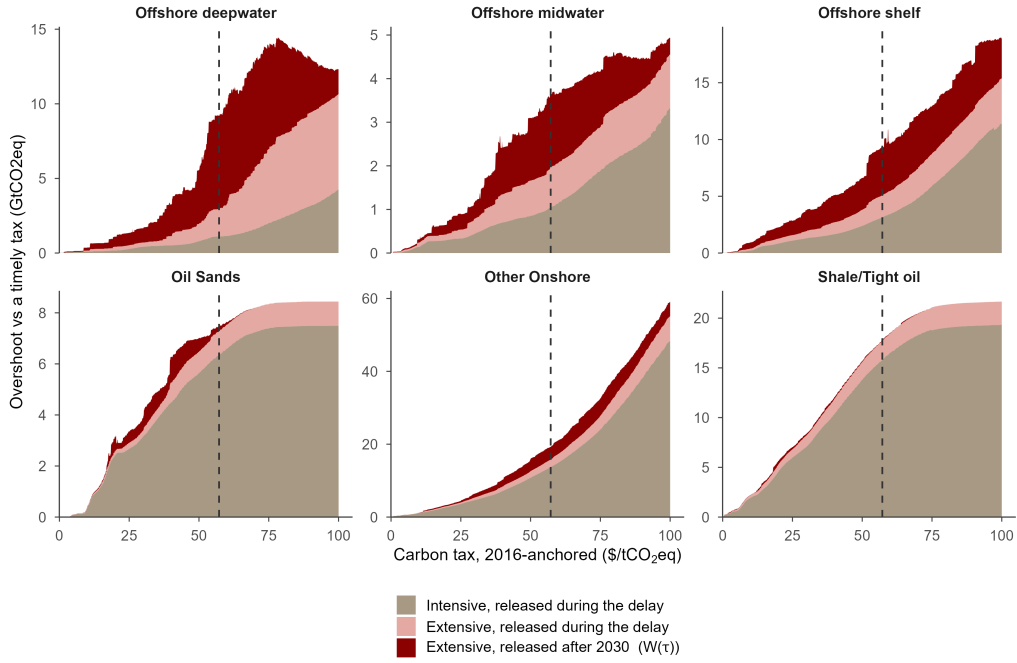


Figure 22: Overshoot caused by the delay by supply segment, margin and date of emission.

Notes: Panels have different vertical scales. The dashed line is the tax that meets the budget under a timely policy. Taxes are 2016-anchored levels of the tax rising at 2% per year.

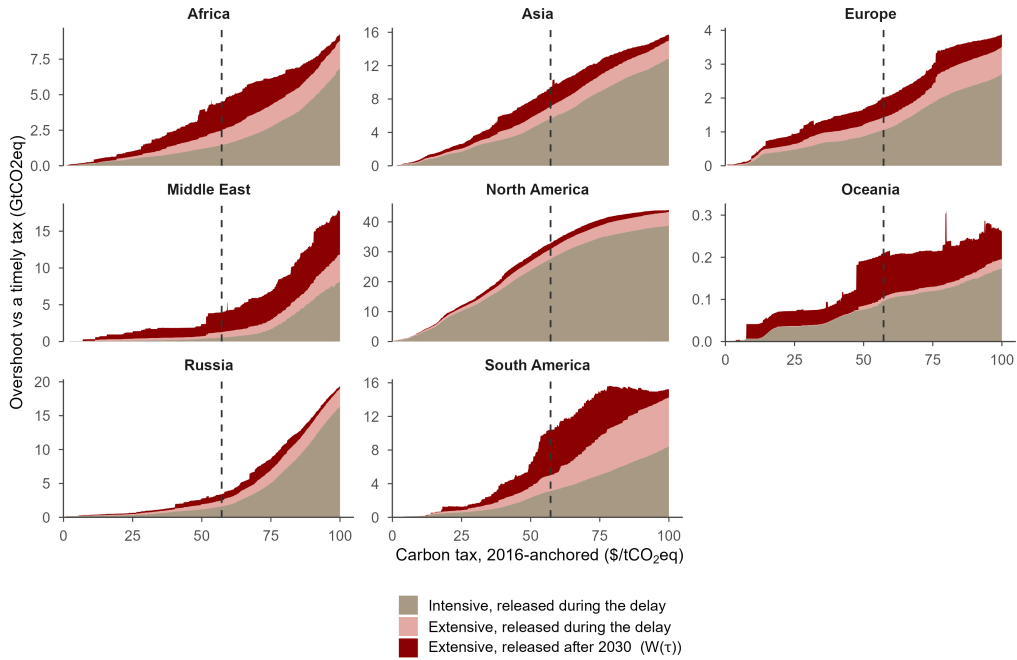


Figure 23: Overshoot caused by the delay by region, margin and date of emission.

Notes: Panels have different vertical scales. The dashed line is the tax that meets the budget under a timely policy. Taxes are 2016-anchored levels of the tax rising at 2% per year.

D.3 Capital intensity and the depth of lock-in, field by field

Figure 24 plots the depth of each field’s lock-in zone against its capital intensity. Panel (a) measures depth in tax units, $\bar{\tau}_P(0) - \bar{\tau}_D$, and shows that fields keep the ordering of Table 2 individually: shale fields, at the left of the distribution, have shallow zones, and offshore fields, at the right, have deep ones, despite the variation in carbon intensity and operating costs across fields. Panel (b) measures depth per barrel, $\theta(\bar{\tau}_P(0) - \bar{\tau}_D)$, which removes carbon intensity, on logarithmic scales. The fitted slope is 0.55, estimated on the 3,023 fields of the delay cohort with a defined capital intensity. The model of Section 2.2 predicts 0.5 without discounting and a higher value at any positive discount rate, so the estimate lies on the side the calibration implies. Because development capital enters both the horizontal axis and the full-cycle breakeven, part of the fit reflects this common term, and the goodness of fit should not be read as evidence for the model; the slope, which describes how the depth of lock-in scales with capital intensity, is the informative statistic.

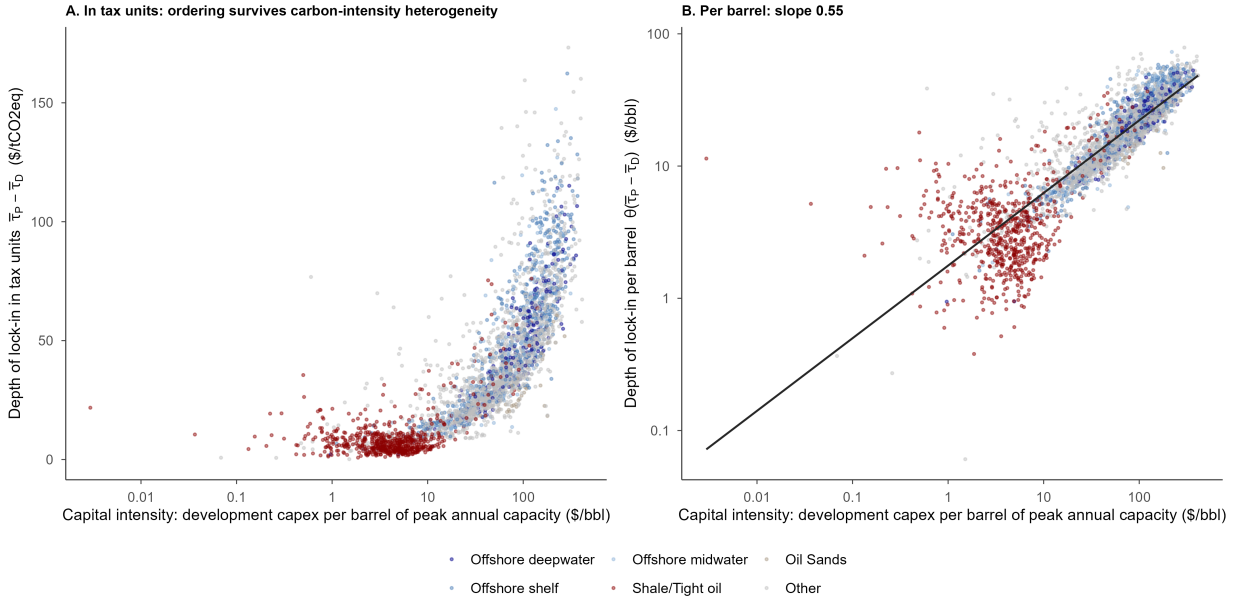


Figure 24: Capital intensity and the depth of lock-in, field by field.

Notes: Each point is one field of the delay cohort. Capital intensity is development capital expenditure per barrel of peak annual capacity. (a) Depth of the lock-in zone in tax units; (b) depth per barrel, logarithmic scales, with the fitted line.

D.4 Constant carbon tax

This appendix reports the results for a constant tax that meets the same budget under a timely policy, \$99.4/tCO₂eq. The delay then overshoots the budget by 119 GtCO₂eq, 34% of

the budget, of which 78 GtCO₂eq on the intensive margin and 41 GtCO₂eq on the extensive margin. The intensive margin is larger than under the increasing tax because a constant tax meeting the same budget is higher in the early years, so the timely policy removes more production during the delay. Table 17 is the counterpart of Table 4. In 2030, the lock-in-adjusted tax is \$133.5, the premium \$3.2 and the residual overshoot 18.9 GtCO₂eq, 13.1% of the remaining budget. Neither the premium nor the residual overshoot rises steadily with the length of the delay, because large fields enter and leave their lock-in zone one at a time, but by 2030 both are of the same order as under the increasing tax. The lock-in component of the cost of delay is therefore of the same size under both designs, while the total cost of the delay is not (Section 4.4).

Year	Budget (GtCO ₂ eq)	Adjusted tax (\$/tCO ₂ eq)	From 2016		From 2025	
			Premium (\$/tCO ₂ eq)	Residual overshoot (GtCO ₂ eq)	Premium (\$/tCO ₂ eq)	Residual overshoot (GtCO ₂ eq)
2016	350.3	99.4	—	—	—	—
2018	320.9	105.2	2.8	7.4 (2.3%)	—	—
2020	291.4	109.7	2.4	9.9 (3.4%)	—	—
2022	264.0	112.8	2.1	12.3 (4.7%)	—	—
2024	235.1	118.0	3.4	15.6 (6.6%)	—	—
2026	205.8	125.6	5.6	15.8 (7.7%)	0.1	2.6 (1.3%)
2028	175.6	129.8	2.5	19.4 (11.0%)	0.0	7.3 (4.2%)
2030	144.9	133.5	3.2	18.9 (13.1%)	2.4	10.3 (7.1%)

Table 17: Lock-in premium and residual overshoot by year of implementation, constant carbon tax.

Notes: Taxes in 2016 dollars. Overshoot percentages are relative to the remaining budget. “From 2016” and “From 2025” indicate the year from which the delay is measured. Taxes, premiums and overshoots move in discrete steps because large fields enter and leave their lock-in zone one at a time. When a large locked-in field is phased out just above the naive tax, the lock-in-adjusted tax falls on the same step and the premium rounds to zero while the residual overshoot, measured at the naive tax, stays positive (2028, from 2025).

D.5 Depth of lock-in

Figure 25 reports, for all delay-induced fields together, the emissions locked in at the naive tax that survive each level of the 2030 tax. The stock falls to 92% of its initial level at the lock-in-adjusted tax, and still stands at 70% at \$80 and 43% at \$90. Section 5.1 reads the same object by supply segment. The curve falls slowly over the range the budget makes relevant and steeply above it, which is the aggregate counterpart of the fact that a locked-in field stops producing only when the tax exceeds what its operating costs can bear.

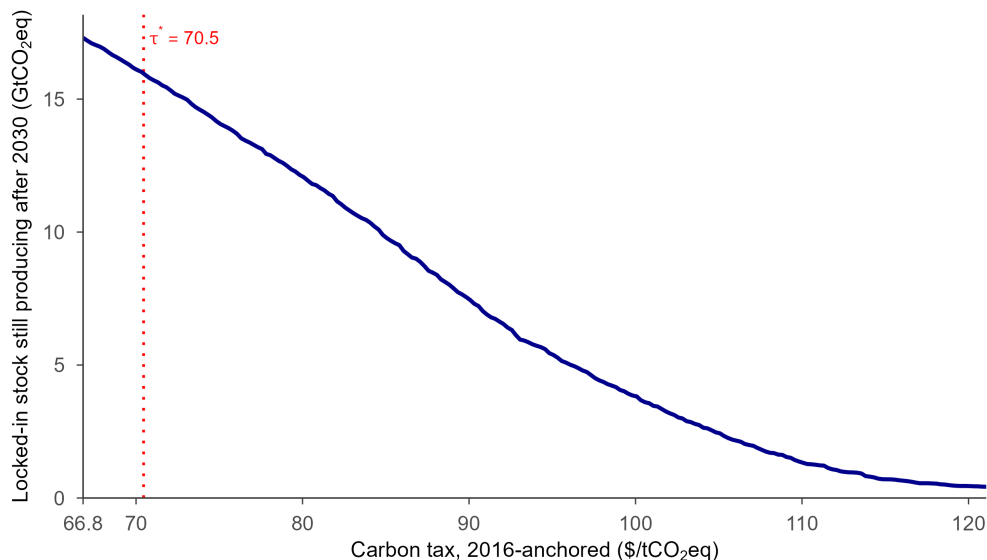


Figure 25: Depth of lock-in, all delay-induced fields.

Notes: Post-2030 emissions locked in at the naive tax that are still released when the tax implemented in 2030 is raised to the level on the horizontal axis, all delay-induced fields, in GtCO₂eq. The horizontal axis starts at the naive tax, $\tau^N = \$66.8$, and the dotted line marks the lock-in-adjusted tax, $\tau^* = \$70.5$. Taxes are 2016-anchored levels of the tax rising at 2% per year.

D.6 The largest locked-in assets

Table 18 lists the twenty assets that carry the most post-2030 emissions at the naive tax, out of the 694 that carry any. They account for 31.9% of the total, and fifty assets account for 56.1%, which is what the size-based bans of Section 5.3 remove. The year of responsibility is the year from which the asset would be completed even if the tax were implemented immediately (Section 3.4).

Asset	Country	Segment	Year of responsibility	Emissions (GtCO ₂ eq)	Share	Cumulative
Marjan (Expansion)	Saudi Arabia	Offshore shelf	2023	0.87	5.0%	5.0%
Atapu (P-84)	Brazil	Offshore deepwater	2027	0.34	1.9%	7.0%
Venus Phase 1	Namibia	Offshore deepwater	2029	0.33	1.9%	8.9%
Tengiz FGP	Kazakhstan	Onshore conventional	2019	0.31	1.8%	10.7%
Satah Al Razboot (SARB)	UAE	Offshore shelf	2016	0.27	1.6%	12.2%
Buzios XI (P-83)	Brazil	Offshore deepwater	2025	0.27	1.5%	13.8%
Sepia (P-85)	Brazil	Offshore deepwater	2028	0.26	1.5%	15.3%
Kirkuk (redevelopment)	Iraq	Onshore conventional	2028	0.26	1.5%	16.8%
Buzios IX (P-80)	Brazil	Offshore deepwater	2026	0.25	1.5%	18.2%
Buzios X (P-82)	Brazil	Offshore deepwater	2026	0.25	1.4%	19.7%
Willow (GMT)	United States	Onshore conventional	2026	0.24	1.4%	21.0%
Bacalhau Phase 2	Brazil	Offshore deepwater	2029	0.23	1.3%	22.4%
Longtail	Guyana	Offshore deepwater	2028	0.23	1.3%	23.7%
Argos (Mad Dog Phase 2)	United States	Offshore deepwater	2020	0.23	1.3%	25.0%
Al Shaheen Ruya 3 (Batch 1)	Qatar	Offshore shelf	2026	0.22	1.3%	26.3%
Hammerhead	Guyana	Offshore midwater	2029	0.21	1.2%	27.5%
Hail (Sour Gas)	UAE	Offshore shelf	2027	0.20	1.1%	28.6%
Bacalhau Phase 1	Brazil	Offshore deepwater	2022	0.20	1.1%	29.7%
Azadegan North Phase 2	Iran	Onshore conventional	2028	0.19	1.1%	30.9%
Trion	Mexico	Offshore deepwater	2027	0.19	1.1%	31.9%

Table 18: The twenty largest locked-in assets.

Notes: Post-2030 emissions of delay-induced fields surviving the naive tax implemented in 2030, no-anticipation scenario, tax rising at 2% per year. Asset names are those of the Rystad database.

D.7 Incidence by country

Table 19 reports the decomposition of Table 5 for the largest producers of each region and for the producers discussed in Section 5.2.

	Overproduction (GtCO ₂ eq)	Lock-in (GtCO ₂ eq)	Net (GtCO ₂ eq)	Net (% of 2016 reserves)
<i>Net winners through overproduction mainly</i>				
United States	18.63	1.40	20.03	17.5
Canada	9.36	−0.10	9.27	16.0
China	3.99	0.67	4.66	17.0
Norway	0.47	0.33	0.80	10.0
United Kingdom	0.50	0.09	0.58	17.6
Mexico	0.38	0.17	0.55	6.8
<i>Net winners through lock-in mainly</i>				
Guyana	0.19	0.73	0.92	16.3
Nigeria	0.04	0.31	0.36	3.9
<i>Sign reversed by lock-in</i>				
Brazil	−0.38	3.04	2.66	13.0
Angola	−0.08	0.24	0.15	2.0
Namibia	−0.13	0.23	0.10	4.6
Senegal	−0.02	0.09	0.07	23.3
Vietnam	−0.00	0.04	0.04	4.2
<i>Net losers despite lock-in gains</i>				
Kazakhstan	−0.83	0.34	−0.50	−4.7
Qatar	−1.21	0.27	−0.94	−5.0
Azerbaijan	−0.43	0.01	−0.41	−11.7
UAE	−3.23	0.09	−3.14	−8.1
<i>Net losers on both counts</i>				
Saudi Arabia	−10.60	−2.53	−13.13	−10.4
Iraq	−7.35	−1.98	−9.34	−17.7
Russia	−5.41	−2.13	−7.54	−11.8
Iran	−3.93	−0.83	−4.76	−10.0
Kuwait	−1.65	−0.95	−2.60	−11.4
Algeria	−0.21	−0.01	−0.22	−4.2

Table 19: Incidence of the delay by country.

Notes: Change in cumulative emissions over 2016–2100 relative to a timely tax, no-anticipation scenario, tax rising at 2% per year. The last column is the net change in production (in barrels) as a share of the country’s 2016 reserves.

D.8 Incidence by firm

Table 20 reports the decomposition of Table 5 for the twenty largest companies by 2016 equity reserves. Production is attributed to equity holders at their share in each year, so a firm keeps what it lifted while it held the asset, and ownership beyond 2026 is held at its 2026 level. Figure 11 plots the same decomposition for the fifty largest firms, normalised by their own reserves.

Company	Type	Headquarters	Reserves (Gtbl)	Overproduction (GtCO ₂ eq)	Lock-in (GtCO ₂ eq)	Net (GtCO ₂ eq)	Net (% res.)
Saudi Aramco	NOC	Saudi Arabia	289.7	-10.62	-2.50	-13.13	-10.4
NIOC	NOC	Iran	99.1	-3.92	-0.82	-4.74	-10.2
ADNOC	NOC	UAE	64.8	-2.08	0.02	-2.06	-7.5
Kuwait Petroleum Corp	NOC	Kuwait	56.1	-1.79	-0.97	-2.77	-11.4
PetroChina	NOC	China	52.7	0.56	-0.40	0.16	0.5
Rosneft	NOC	Russia	52.3	-2.90	-1.37	-4.28	-17.7
ExxonMobil	Major	United States	46.4	1.43	0.44	1.87	7.9
QatarEnergy	NOC	Qatar	37.5	-1.05	0.35	-0.70	-4.9
Shell	Major	United Kingdom	36.2	-0.15	0.15	0.00	-0.6
Petrobras	NOC	Brazil	33.1	-0.09	2.00	1.91	12.1
Chevron	Major	United States	29.9	2.09	0.23	2.32	15.7
Gazprom	NOC	Russia	29.8	-1.11	-0.19	-1.30	-10.3
Lukoil	Integrated	Russia	24.7	-0.47	-0.49	-0.97	-8.2
BP	Major	United Kingdom	24.4	-0.44	0.06	-0.38	-3.4
TotalEnergies	Major	France	22.5	-0.53	0.98	0.45	3.1
CNOOC	NOC	China	20.5	0.95	1.08	2.04	20.1
ConocoPhillips	Independent	United States	16.8	2.23	0.26	2.50	32.3
Basra Oil Company	NOC	Iraq	14.8	-1.64	-0.82	-2.46	-34.8
Pemex	NOC	Mexico	14.2	0.36	-0.07	0.29	4.1
Canadian Natural Resources	Independent	Canada	13.0	2.12	-0.01	2.11	27.6

Table 20: Incidence of the delay by firm.

Notes: Change in cumulative emissions over 2016–2100 relative to a timely tax, no-anticipation scenario, tax rising at 2% per year. The twenty largest companies by 2016 equity reserves, excluding one residual category of unattributed United States partners. The last column is the net change in production (in barrels) as a share of the firm’s own 2016 equity reserves.

D.9 Definition of the targeted instruments

Each instrument of Table 6 restricts or taxes a set of assets from a starting year, and is then evaluated at the tax the remaining budget requires under that restriction.

Perimeters. A segment restriction applies to all assets of the segment, using the classification of Table 2. An ownership restriction applies to the assets that firms headquartered in the perimeter own at least half of *and* operate; Europe is the European Union together with Norway and the United Kingdom. A size restriction applies to the assets that carry the most post-2030 emissions at the naive tax in the delay without restriction, ranked as in Table 18; the twenty largest carry 31.9% of the residual overshoot and the fifty largest 56.1%.

Instruments. A ban on new approvals removes from the analysis every asset of the perimeter approved from the starting year onwards, so assets already approved are left alone and no capital already committed is cancelled. An early carbon tax applies to the assets of the perimeter from the starting year, at the level the budget requires, while the rest of the industry faces the tax only from 2030; it therefore acts on the development of new assets and on the production of assets already in operation. A shock to development capital raises the development cost of every asset of the perimeter by the stated percentage from the start-

ing year, leaving operating costs unchanged. In every case, the naive planner observes the restriction and its effect on development costs when calibrating the 2030 tax.

D.10 Other targeted instruments

Figure 26 reports the naive tax, the lock-in-adjusted tax, the premium between them and the residual overshoot for every instrument evaluated in Section 5.3. It covers the development bans, by segment, by perimeter of ownership and by size of project, the early carbon taxes applied to the same perimeters, and the two shocks to development capital, each of them starting in 2016 or in 2025 and each evaluated at its own budget-consistent tax.

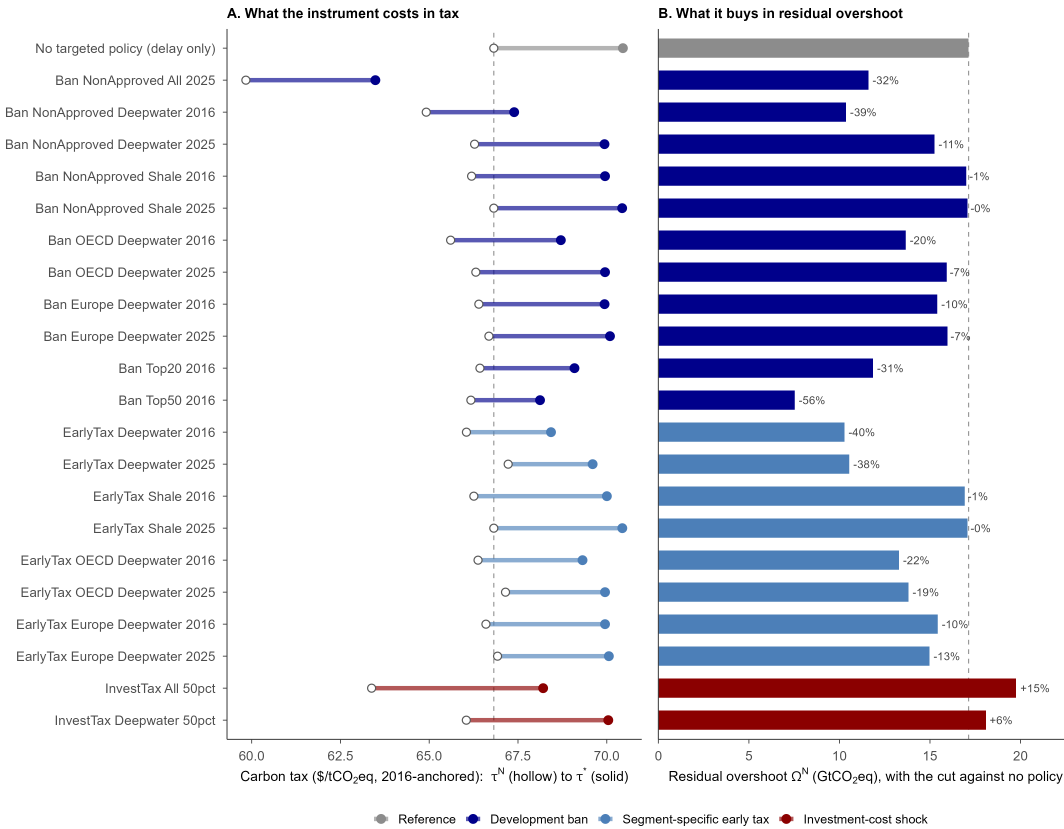


Figure 26: Targeted instruments.

Notes: Panel (a) runs from the naive tax (hollow) to the lock-in-adjusted tax (solid), so that the length of each segment is the premium. Panel (b) gives the residual overshoot and its change relative to the delay without restriction. Each instrument is evaluated at its own budget-consistent tax. Taxes are 2016-anchored levels of the tax rising at 2% per year.

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